INCH-POUND

MIL-M-38510/654A
9 May 2006

SUPERSEDING
MIL-M-38510/654
19 December 1986

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, HIGH-SPEED CMOS, LATCHES, MONOLITHIC SILICON

Reactivated after 9 May 2006 and may be used for new and existing designs and acquisitions.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product herein shall consist of this specification sheet and MIL-PRF 38535.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, high speed CMOS, logic microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete Part or Identifying Number (PIN). For this product, the requirements of MIL-M-38510 have been superseded by MIL-PRF-38535 (see 6.3).

1.2 Part or identifying number (PIN). The PIN is in accordance with MIL-PRF-38535 and as specified herein.

1.2.1 Device types. The device types are as follows:

Device type	Circuit
01	Quad bistable transparent latch
02	8-bit addressable latch
03	Octal D type transparent latch with three-state outputs
04	Octal transparent latch, three-state, inverting
05	Octal transparent latch, three-state, inverting
06	Octal transparent latch, three-state
53	Octal D-type transparent latch with three-state outputs and TTL compatible inputs

1.2.2 Device class. The device class is the product assurance level as defined in MIL-PRF-38535.

1.2.3 Case outlines. The case outlines are as designated in MIL-STD-1835 and as follows:

Outline letter	Descriptive designator	Terminals	Package style
E	GDIP1-T16 or CDIP2-T16	16	Dual-in-line
F	GDFP2-F16 or CDFP3-F16	16	Flat pack
R	GDIP1-T20 or CDIP2-T20	20	Dual-in-line
S	GDFP2-F20 or CDFP3-F20	20	Flat pack
2	CQCC1-N20	20	Square leadless chip carrier

Comments, suggestions, or questions on this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or email CMOS@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil/.

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1.3 Absolute maximum ratings.

Supply voltage range (V_{CC})	-0.5 V dc to +7.0 V dc
DC input voltage range (V_{IN})	-0.5 V dc to V_{CC} + 0.5 V dc
DC output voltage range (V_{OUT})	-0.5 V dc to V_{CC} + 0.5 V dc
Clamp diode current (I_{IK}, I_{OK})	±20 mA
DC output current per pin (I_{OUT}):	
Device types 01, 02	±25 mA
Device types 03-06, 53	±35 mA
DC V_{CC} or GND current per pin (I_{CC}, I_{GND}):	
Device types 01, 02	±50 mA
Device types 03-06, 53	±70 mA
Storage temperature range (T_{STG})	-65°C to +150°C
Maximum power dissipation (P_D)	300 mW
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction to case (θ_{JC})	See MIL-STD-1835
Junction temperature (T_J)	175°C

1.4 Recommended operating conditions.

Device types 01 to 06:

Maximum input low voltage (V_{IL})	0.3 V at V_{CC} = 2.0 V 0.9 V at V_{CC} = 4.5 V 1.2 V at V_{CC} = 6.0 V
Minimum input high voltage (V_{IH})	1.5 V at V_{CC} = 2.0 V 3.15 V at V_{CC} = 4.5 V 4.2 V at V_{CC} = 6.0 V
Supply voltage (V_{CC})	2.0 V dc to 6.0 V dc
Input voltage (V_{IN})	0.0 V to V_{CC}
Operating temperature range (T_A)	-55°C to +125°C
Maximum input rise and fall times (t_r, t_f):	
V_{CC} = 2.0 V	1000 ns
V_{CC} = 4.5 V	500 ns
V_{CC} = 6.0 V	400 ns

Device type 53:

Maximum input low voltage (V_{IL})	0.8 V at V_{CC} = 4.5 V and 5.5 V
Minimum input high voltage (V_{IH})	2.0 V at V_{CC} = 4.5 V and 5.5 V
Supply voltage (V_{CC})	4.5 V dc to 5.5 V dc
Input voltage	0.0 V to V_{CC}
Operating temperature range (T_A)	-55°C to +125°C
Maximum input rise and fall time (t_r, t_f):	
V_{CC} = 4.5 V	500 ns
Width of clock pulse (t_p clock) (V_{CC} = 4.5 V minimum):	
Device types 01, 02, 05, 06	24 ns minimum
Device types 03, 04, 53	30 ns minimum
Device type 02 reset	24 ns minimum
Data setup time before clock (t_{setup}) (V_{CC} = 4.5 V minimum):	
Device types 01, 02, 03, 04, 53	30 ns minimum
Device types 05, 06	23 ns minimum
Data hold time after clock (t_{hold}) (V_{CC} = 4.5 V minimum):	
Device types 01, 02,	8 ns minimum
Device types 03-06, 53	15 ns minimum

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and Standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATION

MIL-PRF-38535 - Integrated Circuits (Microcircuits) Manufacturing, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-883 - Test Method Standard Microcircuits.
MIL-STD-1835 - Interface Standard Electronic Component Case Outlines.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Microcircuits furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.3 and 6.4).

3.2 Item requirements. The individual item requirements shall be in accordance with MIL-PRF-38535 and as specified herein or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not affect the form, fit, or function as described herein.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-PRF-38535 and herein.

3.3.1 Logic diagrams and terminal connections. The logic diagrams and terminal connections shall be as specified on figure 1.

3.3.2 Truth tables. The truth tables shall be as specified on figure 2.

3.3.3 Switching time test circuit and waveforms. The switching time test circuit and waveforms shall be as specified on figure 3.

3.3.4 Schematic circuits. The schematic circuits shall be maintained by the manufacturer and made available to the qualifying activity or preparing activity upon request.

3.3.5 Case outlines. The case outlines shall be as specified in 1.2.3 and MIL-STD-1835.

3.4 Lead material and finish. The lead material and finish shall be in accordance with MIL-PRF-38535 (see 6.6).

3.5 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics are as specified in table I, and apply over the full recommended ambient operating temperature range.

3.6 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.7 Marking. Marking shall be in accordance with MIL-PRF-38535.

3.8 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 38 (see MIL-PRF-38535, appendix A).

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4. VERIFICATION

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-PRF-38535 or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not affect the form, fit, or function as described herein.

4.2 Screening. Screening shall be in accordance with MIL-PRF-38535 and shall be conducted on all devices prior to qualification and conformance inspection. The following additional criteria shall apply:

a. The burn-in test duration, test condition, and test temperature, or approved alternatives shall be as specified in the device manufacturer's QM plan in accordance with MIL-PRF-38535. The burn-in test circuit shall be maintained under document control by the device manufacturer's Technology Review Board (TRB) in accordance with MIL-PRF-38535 and shall be made available to the acquiring or preparing activity upon request. The test circuit shall specify the inputs, outputs, biases, and power dissipation, as applicable, in accordance with the intent specified in method 1015 of MIL-STD-883.

b. Delete the sequence specified as interim (pre-burn-in) electrical parameters through interim (post-burn-in) electrical parameters of table IA of MIL-PRF-38535 and substitute lines 1 through 7 of table II herein.

c. Burn-in (method 1015 of MIL-STD-883).

(1) Unless otherwise specified in the manufacturers QM plan for static tests (test condition A), ambient temperature (T_A) shall be +125°C minimum. Test duration for each static test shall be 24 hours minimum for class S devices and in accordance with table I of method 1015 for class B devices.

i. For static burn-in I, all inputs shall be connected to GND. Outputs shall be open or connected to $V_{CC}/2$. Resistors are optional on outputs if open. Resistors are required on inputs and on outputs connected to $V_{CC}/2$. R = 680Ω to 47 kΩ.

ii. For static burn-in II, all inputs shall be connected through a resistor to V_{CC}. Output shall be open or connected to $V_{CC}/2$. Resistors are optional on outputs if open. Resistors are required on inputs and on outputs connected to $V_{CC}/2$. R = 680Ω to 47 kΩ.

(2) Unless otherwise specified in the manufacturers QM plan for dynamic test (test condition D), ambient temperature shall be +125°C minimum. Test duration shall be in accordance with table I of method 1015.

i. For dynamic burn-in, all inputs shall be connected through the resistors in parallel to a common CP. Outputs shall be connected to $V_{CC}/2$ ±0.5 V through the resistors. R = 1 kΩ ±5% for outputs for device types 01 and 02. R = 680Ω ±5% for outputs for device types 03-06 and 53. R = 1 kΩ to 47 kΩ for inputs for device types 01 and 02, and R = 680Ω to 47 kΩ for inputs of device types 03-06 and 53.

ii. CP = 25 kHz to 1 MHz square wave; duty cycle = 50% ±15%; V_{IH} = 4.5 V to V_{CC}; V_{IL} = 0.0 V ±0.5 V; transition time ≤ 0.5 μs.

iii. For device types 01-06, V_{CC} = 6.0 V ±0.5 V. For device type 53, V_{CC} = 5.5 V ±0.5 V.

d. Interim and final electrical test parameters shall be as specified in table II.

e. For class S devices, post dynamic burn-in, or class B devices, post static burn-in, electrical parameter measurements may, at the manufacturer’s option, be performed separately or included in the final electrical parameter requirements.

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4.2.1 Percent defective allowable (PDA).

a. The PDA for class S devices shall be 5 percent for static burn-in and 5 percent for dynamic burn-in, based on the exact number of devices submitted to each separate burn-in.

b. Static burn-in I and II failure shall be cumulative for determining the PDA.

c. The PDA for class B devices shall be in accordance with MIL-PRF-38535 for static burn-in. Dynamic burn-in is not required.

d. Those devices whose measured characteristics, after burn-in, exceed the specified delta (Δ) limits or electrical parameter limits specified in table III, subgroup 1, are defective and shall be removed from the lot. The verified failures divided by the total number of devices in the lot initially submitted to burn-in shall be used to determine the percent defective for the lot and the lot shall be accepted or rejected based on the specified PDA.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-38535.

4.4 Technology Conformance inspection (TCI). Technology conformance inspection shall be in accordance with MIL-PRF-38535 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table III of MIL-PRF-38535 and as follows:

a. Tests shall be performed in accordance with table II herein.

b. Subgroups 5 and 6 shall be omitted.

c. Subgroup 4 (C_{in} and C_O measurements) shall be measured only for initial qualification and after process or design changes that may affect capacitance. Capacitance shall be measured between the designated terminal and V_{SS} at a frequency of 1 MHz.

d. Subgroups 9 and 11 shall be measured only for initial qualification and after process or design changes which may affect dynamic performance.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of MIL-PRF-38535.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table IV of MIL-PRF-38535 and as follows:

a. End-point electrical parameters shall be as specified in table II herein. Delta limits shall apply only to subgroup 1 of group C inspection and shall consist of tests specified in table IV herein.

b. The steady-state life test duration, test condition, and test temperature, or approved alternatives shall be as specified in the device manufacturer's QM plan in accordance with MIL-PRF-38535. The burn-in test circuit shall be maintained under document control by the device manufacturer's Technology Review Board (TRB) in accordance with MIL-PRF-38535 and shall be made available to the acquiring or preparing activity upon request. The test circuit shall specify the inputs, outputs, biases, and power dissipation, as applicable, in accordance with the intent specified in method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table V of MIL-PRF-38535. End-point electrical parameters shall be as specified in table II herein.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions 1/ T_C = +125°C unless otherwise specified		Device type	V_{CC}	Limits		Unit
						Min	Max	
High-level output voltage	V_{OH1} 2/	V_{IH} = 1.5 V V_{IL} = 0.3 V I_{OH} = -20 µA		01-06	2.0 V	1.95		V
	V_{OH2} 2/	V_{IH} = 3.15 V V_{IL} = 0.9 V I_{OH} = -20 µA		01-06	4.5 V	4.45		V
	V_{OH3}	V_{IH} = 4.2 V V_{IL} = 1.2 V I_{OH} = -20 µA		01-06	6.0 V	5.95		V
	V_{OH4} 2/	V_{IH} = 3.15 V V_{IL} = 0.9 V	I_{OH} = -6.0 mA I_{OH} = -4.0 mA	03-06 01,02	4.5 V	3.7		V
	V_{OH5}	V_{IH} = 4.2 V V_{IL} = 1.2 V	I_{OH} = -5.2 mA I_{OH} = -7.8 mA	01,02 03-06	6.0 V	5.2		V
	V_{OH6}	V_{IH} = 2.0 V V_{IL} = 0.8 V I_{OH} = -20 µA		53	4.5 V	4.4		V
	V_{OH7}	V_{IH} = 2.0 V V_{IL} = 0.8 V I_{OH} = -6.0 mA		53	5.5 V	3.7		V
Low-level output voltage	V_{OL1} 2/	V_{IL} = 0.3 V V_{IH} = 1.5 V I_{OL} = 20.0 µA		01-06	2.0 V		0.05	V
	V_{OL2} 2/	V_{IL} = 0.9 V V_{IH} = 3.15 V I_{OL} = 20.0 µA		01-06	4.5 V		0.05	V
	V_{OL3}	V_{IL} = 1.2 V V_{IH} = 4.2 V I_{OL} = 20.0 µA		01-06	6.0 V		0.05	V
	V_{OL4} 2/	V_{IL} = 0.9 V V_{IH} = 3.15 V	I_{OL} = 4.0 mA I_{OL} = 6.0 mA	01,02 03-06	4.5 V		0.4	V
	V_{OL5}	V_{IL} = 1.2 V V_{IH} = 4.2 V	I_{OL} = 5.2 mA I_{OL} = 7.8 mA	01,02 03-06	6.0 V		0.4	V
	V_{OL6}	V_{IH} = 2.0 V V_{IL} = 0.8 V I_{OL} = 20 µA		53	4.5 V		0.1	V
	V_{OL7}	V_{IH} = 2.0 V V_{IL} = 0.8 V I_{OL} = 6.0 mA		53	5.5 V		0.4	V
Positive input clamp voltage	V_{IC+}	I_{IN} = 1 mA T_C = +25°C		All	GND		1.5	V
Negative input clamp voltage	V_{IC-}	I_{IN} = -1 mA T_C = +25°C		All	OPEN		-1.5	V
Input current low	I_{IL}	V_{IN} = GND		01-06 53	6.0 V 5.5 V		-0.1	µA
Input current high	I_{IH}	V_{IN} = V_{CC}		01-05 53	6.0 V 5.5 V		0.1	µA

See footnotes at end of table.

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TABLE I. Electrical performance characteristics – Continued.

Test	Symbol	Conditions 1/ T_C = +125°C unless otherwise specified	Device type	V_{CC}	Limits		Unit
					Min	Max	
Short circuit output current	I_{OS1} 2/	V_O = GND V_I = GND or V_{CC}	01-02	2.0 V	-2	-50	mA
			03-06			-60	
	I_{OS2} 2/		01-02	4.5 V	-15	-150	
			03-06			-165	
	I_{OS3} 2/		01-02	6.0 V	-25	-180	
			03-06			-210	
	I_{OS4}		01-02	4.0 V	-10	-120	
			03-06, 53			-135	
	I_{OS5} 2/		53	5.5 V	-25	-210	
Quiescent supply current	I_{CC}	V_I = 6.0 V or GND	01	6.0 V		10	µA
			02-06			20	
	I_{CC}	V_I = 5.5 V or GND	53	5.5 V		20	
	I_{CCZ}	V_I = 6.0 V	03-06	6.0 V		10	
		V_I = 5.5 V	53	5.5 V			
Additional quiescent supply current per input pin (one unit load)	$I_{CC\Delta}$ 3/	V_{IL} = 0.8 V V_{IH} = 2.4 V, I_O = 0 Pin under test at 2.4 V, other pins at GND, 0.8 V, or V_{CC}	53	5.5 V		3.0	µA
Power dissipation capacitance	C_{PD} 2/ 4/	T_C = 25°C	01			46	pF
			02			33	
			03			51	
			53			53	
			04-06			50	
Three-state output leakage current low	I_{OZL} 5/	V_{OUT} = GND OC = V_{IH}	03-06	6.0 V		-2.0	µA
			53	5.5 V			
Three-state output leakage current high	I_{OZH} 5/	V_{OUT} = V_{CC} OC = V_{IH}	03-06	6.0 V		+2.0	µA
			53	5.5 V			
Input capacitance	C_{IN}	T_C = 25°C	All			10	pF
Three-state output capacitance	C_O		03-06, 53			20	
Propagation delay time, data to outputs	t_{PHL1}, t_{PLH1} 6/ 7/	C_L = 50 pF ± 10%	01	4.5 V	3	30	ns
			02		5	43	
			03		6	35	
			04		6	39	
			05,06		4	41	
			53		6	41	
Propagation delay time, enable or clock to outputs	t_{PHL2}, t_{PLH2} 6/ 7/		01		4	34	
			02		7	47	
			03,53		6	41	
			04		6	41	
			05		4	41	
			06		4	41	
Propagation delay time, A to outputs	t_{PHL3}, t_{PLH3} 6/, 7/		02		5	50	

See footnotes at end of table.

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TABLE I. Electrical performance characteristics – Continued.

Test	Symbol	Conditions 1/ T_C = +125°C unless otherwise specified	Device type	V_{CC}	Limits		Unit
					Min	Max	
Propagation delay time, clear to outputs	t_{PHL4} 6/, 7/	C_L = 50 pF ± 10%	02	4.5 V	5	36	ns
Output enable time	t_{PZH}, t_{PZL} 6/, 7/	C_L = 50 pF ± 10% R_L = 1kΩ ± 10%	03		5	35	
			04		5	35	
			05,06		4	35	
			53		5	41	
Output disable time	t_{PHZ}, t_{PLZ} 6/, 7/		03		5	35	
			04		5	35	
			05,06		3	35	
			53		5	41	
Transition delay time	t_{THL}, t_{TLH} 6/, 7/	C_L = 50 pF ± 10%	01		3	20	
			02		3	20	
			03,04, 53		2	16	
			05		2	16	
			06		2	16	

1/ Complete terminal conditions shall be as specified in table III.
2/ Guaranteed but not tested.
3/ I_{CC} + $I_{CC}\Delta$ = Total supply current.
4/ Power dissipation capacitance (C_{PD}) per latch.
5/ I_{OZL}: Set output to high state.
I_{OZH}: Set output to low state.
6/ Tested at V_{CC} = 4.5 V at 125°C for sample testing and at V_{CC} = 4.5 V at 25°C for screening. Guaranteed at other V_{CC} voltages and temperatures. See table IA and the exception in 4.4.1d.
7/ For propagation and transition delay times at V_{CC} = 2.0 V, increase limit by a factor of 5. For propagation and transition delay times at V_{CC} = 6.0 V, decrease limit by a factor of .85.

TABLE IA. Calculated dynamic values at -55°C /+25°C case temperature.

V_{CC}	T_C = °C	
	+125	-55/+25
2.0 V	5 ↑	5 x 0.75 ↑
4.5 V	1→ ↓	→0.75 ↓
6.0 V	0.85	0.85 x 0.75

NOTE: Normalized numbers (+125°C = 1). The 2.0 V and 6.0 V numbers are derived from their 4.5 V integer value (rounding off according 5/4).

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Line no.	MIL-PRF-38535 test requirements	Class S device 1/			Class B device 1/		
		Ref. par.	Table III Subgroups 2/	Table IV delta limits 3/	Ref. par.	Table III subgroups 2/	Table IV delta 3/
1	Interim electrical parameters		1			1	
2	Static burn-in I (method 1015)	4.2c 4.5.2	Req'd			Not req'd	
3	Same as line 1		*1	Δ			
4	Static burn-in II (method 1015)	4.2c 4.5.2	Req'd		4.2c 4.5.2	req'd 4/	
5	Same as line 1	4.2e	*1	Δ	4.2e	1*	Δ
6	Dynamic burn-in (method 1015)	4.2c 4.5.2	Req'd			Not req'd	
7	Same as line 1	4.2e	1	Δ			
8	Final electrical parameters		* 1, 2, 3, 7, 8, 9			1*, 2, 7, 9 4/	
9	Group A test requirements	4.4.1	1, 2, 3, 4, 7, 8, 9, 10, 11		4.4.1	1, 2, 3, 4, 7, 8, 9, 10, 11	
10	Group B test when using method 5005 QCI option	4.4.2	1, 2, 3, 9, 10, 11	Δ			
11	Group C end-point electrical parameters	4.4.2	1, 2, 3, 9, 10, 11	Δ	4.4.3	1, 2	Δ
12	Group D end-point electrical parameters	4.4.4	1, 2, 3		4.4.4	1, 2	

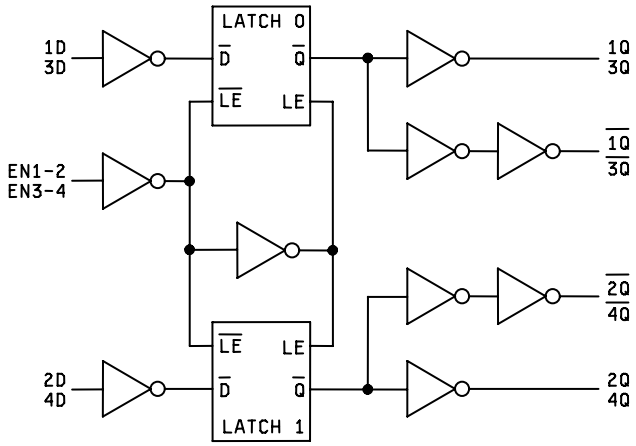
1/ Blank spaces indicate tests are not applicable.

2/ * indicates PDA applies to subgroup 1 (see 4.2.1).

3/ Δ indicates delta limits shall be required only on table III subgroup 1, where specified, and the delta values shall be computed with reference to the previous interim electrical parameters.

4/ The device manufacturer may, at his option, either complete subgroup 1 electrical parameter measurements, including delta measurements, within 96 hours after burn-in completion (removal of bias); or may complete subgroup 1 electrical measurements without delta measurements within 24 hours after burn-in completion (removal of bias).

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DEVICE TYPE 01
CASES E AND F

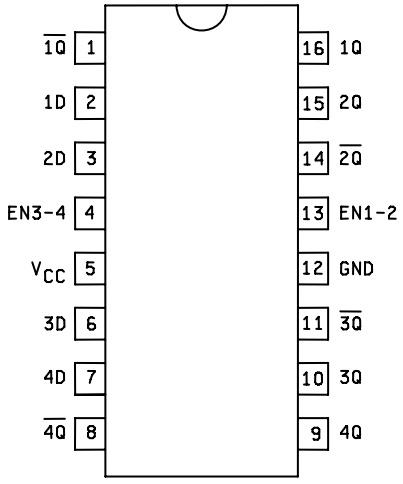


FIGURE 1. Logic diagrams and terminal connections.

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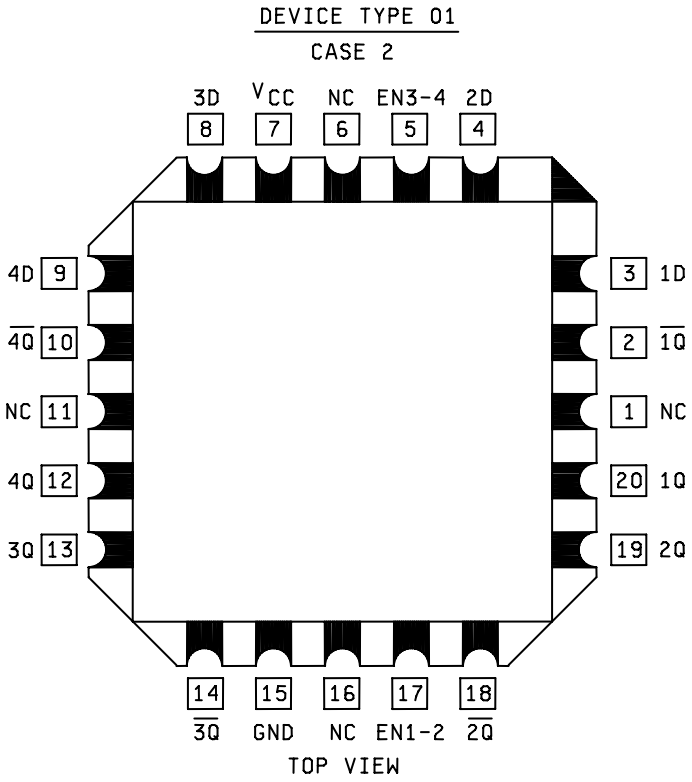
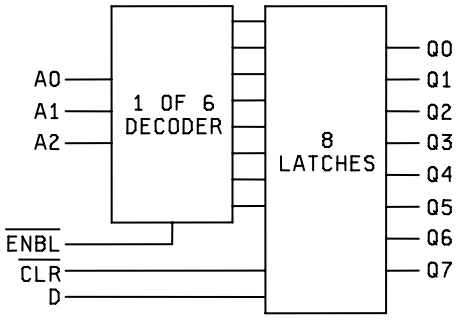


FIGURE 1. Logic diagrams and terminal connections – Continued.

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DEVICE TYPE 02

CASES E AND F

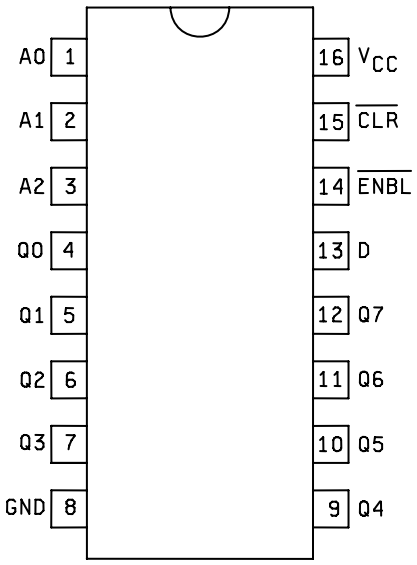


FIGURE 1. Logic diagrams and terminal connections – Continued.

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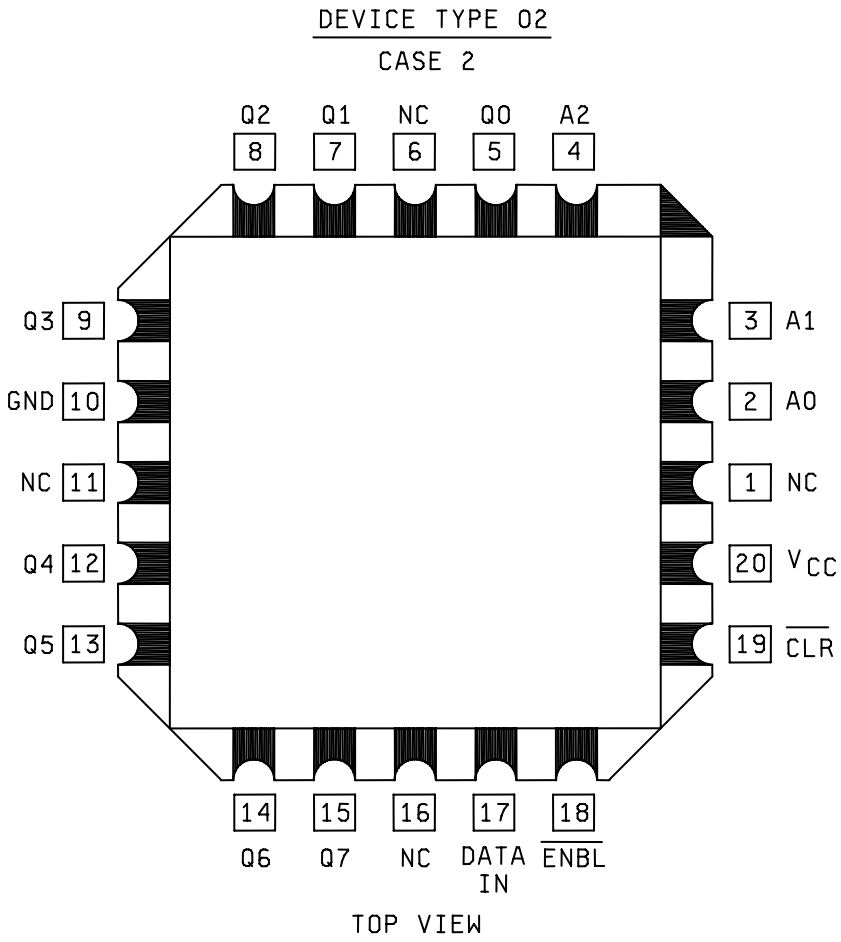


FIGURE 1. Logic diagrams and terminal connections – Continued.

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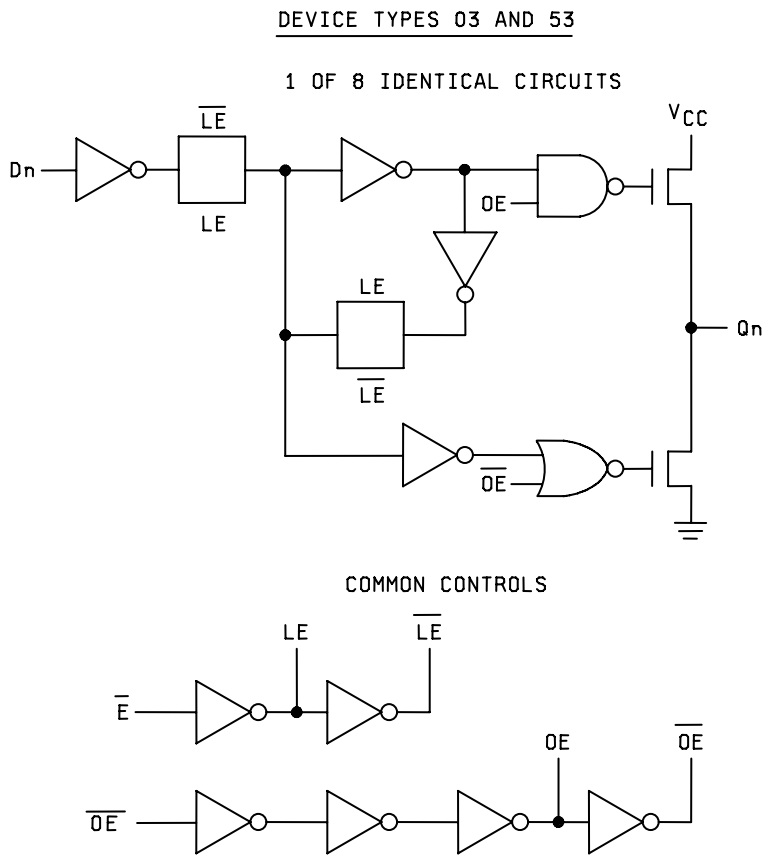


FIGURE 1. Logic diagrams and terminal connections – Continued.

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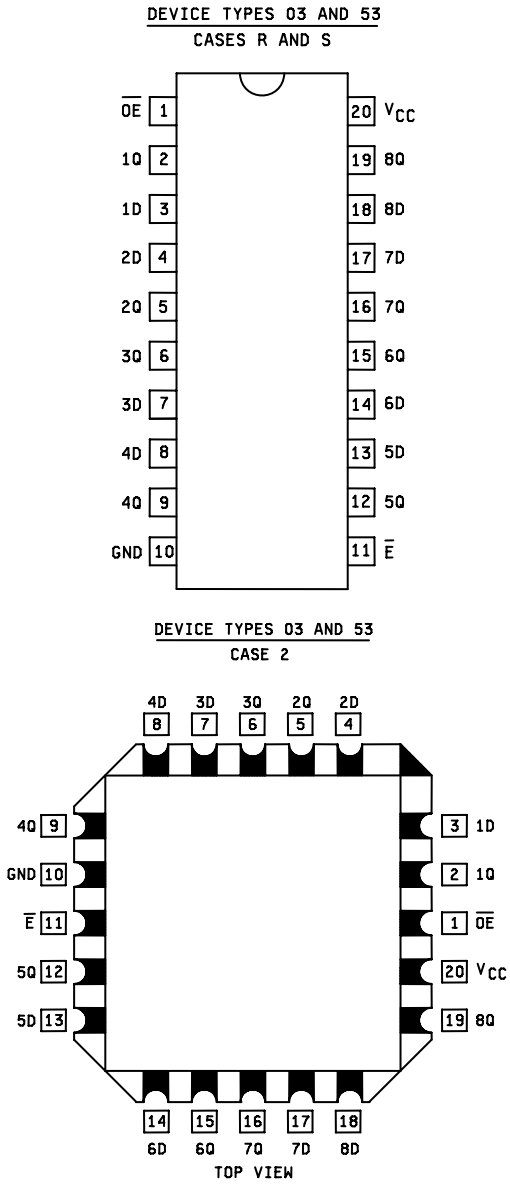


FIGURE 1. Logic diagrams and terminal connections – Continued.

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DEVICE TYPE 04

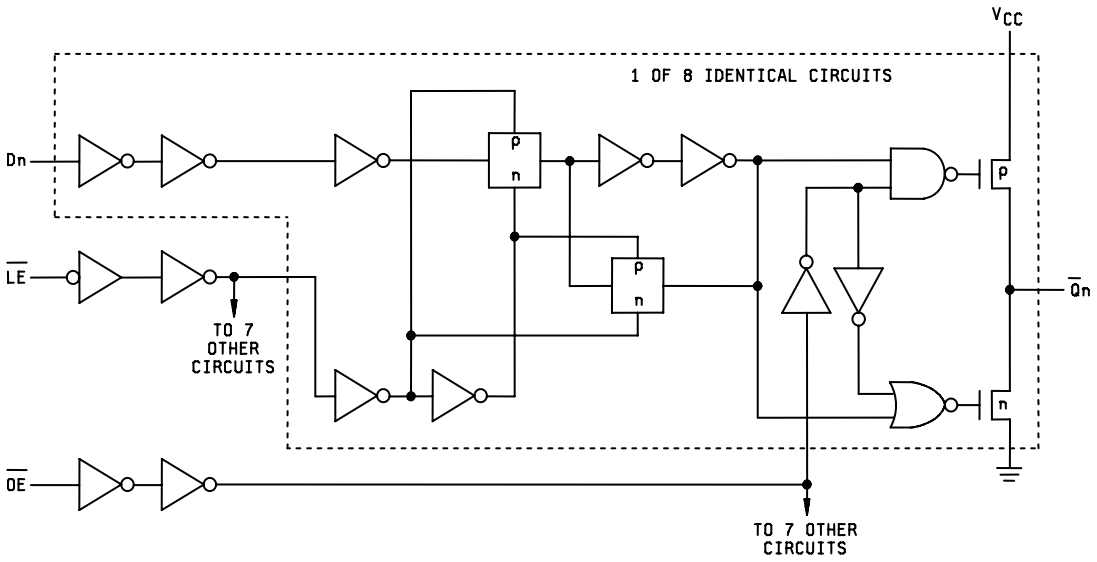


FIGURE 1. Logic diagrams and terminal connections – Continued.

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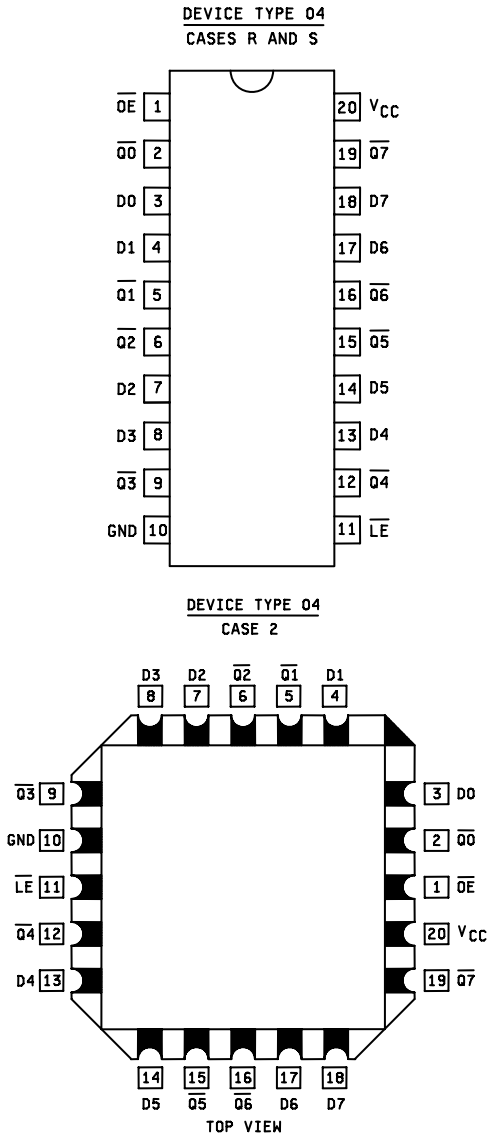


FIGURE 1. Logic diagrams and terminal connections – Continued.

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DEVICE TYPE 05

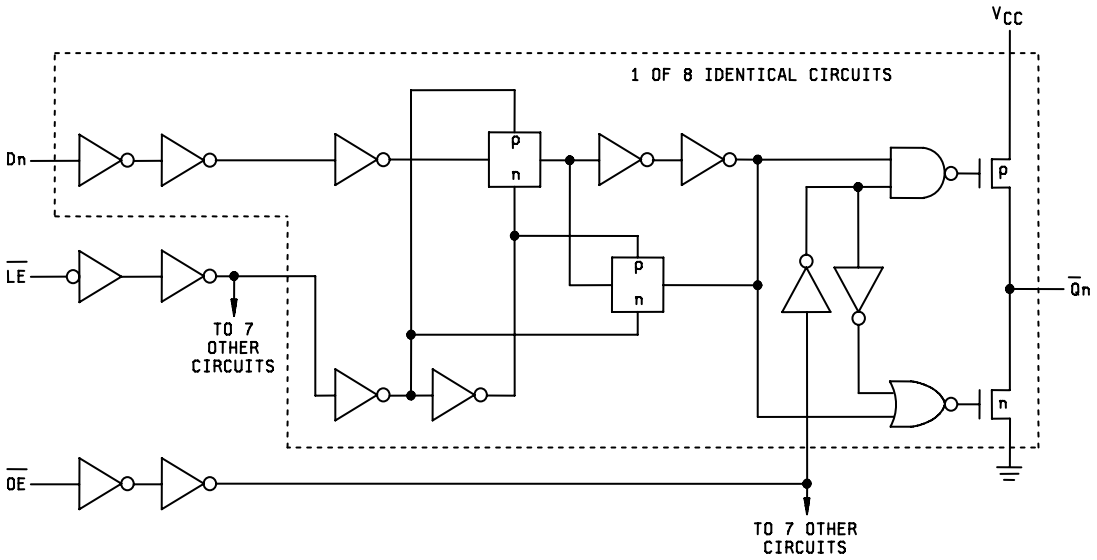
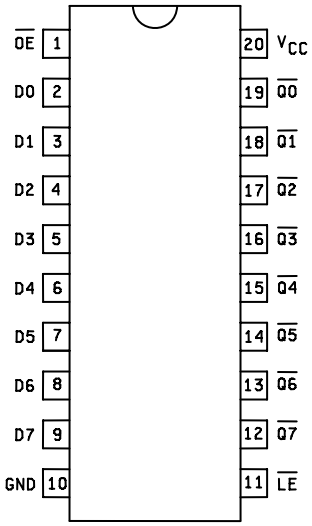


FIGURE 1. Logic diagrams and terminal connections – Continued.

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DEVICE TYPE 05
CASES R AND S



DEVICE TYPE 05
CASE 2

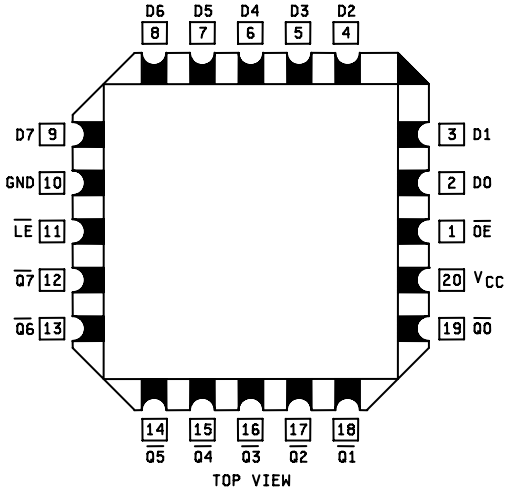


FIGURE 1. Logic diagrams and terminal connections – Continued.

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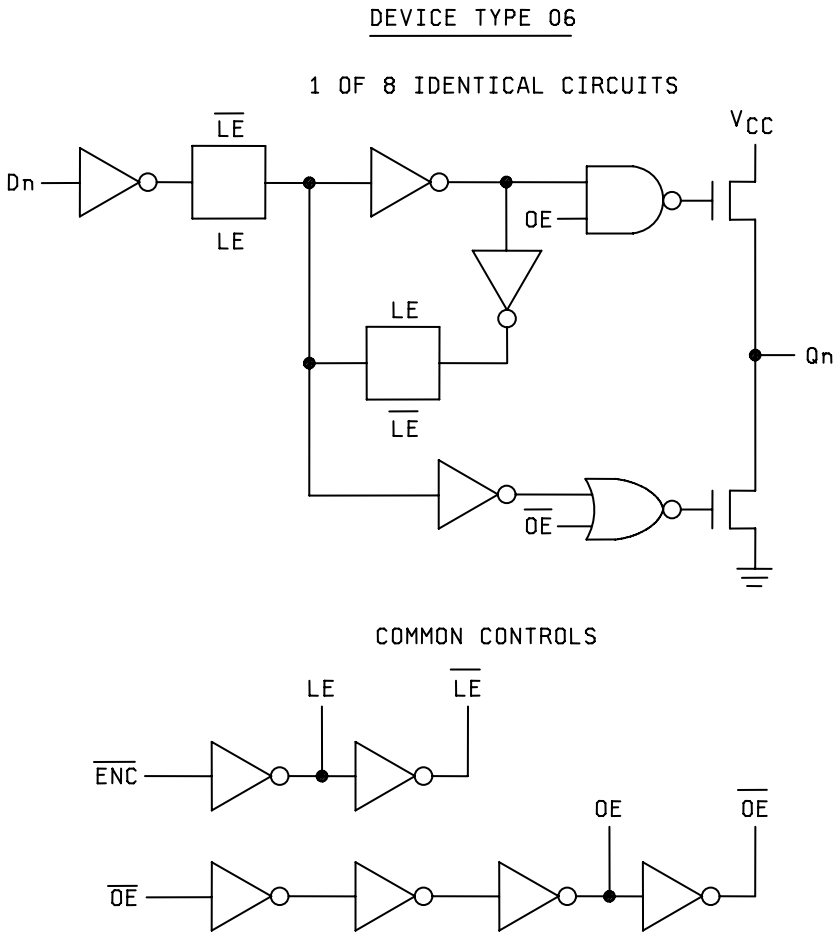


FIGURE 1. Logic diagrams and terminal connections – Continued.

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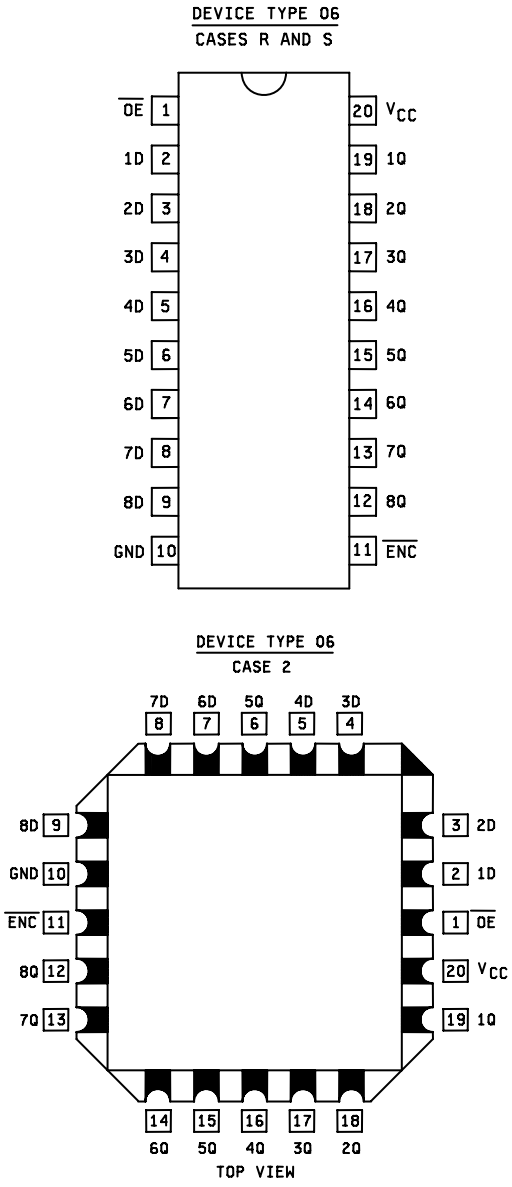


FIGURE 1. Logic diagrams and terminal connections – Continued.

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Device type 01

Inputs		Outputs	
D	$\overline{EN}$	Q	$\overline{Q}$
L	H	L	H
H	H	H	L
X	L	Qo	$\overline{Qo}$

H = High level voltage
L = Low level voltage
X = Don't care
Q0 ($\overline{Q0}$) = The level of Q ($\overline{Q}$) before the transition of $\overline{EN}$

Device type 02

Inputs		Ouput of addressed latch	Each other output	Function
$\overline{CLR}$	$\overline{EN}$			
H	L	D	$\overline{Qio}$	Addressable latch
H	H	Qio	Qio	Memory
L	L	D	L	8-line demultiplexer
L	H	L	L	Reset

H = High level voltage
L = Low level voltage
D = The level at the data input
Qio = The level of Q (i = 0, 1,...,7) as appropriate before the indicated steady-state input conditions were established

Latch selection table, device type 02

Select inputs			Latch addressed
A2	A1	A0	
L	L	L	0
L	L	H	1
L	H	L	2
L	H	H	3
H	L	L	4
H	L	H	5
H	H	L	6
H	H	H	7

FIGURE 2. Truth tables.

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Device types 03 and 53

$\overline{OE}$	$\overline{E}$	D data	Q output
L	H	H	H
L	H	L	L
L	L	l	L
L	L	h	H
H	X	X	Z

Device type 04

Output enable $\overline{OE}$	Latch enable $\overline{LE}$	D data	$\overline{Q}$ output
L	H	H	L
L	H	L	H
L	L	l	H
L	L	h	L
H	X	X	Z

Device type 05

Output enable $\overline{OE}$	Latch enable $\overline{LE}$	D data	$\overline{Q}$ output
L	H	H	L
L	H	L	H
L	L	l	H
L	L	h	L
H	X	X	Z

Device type 06

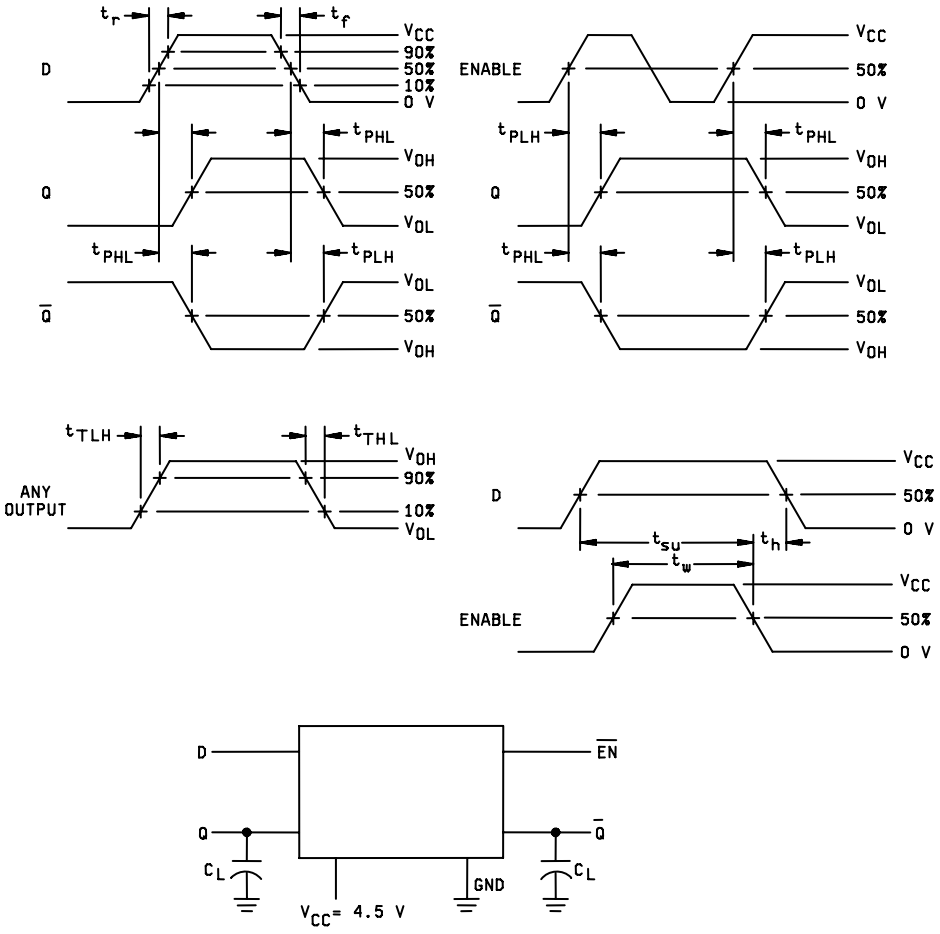
$\overline{OE}$	$\overline{ENC}$	D data	Q output
L	H	H	H
L	H	L	L
L	L	l	L
L	L	h	H
H	X	X	Z

NOTES:
L = Low level voltage
H = High level voltage
l = Low voltage level one setup time prior to the high-to-low latch enable transition
h = High voltage level one setup time prior to the high-to-low latch enable transition
X = Don't care
Z = High impedance state

FIGURE 2. Truth tables – Continued.

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DEVICE TYPE 01



NOTES:

1. Clock input pulse characteristics: $t_r = t_f \leq 6$ ns.
2. Data or enable input characteristics: $t_r = t_f = 6$ ns; $t_{SETUP} \leq 30$ ns; $t_{HOLD} = 8$ ns, and $t_{PULSE\ WIDTH} = 24$ ns.
3. $C_L = 50$ pF.
4. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 3. Switching time test circuit and waveforms.

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DEVICE TYPE 02

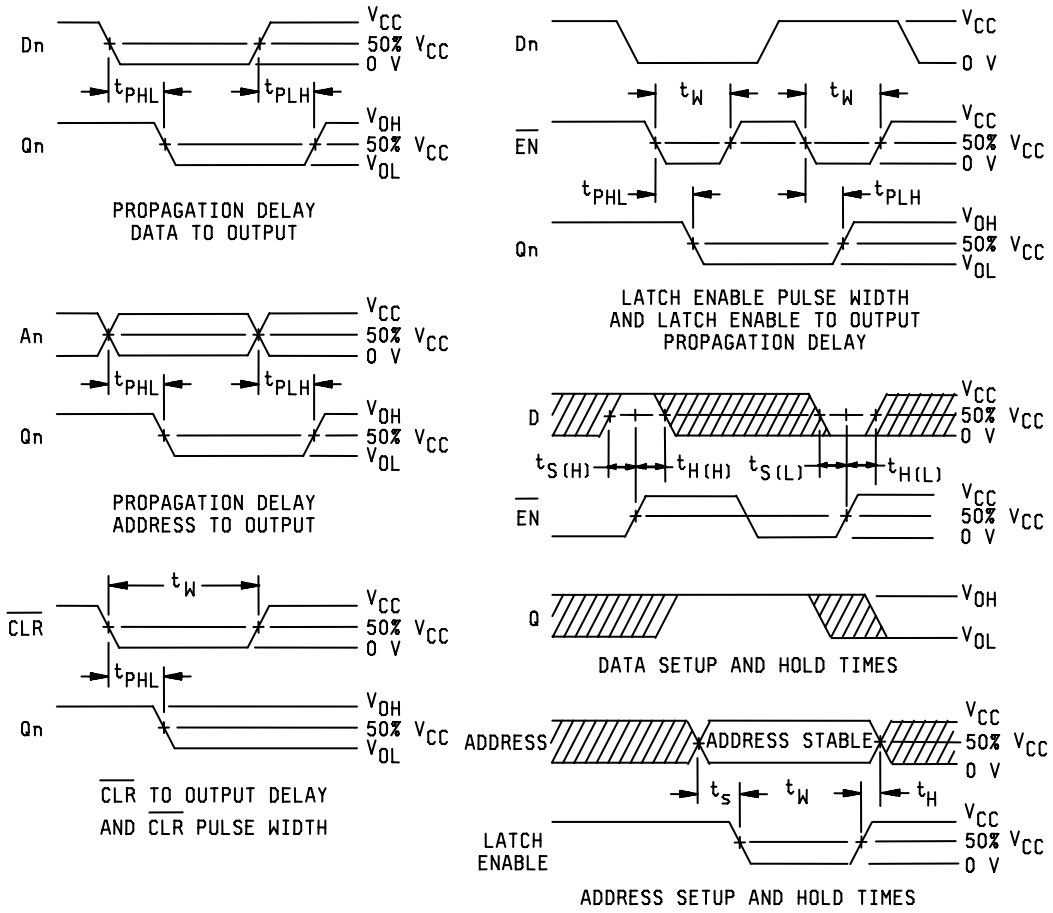
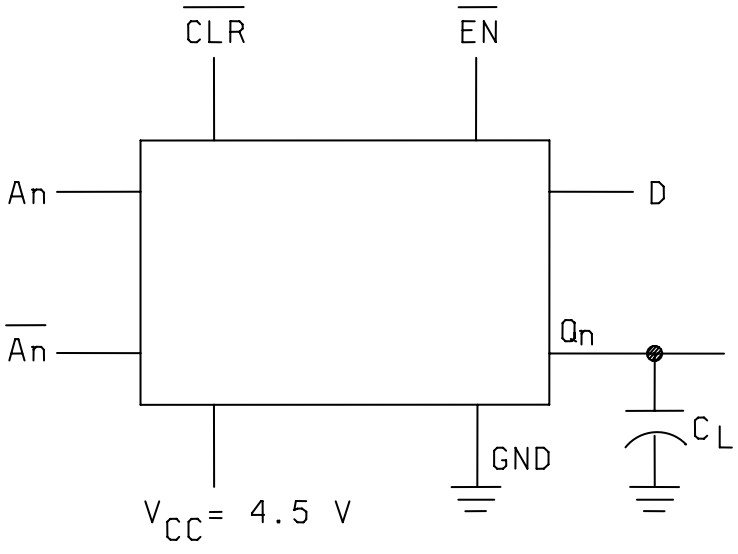


FIGURE 3. Switching time test circuit and waveforms – Continued.

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DEVICE TYPE 02



NOTES:
1. Clock input pulse characteristics: $t_r = t_f \leq 6$ ns.
2. Data, address, latch enable, or clear input pulse characteristics: $t_r = t_f = 6$ ns; $t_{SETUP} \leq 30$ ns; $t_{PULSE\ WIDTH} \leq 24$ ns and $t_{HOLD} \leq 24$ ns.
3. $C_L = 50$ pF.
4. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 3. Switching time test circuit and waveforms – Continued.

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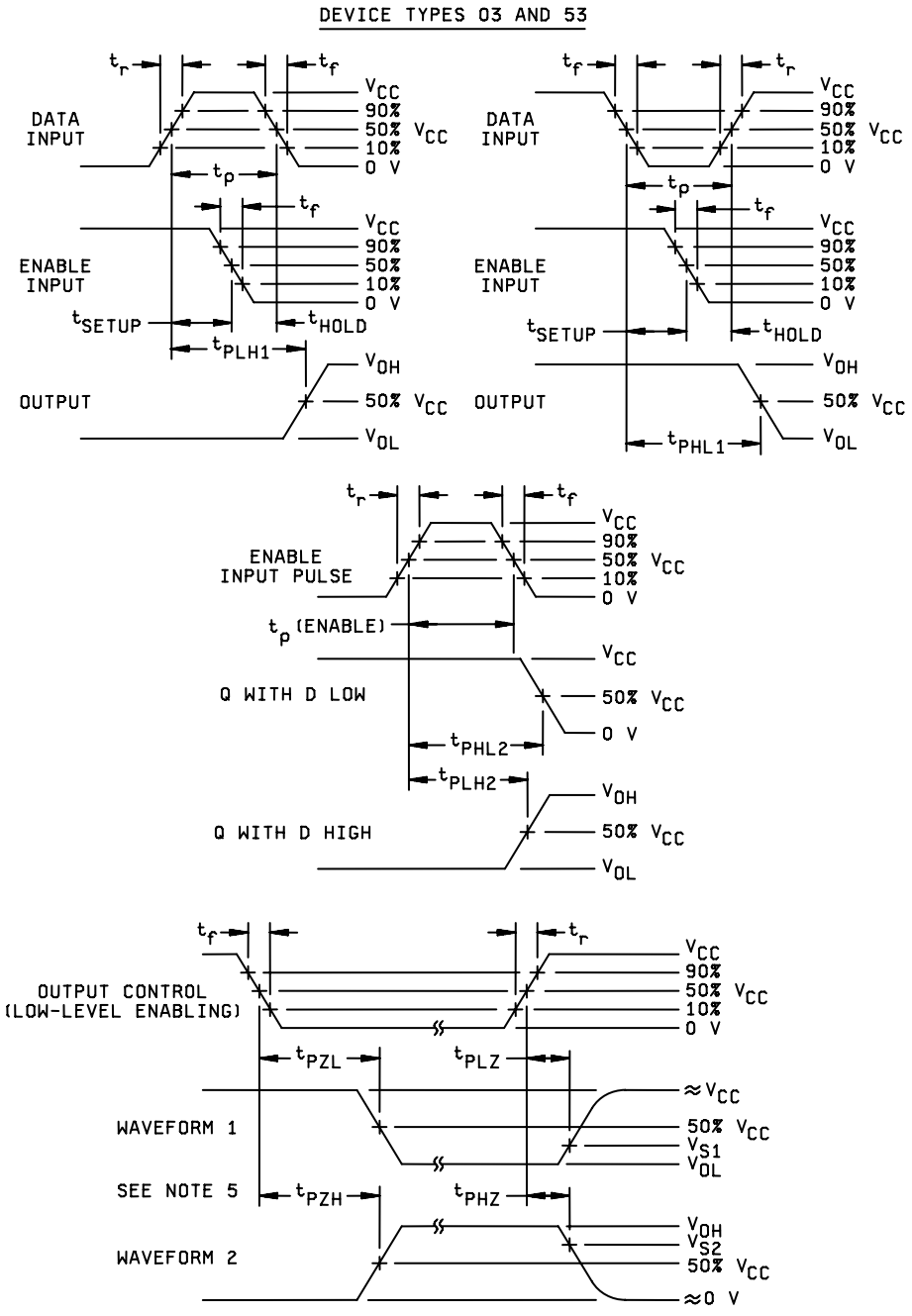
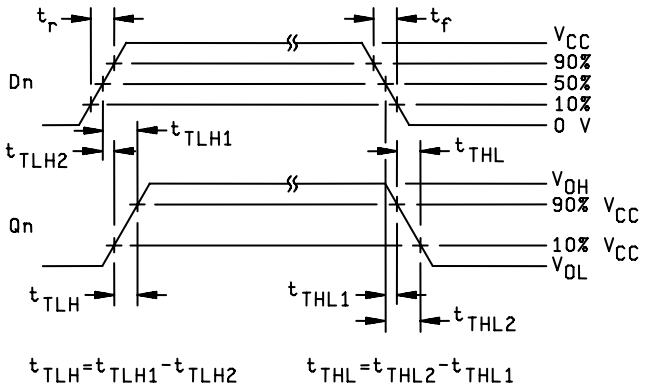
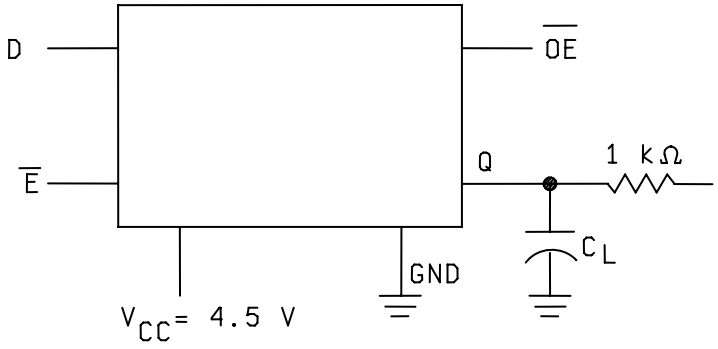


FIGURE 3. Switching time test circuit and waveforms – Continued.

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DEVICE TYPES 03 AND 53





NOTES:

1. Clock input pulse characteristics: $t_r = t_f \leq 6$ ns.
2. Data, latch enable, and output enable input pulse characteristics: $t_r = t_f \leq 6$ ns; $t_{SETUP} \leq 30$ ns; $t_{PULSE\ WIDTH} \leq 30$ ns and $t_{HOLD} \leq 15$ ns.
3. $C_L = 50$ pF and $R_L = 1000\Omega$.
4. Voltage measurements are to be made with respect to the network ground terminal.
5. For t_{PHZ} and t_{PZH}, a 1 kΩ resistor is connected between the output and GND terminal. For t_{PZL} and t_{PLZ}, a 1 kΩ resistor is connected between the output and V_{CC} terminal. $V_{S1} = V_{OL} + 0.1$ V ($V_{OH} - V_{OL}$). $V_{S2} = V_{OH} - 0.1$ V ($V_{OH} - V_{OL}$).
6. The input signal for HCT device type 53 will be 0 – 3 V; however, the 50% V_{CC} measure point is 1.3 V for input and output signals. The 10% V_{CC} and 90% V_{CC} points are 0.3 V and 2.7 V, respectively.

FIGURE 3. Switching time test circuit and waveforms – Continued.

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DEVICE TYPE 04

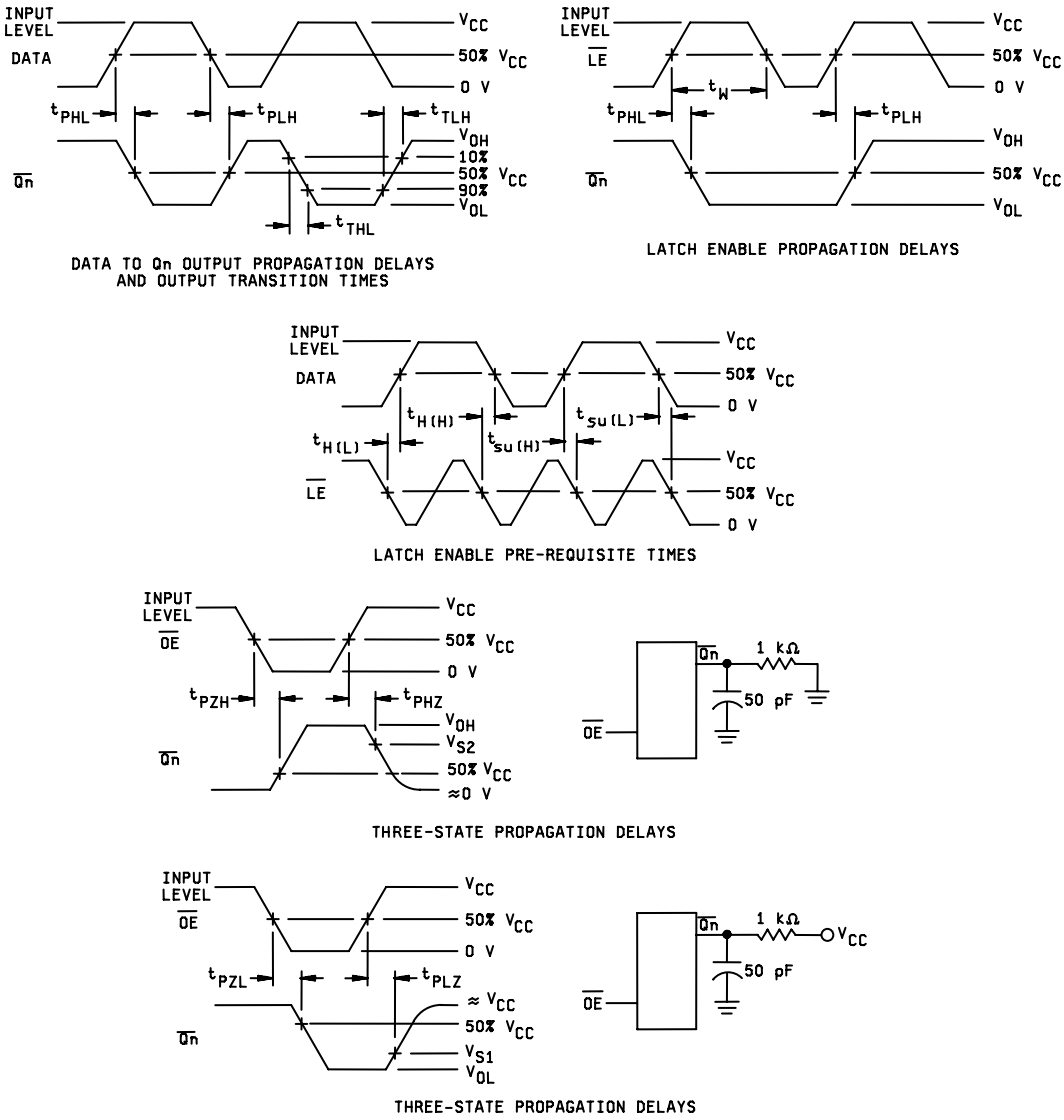
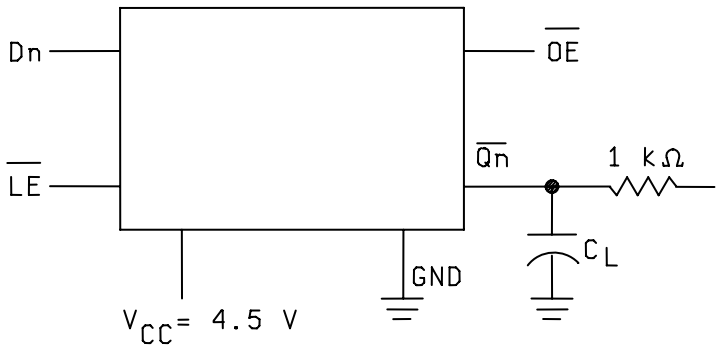


FIGURE 3. Switching time test circuit and waveforms – Continued.

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DEVICE TYPE 04



NOTES:

1. Clock input pulse characteristics: $t_r = t_f \leq 6$ ns.
2. Data, latch enable, and output enable input pulse characteristics: $t_r = t_f \leq 6$ ns; $t_{SETUP} \leq 30$ ns; $t_{PULSE\ WIDTH} \leq 30$ ns and $t_{HOLD} \leq 15$ ns.
3. $C_L = 50$ pF and $R_L = 1000\Omega$.
4. Voltage measurements are to be made with respect to the network ground terminal.
5. For t_{PHZ} and t_{PZH}, a 1 kΩ resistor is connected between the output and ground terminal. For t_{PZL} and t_{PLZ}, a 1 kΩ resistor is connected between the output and V_{CC} terminal. $V_{S1} = V_{OL} + 0.1$ V ($V_{OH} - V_{OL}$). $V_{S2} = V_{OH} - 0.1$ V ($V_{OH} - V_{OL}$).

FIGURE 3. Switching time test circuit and waveforms – Continued.

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DEVICE TYPE 05

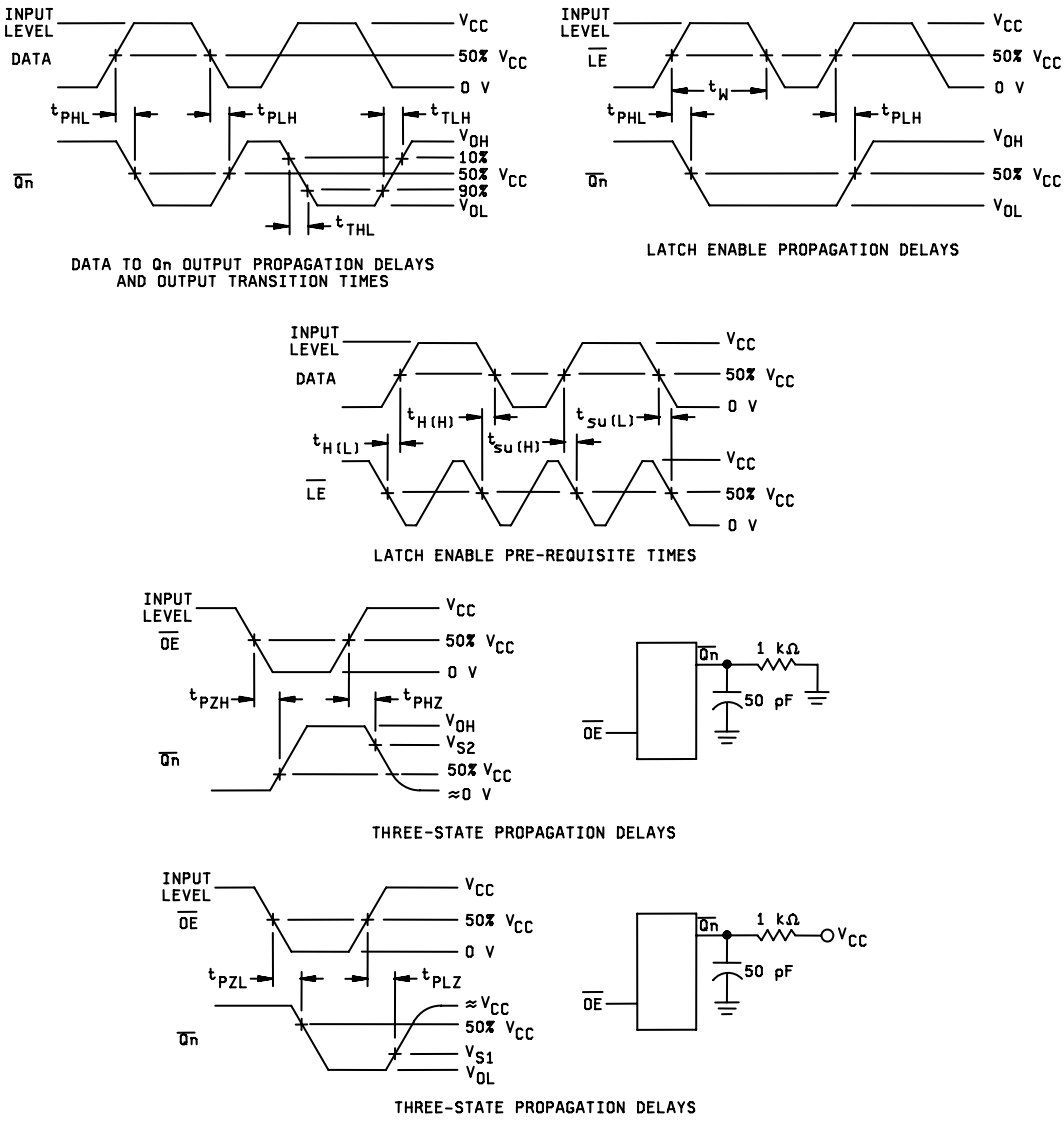
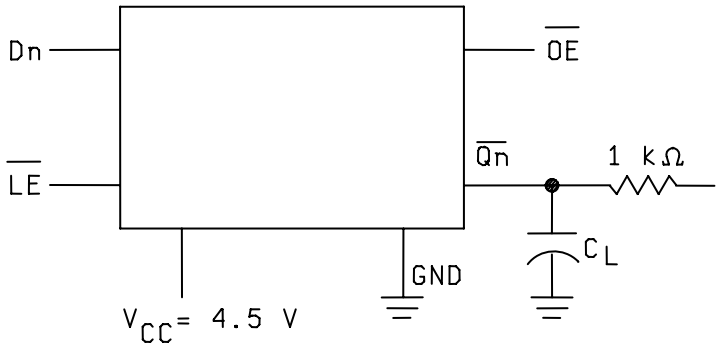


FIGURE 3. Switching time test circuit and waveforms – Continued.

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DEVICE TYPE 05



NOTES:

1. Clock input pulse characteristics: $t_r = t_f \leq 6$ ns.
2. Data, latch enable, and output enable input pulse characteristics: $t_r = t_f \leq 6$ ns; $t_{SETUP} \leq 23$ ns; $t_{PULSE\ WIDTH} \leq 24$ ns and $t_{HOLD} \leq 15$ ns.
3. $C_L = 50$ pF and $R_L = 1000\Omega$.
4. Voltage measurements are to be made with respect to the network ground terminal.
5. For t_{PHZ} and t_{PZH}, a 1 kΩ resistor is connected between the output and ground terminal. For t_{PZL} and t_{PLZ}, a 1 kΩ resistor is connected between the output and V_{CC} terminal. $V_{S1} = V_{OL} + 0.1$ V ($V_{OH} - V_{OL}$). $V_{S2} = V_{OH} - 0.1$ V ($V_{OH} - V_{OL}$).
6. All trip points are referenced to V_{CC}.

FIGURE 3. Switching time test circuit and waveforms – Continued.

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DEVICE TYPE 06

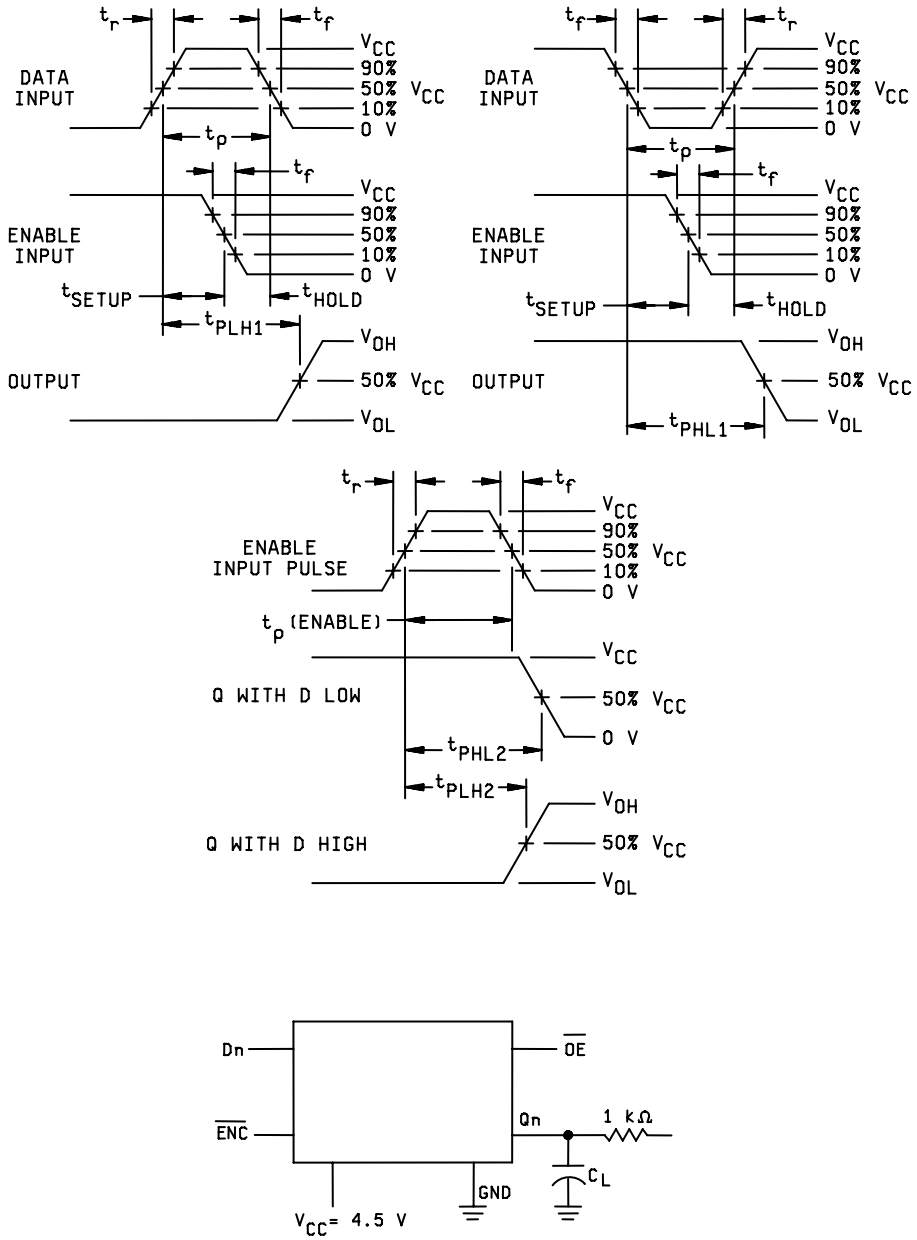
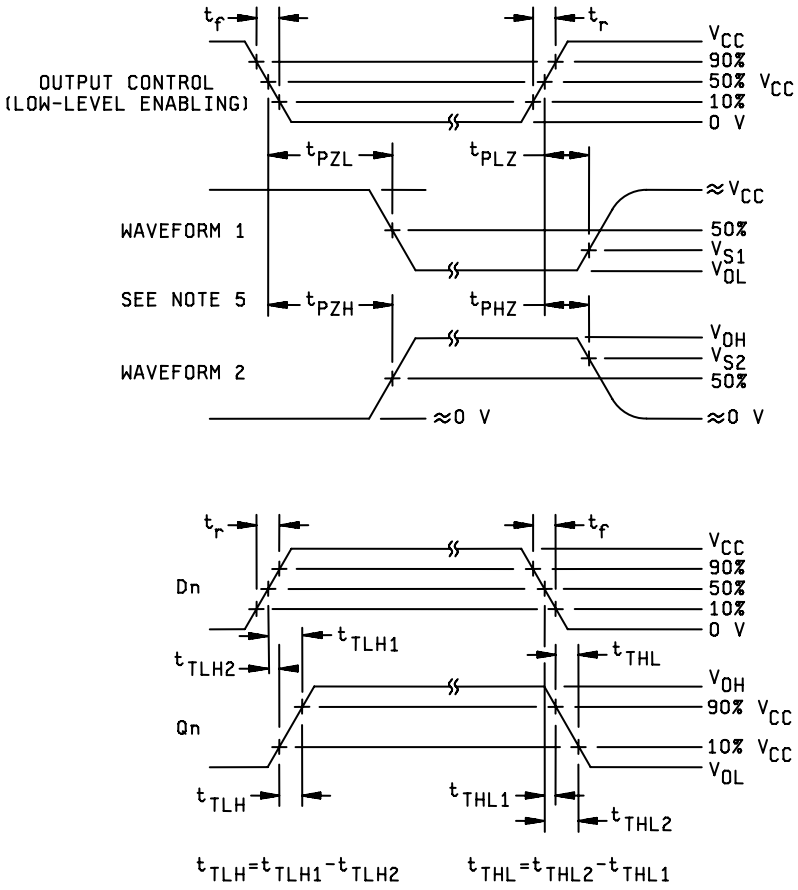


FIGURE 3. Switching time test circuit and waveforms – Continued.

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DEVICE TYPE 06



NOTES:

1. Clock input pulse characteristics: $t_r = t_f \leq 6$ ns.
2. Data, latch enable, and output enable input pulse characteristics: $t_r = t_f \leq 6$ ns; $t_{SETUP} \leq 23$ ns; $t_{PULSE\ WIDTH} \leq 24$ ns and $t_{HOLD} \leq 15$ ns.
3. $C_L = 50$ pF and $R_L = 1000\Omega$.
4. Voltage measurements are to be made with respect to the network ground terminal.
5. For t_{PHZ} and t_{PZH}, a 1 kΩ resistor is connected between the output and ground terminal. For t_{PZL} and t_{PLZ}, a 1 kΩ resistor is connected between the output and V_{CC} terminal. $V_{S1} = V_{OL} + 0.1$ V ($V_{OH} - V_{OL}$). $V_{S2} = V_{OH} - 0.1$ V ($V_{OH} - V_{OL}$).
6. All trip points are referenced to V_{CC}.

FIGURE 3. Switching time test circuit and waveforms – Continued.

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TABLE III. Group A inspection for device type 01.

Symbol	MIL-STD-883 method	Case 2	Terminal conditions 1/																Measured terminal	Test limits						Unit
			2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Cases E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
		Test no.	$\overline{1Q}$	1D	2D	EN3-4	V_{CC}	3D	4D	$\overline{4Q}$	4Q	3Q	$\overline{3Q}$	GND	EN1-2	$\overline{2Q}$	2Q	1Q		Min	Max	Min	Max	Min	Max	
V_{IC} (pos)		1		1mA			GND							1/					1D		1.5					V
		2			1mA		"							"					2D		"					"
		3				1mA	"							"					EN3-4		"					"
		4					"	1mA						"					3D		"					"
		5					"		1mA					"					4D		"					"
		6					"							"	1mA				EN1-2		"					"
V_{IC} (neg)		7		-1mA			1/							GND					1D		-1.5					"
		8			-1mA		"							"					2D		"					"
		9				-1mA	"							"					EN3-4		"					"
		10					"	-1mA						"					3D		"					"
		11					"		-1mA					"					4D		"					"
		12					"							"	-1mA				EN1-2		"					"
I_{CC}	3005	13		6.0V	6.0V	6.0V	6.0V	6.0V	6.0V					"	6.0V				V_{CC}		.1		10			µA
		14		GND	GND	GND	"	GND	GND					"	GND				V_{CC}		.1		10			µA
V_{OH3}	3006	15		4.2V			"							"	4.2V			-20µA	1Q	5.95		5.95		5.95		V
		16			4.2V		"							"	4.2V		-20µA		2Q	"		"		"		"
		17				4.2V	"	4.2V				-20µA		"					3Q	"		"		"		"
		18				4.2V	"		4.2V		-20µA			"					4Q	"		"		"		"
		19	-20µA	1.2V			"							"	4.2V				$\overline{1Q}$	"		"		"		"
		20			1.2V		"							"	4.2V	-20µA			$\overline{2Q}$	"		"		"		"
		21				4.2V	"	1.2V					-20µA	"					$\overline{3Q}$	"		"		"		"
		22				4.2V	"		1.2V	-20µA				"					4Q	"		"		"		"
V_{OH5}	3006	23-30	Same tests and terminal conditions as V_{OH3} except I_{OH} = -5.2 mA.																	5.48		5.2		5.48		"
V_{OL3}	3007	31		1.2V			6.0 V							GND	4.2V			20µA	1Q		.05		.05		.05	"
		32			1.2V		"							"	4.2V		20µA		2Q		"		"		"	"
		33				4.2V	"	1.2V				20µA		"					3Q		"		"		"	"
		34				4.2V	"		1.2V		20µA			"					4Q		"		"		"	"
		35	20µA	4.2V			"							"	4.2V				$\overline{1Q}$		"		"		"	"
		36			4.2V		"							"	4.2V	20µA			$\overline{2Q}$		"		"		"	"
		37				4.2V	"	4.2V					20µA	"					$\overline{3Q}$		"		"		"	"
		38				4.2V	"		4.2V	20µA				"					4Q		"		"		"	"
V_{OL5}	3007	39-46	Same tests and terminal conditions as V_{OL3} except I_{OL} = 5.2 mA.																		.26		.4		.26	"
I_{OS4}	3011	47		4.0V			4.0V							GND	4.0V			GND	1Q	-10	-120	-10	-120	-10	-120	mA
		48			4.0V		"							"	4.0V		GND		2Q	"	"	"	"	"	"	"
		49				4.0V	"	4.0V				GND		"					3Q	"	"	"	"	"	"	"
		50				4.0V	"		4.0V		GND			"					4Q	"	"	"	"	"	"	"
		51	GND	GND			"							"	4.0V				$\overline{1Q}$	"	"	"	"	"	"	"
		52			GND		"							"	4.0V	GND			$\overline{2Q}$	"	"	"	"	"	"	"
		53				4.0V	"	GND					GND	"					$\overline{3Q}$	"	"	"	"	"	"	"
		54				4.0V	"		GND	GND				"					4Q	"	"	"	"	"	"	"
I_{IH}	3010	55		6.0V			6.0V							"					1D		.05		.1			µA
		56			6.0V		"							"					2D		"		"			"
		57				6.0V	"							"					EN3-4		"		"			"
		58					"	6.0V						"					3D		"		"			"
		59					"		6.0V					"					4D		"		"			"
		60					"							"	6.0V				EN1-2		"		"			"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 – Continued.

Symbol	MIL-STD-883 method	Case 2	Terminal conditions 1/																Measured terminal	Test limits						Unit
			2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Cases E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
		Test no.	$\overline{1Q}$	1D	2D	EN3-4	V_{CC}	3D	4D	$\overline{4Q}$	4Q	3Q	$\overline{3Q}$	GND	EN1-2	$\overline{2Q}$	2Q	1Q		Min	Max	Min	Max	Min	Max	
I_{IL}	3009	61		GND			6.0V							GND					1D		-.05		-.1			µA
		62			GND		"							"					2D		"		"			"
		63				GND	"							"					EN3-4		"		"			"
		64					"	GND						"					3D		"		"			"
		65					"		GND					"					4D		"		"			"
		66					"							"	GND				EN1-2		"		"			"
																				Subgroup 4 T_C = +25°C						
																				Min	Max					
C_{in}	3012	67		2/			GND							GND					1D		10					pF
		68			2/		"							"					2D		"					"
		69					"	2/						"					3D		"					"
		70					"		2/					"					4D		"					"
		71				2/	"							"					EN3-4		"					"
		72					"							"	2/				EN1-2		"					"
																				Subgroup 7 T_C = +25°C		Subgroup 8 T_C = +125°C		Subgroup 8 T_C = -55°C		
																				Min	Max	Min	Max	Min	Max	
Truth table 3/	3014	73	L	A	A	A	4.5V	A	A	L	H	H	L	GND	A	L	H	H	All outputs 4/							
		74	L	A	A	B	"	A	A	L	H	H	L	"	B	L	H	H	"							
		75	L	B	B	B	"	B	B	L	H	H	L	"	B	L	H	H	"							
		76	H	B	B	A	"	B	B	H	L	L	H	"	A	H	L	L	"							
																				Subgroup 9 T_C = +25°C		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C		
																				Min	Max	Min	Max	Min	Max	
t_{PLH1}	3003 (Fig. 3)	77		IN			4.5V							GND	4.5V			OUT	1D to 1Q	3	23	3	30	3	23	ns
		78			IN		"							"	4.5V		OUT		2D to 2Q	"	"	"	"	"	"	"
		79				4.5V	"	IN				OUT		"					3D to 3Q	"	"	"	"	"	"	"
		80				4.5V	"		IN		OUT			"					4D to 4Q	"	"	"	"	"	"	"
t_{PHL1}	3003 (Fig. 3)	81		IN			"							"	4.5V			OUT	1D to 1Q	"	"	"	"	"	"	"
		82			IN		"							"	4.5V		OUT		2D to 2Q	"	"	"	"	"	"	"
		83				4.5V	"	IN				OUT		"					3D to 3Q	"	"	"	"	"	"	"
		84				4.5V	"		IN		OUT			"					4D to 4Q	"	"	"	"	"	"	"
t_{PLH1}	3003 (Fig. 3)	85	OUT	IN			"							"	4.5V				1D to $\overline{1Q}$	"	"	"	"	"	"	"
		86			IN		"							"	4.5V	OUT			2D to $\overline{2Q}$	"	"	"	"	"	"	"
		87				4.5V	"	IN					OUT	"					3D to $\overline{3Q}$	"	"	"	"	"	"	"
		88				4.5V	"		IN	OUT				"					4D to $\overline{4Q}$	"	"	"	"	"	"	"
t_{PHL1}	3003 (Fig. 3)	89	OUT	IN			"							"	4.5V				1D to $\overline{1Q}$	"	"	"	"	"	"	"
		90			IN		"							"	4.5V	OUT			2D to $\overline{2Q}$	"	"	"	"	"	"	"
		91				4.5V	"	IN					OUT	"					3D to $\overline{3Q}$	"	"	"	"	"	"	"
		92				4.5V	"		IN	OUT				"					4D to 4Q	"	"	"	"	"	"	"

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TABLE III. Group A inspection for device type 01 – Continued.

Symbol	MIL-STD-883 method	Case 2	Terminal conditions 1/															Measured terminal	Test limits						Unit
			2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Subgroup 9 T_C = +25°C		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C		
		Cases E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
		Test no.	$\overline{1Q}$	1D	2D	EN3-4	V_{CC}	3D	4D	$\overline{4Q}$	4Q	3Q	$\overline{3Q}$	GND	EN1-2	$\overline{2Q}$	2Q	1Q		Min	Max	Min	Max	Min	Max	
t_{PHL2}	3003 (Fig. 3)	93		4.5V			4.5V							GND	IN			OUT	EN to 1Q	4	26	4	34	4	26	ns
		94			4.5V		"							"	IN		OUT		EN to 2Q	"	"	"	"	"	"	"
		95				IN	"	4.5V				OUT		"					EN to 3Q	"	"	"	"	"	"	"
		96				IN	"		4.5V		OUT			"					EN to 4Q	"	"	"	"	"	"	"
		97		GND			"							"	IN			OUT	EN to 1Q	"	"	"	"	"	"	"
		98			GND		"							"	IN		OUT		EN to 2Q	"	"	"	"	"	"	"
		99				IN	"	GND				OUT		"					EN to 3Q	"	"	"	"	"	"	"
		100				IN	"		GND		OUT			"					EN to 4Q	"	"	"	"	"	"	"
t_{PLH2}	3003 (Fig. 3)	101	OUT	GND			"							"	IN				EN to $\overline{1Q}$	"	"	"	"	"	"	"
		102			GND		"							"	IN	OUT			EN to 2Q	"	"	"	"	"	"	"
		103				IN	"	GND					OUT	"					EN to 3Q	"	"	"	"	"	"	"
		104				IN	"		GND	OUT				"					EN to 4Q	"	"	"	"	"	"	"
		105	OUT	4.5V			"							"	IN				EN to 1Q	"	"	"	"	"	"	"
		106			4.5V		"							"	IN	OUT			EN to 2Q	"	"	"	"	"	"	"
		107				IN	"	4.5V					OUT	"					EN to 3Q	"	"	"	"	"	"	"
		108				IN	"		4.5V	OUT				"					EN to 4Q	"	"	"	"	"	"	"
t_{THL}	3004 (Fig. 3)	109		IN			"							"	4.5V			OUT	1Q	3	15	3	20	3	15	"
		110			IN		"							"	4.5V		OUT		2Q	"	"	"	"	"	"	"
		111				4.5V	"	IN				OUT		"					3Q	"	"	"	"	"	"	"
		112				4.5V	"		IN		OUT			"					4Q	"	"	"	"	"	"	"
t_{TLH}	3004 (Fig. 3)	113		IN			"							"	4.5V			OUT	1Q	"	"	"	"	"	"	"
		114			IN		"							"	4.5V		OUT		2Q	"	"	"	"	"	"	"
		115				4.5V	"	IN				OUT		"					3Q	"	"	"	"	"	"	"
		116				4.5V	"		IN		OUT			"					4Q	"	"	"	"	"	"	"

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TABLE III. Group A inspection for device type 02.

Symbol	MIL-STD-883 method	Case 2	Terminal conditions 1/																Measured terminal	Test limits						Unit
			2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Cases E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
		Test no.	A0	A1	A2	Q0	Q1	Q2	Q3	GND	Q4	Q5	Q6	Q7	Data IN	$\overline{EN}$	$\overline{CLR}$	V_{CC}		Min	Max	Min	Max	Min	Max	
V_{IC} (pos)		1	1mA							1/								GND	A0		1.5					V
		2		1mA						"								"	A1		"					"
		3			1mA					"								"	A2		"					"
		4								"					1mA			"	Data		"					"
		5								"						1mA		"	$\overline{EN}$		"					"
		6								"							1mA	"	$\overline{CLR}$		"					"
V_{IC} (neg)		7	-1mA							GND								1/	A0		-1.5					"
		8		-1mA						"								"	A1		"					"
		9			-1mA					"								"	A2		"					"
		10								"					-1mA			"	Data		"					"
		11								"						-1mA		"	$\overline{EN}$		"					"
		12								"							-1mA	"	$\overline{CLR}$		"					"
I_{CC}	3005	13	6.0V	6.0V	6.0V					"					6.0V	6.0V	6.0V	6.0V	V_{CC}		.2		20			µA
		14	GND	GND	GND					"					GND	GND	GND	6.0V	V_{CC}		.2		20			µA
V_{OH3}	3006	15	1.2V	1.2V	1.2V	-20µA				"					4.2V	1.2V	4.2V	6.0V	Q0	5.95		5.95		5.95		V
		16	4.2V	1.2V	1.2V		-20µA			"					"	"	"	"	Q1	"		"		"		"
		17	1.2V	4.2V	1.2V			-20µA		"					"	"	"	"	Q2	"		"		"		"
		18	4.2V	4.2V	1.2V				-20µA	"					"	"	"	"	Q3	"		"		"		"
		19	1.2V	1.2V	4.2V					"	-20µA				"	"	"	"	Q4	"		"		"		"
		20	4.2V	1.2V	4.2V					"		-20µA			"	"	"	"	Q5	"		"		"		"
		21	1.2V	4.2V	4.2V					"			-20µA		"	"	"	"	Q6	"		"		"		"
		22	4.2V	4.2V	4.2V					"				-20µA	"	"	"	"	Q7	"		"		"		"
V_{OH5}	3006	23-30	Same tests and terminal conditions as V_{OH3}, except I_{OH} = -5.2 mA.																	5.48		5.2		5.48		"
V_{OL3}	3007	31	1.2V	1.2V	1.2V	20mA				GND					1.2V	1.2V	4.2V	6.0V	Q0		.05		.05		.05	"
		32	4.2V	1.2V	1.2V		20mA			"					"	"	"	"	Q1		"		"		"	"
		33	1.2V	4.2V	1.2V			20mA		"					"	"	"	"	Q2		"		"		"	"
		34	4.2V	4.2V	1.2V				20mA	"					"	"	"	"	Q3		"		"		"	"
		35	1.2V	1.2V	4.2V					"	20mA				"	"	"	"	Q4		"		"		"	"
		36	4.2V	1.2V	4.2V					"		20mA			"	"	"	"	Q5		"		"		"	"
		37	1.2V	4.2V	4.2V					"			20mA		"	"	"	"	Q6		"		"		"	"
		38	4.2V	4.2V	4.2V					"				20mA	"	"	"	"	Q7		"		"		"	"
V_{OL5}	3007	39-46	Same tests and terminal conditions as V_{OL3}, except I_{OL} = 5.2 mA.																		.26		.4		.26	"
I_{OS4}	3011	47	GND	GND	GND	GND				GND						GND	4.0V	4.0V	Q0	-10	-120	-10	-120	-10	-120	mA
		48	4.0V	GND	GND		GND			"						"	"	"	Q1	"	"	"	"	"	"	"
		49	GND	4.0V	GND			GND		"						"	"	"	Q2	"	"	"	"	"	"	"
		50	4.0V	4.0V	GND				GND	"						"	"	"	Q3	"	"	"	"	"	"	"
		51	GND	GND	4.0V					"	GND					"	"	"	Q4	"	"	"	"	"	"	"
		52	4.0V	GND	4.0V					"		GND				"	"	"	Q5	"	"	"	"	"	"	"
		53	GND	4.0V	4.0V					"			GND			"	"	"	Q6	"	"	"	"	"	"	"
		54	4.0V	4.0V	4.0V					"				GND		"	"	"	Q7	"	"	"	"	"	"	"
I_{IH}	3010	55	6.0V							GND								6.0V	A0		.05		.1			µA
		56		6.0V						"								"	A1		"		"			"
		57			6.0V					"								"	A2		"		"			"
		58								"					6.0V			"	Data		"		"			"
		59								"						6.0V		"	$\overline{EN}$		"		"			"
		60								"							6.0V	"	$\overline{CLR}$		"		"			"

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TABLE III. Group A inspection for device type 02 – Continued.

Symbol	MIL-STD-883 method	Case 2	Terminal conditions 1/																Measured terminal	Test limits						Unit
			2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Cases E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
		Test no.	A0	A1	A2	Q0	Q1	Q2	Q3	GND	Q4	Q5	Q6	Q7	Data IN	$\overline{EN}$	$\overline{CLR}$	V_{CC}		Min	Max	Min	Max	Min	Max	
I_{IL}	3009	61	GND	GND	GND					GND					GND	GND	GND	6.0V	A0		-.05		-.1			µA
		62								"								"	A1		"		"			"
		63								"								"	A2		"		"			"
		64								"								"	Data		"		"			"
		65								"								"	$\overline{EN}$		"		"			"
		66								"								"	$\overline{CLR}$		"		"			"
																				Subgroup 4 T_C = +25°C						
																				Min	Max					
C_{in}	3012	67	2/	2/	2/					"					2/	2/	2/	GND	A0		10					pF
		68								"								"	A1		"					"
		69								"								"	A2		"					"
		70								"								"	Data		"					"
		71								"								"	$\overline{EN}$		"					"
		72								"								"	$\overline{CLR}$		"					"
																				Subgroup 7 T_C = +25°C		Subgroup 8 T_C = +125°C		Subgroup 8 T_C = -55°C		
																				Min	Max	Min	Max	Min	Max	
Truth table 3/	3014	73	B	B	B	L	L	L	L	GND	L	L	L	L	A	A	B	4.5V	All Outputs 4/							
		74	B	"	"	H	"	"	"	"	"	"	"	"	"	B	"	"	"							
		75	A	"	"	L	"	"	"	"	"	"	"	"	"	A	"	"	"							
		76	A	"	"	L	H	"	"	"	"	"	"	"	"	B	"	"	"							
		77	B	A	"	"	L	"	"	"	"	"	"	"	"	A	"	"	"							
		78	B	"	"	"	"	H	"	"	"	"	"	"	"	B	"	"	"							
		79	A	"	"	"	"	L	"	"	"	"	"	"	"	A	"	"	"							
		80	A	"	"	"	"	"	H	"	"	"	"	"	"	B	"	"	"							
		81	B	B	A	"	"	"	L	"	"	"	"	"	"	A	"	"	"							
		82	B	"	"	"	"	"	"	"	H	"	"	"	"	B	"	"	"							
		83	A	"	"	"	"	"	"	"	L	"	"	"	"	A	"	"	"							
		84	A	"	"	"	"	"	"	"	"	H	"	"	"	B	"	"	"							
		85	B	A	"	"	"	"	"	"	"	L	"	"	"	A	"	"	"							
		86	B	"	"	"	"	"	"	"	"	"	H	"	"	B	"	"	"							
		87	A	"	"	"	"	"	"	"	"	"	L	"	"	A	"	"	"							
		88	A	"	"	"	"	"	"	"	"	"	"	H	"	B	"	"	"							
		89	A	"	"	"	"	"	"	"	"	"	"	"	"	A	"	"	"							
		90	B	"	"	"	"	"	"	"	"	"	"	"	B	"	"	"	"							
		91	A	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"							
		92	B	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"							
		93	A	A	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"							
		94	B	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"							
		95	A	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"							
		96	B	"	"	"	"	"	"	"	"	"	"	"	"	A	"	"	"							
		97	B	"	"	H	"	"	"	"	"	"	"	"	"	"	B	"	"							
		98	B	"	"	"	"	"	"	"	"	"	"	"	"	"	A	"	"							
		99	A	"	"	"	"	"	"	"	"	"	"	"	"	"	A	"	"							
		100	A	"	"	"	H	"	"	"	"	"	"	"	"	"	B	"	"							

See footnotes at end of table.

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TABLE III. Group A inspection for device type 02 – Continued.

Symbol	MIL-STD-883 method	Case 2	Terminal conditions 1/																Measured terminal	Test limits						Unit
			2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Subgroup 7 T_C = +25°C		Subgroup 8 T_C = +125°C		Subgroup 8 T_C = -55°C		
		Cases E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
		Test no.	A0	A1	A2	Q0	Q1	Q2	Q3	GND	Q4	Q5	Q6	Q7	Data IN	$\overline{EN}$	$\overline{CLR}$	V_{CC}		Min	Max	Min	Max	Min	Max	
Truth table 3/	3014	101	A	B	B	H	H	L	L	GND	L	L	L	H	A	A	A	4.5V	All Outputs 4/							
		102	B	A	"	"	"	"	"	"	"	"	"	"	"	A	"	"									
		103	B	"	"	"	"	H	"	"	"	"	"	"	"	B	"	"									
		104	B	"	"	"	"	"	"	"	"	"	"	"	"	A	"	"									
		105	A	"	"	"	"	"	"	"	"	"	"	"	"	A	"	"									
		106	A	"	"	"	"	"	H	"	"	"	"	"	"	B	"	"									
		107	A	"	"	"	"	"	"	"	"	"	"	"	"	A	"	"									
		108	B	B	A	"	"	"	"	"	"	"	"	"	"	A	"	"									
		109	B	"	"	"	"	"	"	"	H	"	"	"	"	B	"	"									
		110	B	"	"	"	"	"	"	"	"	"	"	"	"	A	"	"									
		111	A	"	"	"	"	"	"	"	"	"	"	"	"	A	"	"									
		112	A	"	"	"	"	"	"	"	"	H	"	"	"	B	"	"									
		113	A	"	"	"	"	"	"	"	"	"	"	"	"	A	"	"									
		114	B	A	"	"	"	"	"	"	"	"	"	"	"	A	"	"									
		115	B	"	"	"	"	"	"	"	"	"	H	"	"	B	"	"									
		116	B	"	"	"	"	"	"	"	"	"	"	"	"	A	"	"									
		117	A	"	"	"	"	"	"	"	"	"	"	"	"	A	"	"									
		118	A	"	"	"	"	"	"	"	"	"	"	"	"	B	"	"									
																				Subgroup 9 T_C = +25°C		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C		
																				Min	Max	Min	Max	Min	Max	
t_{PHL1}	3003 (Fig. 3)	119	GND	GND	GND	OUT				GND					IN	GND 5/	4.5V	4.5V	Data to Q0	5	37	5	43	5	37	ns
		120	4.5V	GND	GND		OUT			"					"	"	"	"	Data to Q1	"	"	"	"	"	"	"
		121	GND	4.5V	GND			OUT		"					"	"	"	"	Data to Q2	"	"	"	"	"	"	"
		122	4.5V	4.5V	GND				OUT	"					"	"	"	"	Data to Q3	"	"	"	"	"	"	"
		123	GND	GND	4.5V					"	OUT				"	"	"	"	Data to Q4	"	"	"	"	"	"	"
		124	4.5V	GND	4.5V					"		OUT			"	"	"	"	Data to Q5	"	"	"	"	"	"	"
		125	GND	4.5V	4.5V					"			OUT		"	"	"	"	Data to Q6	"	"	"	"	"	"	"
		126	4.5V	4.5V	4.5V					"				OUT	"	"	"	"	Data to Q7	"	"	"	"	"	"	"
t_{PLH1}	3003 (Fig. 3)	127-134	Same tests and terminal conditions and limits as specified above for t_{PHL1}.																							
t_{PHL2}	3003 (Fig. 3)	135	GND	GND	GND	OUT				GND					4.5V	IN	4.5V	4.5V	$\overline{EN}$ to Q0	7	35	7	47	7	35	ns
		136	4.5V	GND	GND		OUT			"					"	"	"	"	$\overline{EN}$ to Q1	"	"	"	"	"	"	"
		137	GND	4.5V	GND			OUT		"					"	"	"	"	$\overline{EN}$ to Q2	"	"	"	"	"	"	"
		138	4.5V	4.5V	GND				OUT	"					"	"	"	"	$\overline{EN}$ to Q3	"	"	"	"	"	"	"
		139	GND	GND	4.5V					"	OUT				"	"	"	"	$\overline{EN}$ to Q4	"	"	"	"	"	"	"
		140	4.5V	GND	4.5V					"		OUT			"	"	"	"	$\overline{EN}$ to Q5	"	"	"	"	"	"	"
		141	GND	4.5V	4.5V					"			OUT		"	"	"	"	$\overline{EN}$ to Q6	"	"	"	"	"	"	"
		142	4.5V	4.5V	4.5V					"				OUT	"	"	"	"	EN to Q7	"	"	"	"	"	"	"
t_{PLH2}	3003 (Fig. 3)	143-150	Same tests and terminal conditions and limits as specified above for t_{PHL2}, with the exception that Data IN is at GND.																							

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TABLE III. Group A inspection for device type 02 – Continued.

Symbol	MIL-STD-883 method	Case 2	Terminal conditions 1/															Measured terminal	Test limits						Unit
			2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Subgroup 9 T_C = +25°C		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C		
		Cases E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
		Test no.	A0	A1	A2	Q0	Q1	Q2	Q3	GND	Q4	Q5	Q6	Q7	Data IN	$\overline{EN}$	$\overline{CLR}$	V_{CC}		Min	Max	Min	Max	Min	Max	
t_{PHL3}	3003	151	IN	GND	GND	OUT				GND					4.5V	IN	4.5V	4.5V	A0 to Q0	5	38	5	50	5	38	ns
		152	IN	GND	GND		OUT			"					"	"	"	"	A0 to Q1	"	"	"	"	"	"	"
		153	GND	IN	GND			OUT		"					"	"	"	"	A1 to Q2	"	"	"	"	"	"	"
		154	IN	4.5V	GND				OUT	"					"	"	"	"	A0 to Q3	"	"	"	"	"	"	"
		155	GND	GND	IN					"	OUT				"	"	"	"	A2 to Q4	"	"	"	"	"	"	"
		156	IN	GND	4.5V					"		OUT			"	"	"	"	A0 to Q5	"	"	"	"	"	"	"
		157	GND	IN	4.5V					"			OUT		"	"	"	"	A1 to Q6	"	"	"	"	"	"	"
		158	IN	4.5V	4.5V					"				OUT	"	"	"	"	A0 to Q7	"	"	"	"	"	"	"
t_{PLH3}	3003 (Fig. 3)	159-166	Same tests and terminal conditions and limits as specified above for t_{PHL3}, with the exception that Data IN is at GND.																							
t_{PHL4}	3003 (Fig. 3)	167	GND	GND	GND	OUT				GND					4.5V	5/	IN	4.5V	$\overline{CLR}$ to Q0	5	26	5	36	5	26	ns
		168	4.5V	GND	GND		OUT			"					"	"	"	"	$\overline{CLR}$ to Q1	"	"	"	"	"	"	"
		169	GND	4.5V	GND			OUT		"					"	"	"	"	$\overline{CLR}$ to Q2	"	"	"	"	"	"	"
		170	4.5V	4.5V	GND				OUT	"					"	"	"	"	$\overline{CLR}$ to Q3	"	"	"	"	"	"	"
		171	GND	GND	4.5V					"	OUT				"	"	"	"	$\overline{CLR}$ to Q4	"	"	"	"	"	"	"
		172	4.5V	GND	4.5V					"		OUT			"	"	"	"	$\overline{CLR}$ to Q5	"	"	"	"	"	"	"
		173	GND	4.5V	4.5V					"			OUT		"	"	"	"	$\overline{CLR}$ to Q6	"	"	"	"	"	"	"
		174	4.5V	4.5V	4.5V					"				OUT	"	"	"	"	$\overline{CLR}$ to Q7	"	"	"	"	"	"	"
t_{THL}	3004 (Fig. 3)	175	GND	GND	GND	OUT				GND					IN	GND	4.5V	4.5V	Q0	3	15	3	20	3	15	ns
		176	4.5V	GND	GND		OUT			"					"	"	"	"	Q1	"	"	"	"	"	"	"
		177	GND	4.5V	GND			OUT		"					"	"	"	"	Q2	"	"	"	"	"	"	"
		178	4.5V	4.5V	GND				OUT	"					"	"	"	"	Q3	"	"	"	"	"	"	"
		179	GND	GND	4.5V					"	OUT				"	"	"	"	Q4	"	"	"	"	"	"	"
		180	4.5V	GND	4.5V					"		OUT			"	"	"	"	Q5	"	"	"	"	"	"	"
		181	GND	4.5V	4.5V					"			OUT		"	"	"	"	Q6	"	"	"	"	"	"	"
		182	4.5V	4.5V	4.5V					"				OUT	"	"	"	"	Q7	"	"	"	"	"	"	"
t_{TLH}	3004 (Fig. 3)	183-190	Same tests and terminal conditions and limits as specified above for as t_{THL}.																							

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TABLE III. Group A inspection for device type 03.

Symbol	MIL-STD-883 method	Cases R, S, and 2	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	$\overline{OE}$	1Q	1D	2D	2Q	3Q	3D	4D	4Q	GND	$\overline{E}$	5Q	5D	6D	6Q	7Q	7D	8D	8Q	V_{CC}		Min	Max	Min	Max	Min	Max	
V_{IC} (pos)		1	1mA		1mA							1/										GND	$\overline{OE}$	.4 1/	1.5					V
		2				1mA						"										"	1D	"	"					"
		3							1mA			"										"	2D	"	"					"
		4								1mA		"										"	3D	"	"					"
		5										"										"	4D	"	"					"
		6										"	1mA									"	$\overline{E}$	"	"					"
		7										"			1mA							"	5D	"	"					"
		8										"				1mA						"	6D	"	"					"
		9										"							1mA			"	7D	"	"					"
		10										"								1mA			"	8D	"	"					"
V_{IC} (neg)		11	-1mA		-1mA							GND										1/	$\overline{OE}$		-1.5					"
		12				-1mA						"										"	1D		"					"
		13							-1mA			"										"	2D		"					"
		14								-1mA		"										"	3D		"					"
		15										"										"	4D		"					"
		16										"	-1mA									"	$\overline{E}$		"					"
		17										"			-1mA							"	5D		"					"
		18										"				-1mA						"	6D		"					"
		19										"							-1mA			"	7D		"					"
		20										"								-1mA			"	8D		"					"
I_{CC}	3005	21	GND		6.0V	6.0V			6.0V	6.0V		"	6.0V		6.0V	6.0V			6.0V	6.0V		6.0V	V_{CC}		.2		20			µA
		22	GND		GND	GND			GND	GND		"	6.0V		GND	GND			GND	GND		"	V_{CC}		.2		20			µA
I_{CCZ}	3005	23	6.0V		6.0V	6.0V			6.0V	6.0V		"	6.0V		6.0V	6.0V			6.0V	6.0V		"	V_{CC}		.1		10			µA
V_{OH3}	3006	24	1.2V	-20µA	4.2V							"	4.2V									"	1Q	5.95		5.95		5.95		V
		25	"			4.2V	-20µA					"	"									"	2Q	"		"		"		"
		26	"					-20µA	4.2V			"	"									"	3Q	"		"		"		"
		27	"							4.2V	-20µA	"	"									"	4Q	"		"		"		"
		28	"									"	"	-20µA	4.2V							"	5Q	"		"		"		"
		29	"									"	"			4.2V	-20µA					"	6Q	"		"		"		"
		30	"									"	"					-20µA	4.2V			"	7Q	"		"		"		"
		31	"									"	"							4.2V	-20µA	"	8Q	"		"		"		"
V_{OH5}	3006	32-39	Same tests and terminal conditions as V_{OH3}, except I_{OH} = -7.8 mA.																					5.36		5.2		5.36		"
V_{OL3}	3007	40	1.2V	20µA	1.2V							GND	4.2V									6.0V	1Q		.05		.05		.05	"
		41	"			1.2V	20µA					"	"									"	2Q		"		"		"	"
		42	"					20µA	1.2V			"	"									"	3Q		"		"		"	"
		43	"							1.2V	20µA	"	"									"	4Q		"		"		"	"
		44	"									"	"	20µA	1.2V							"	5Q		"		"		"	"
		45	"									"	"			1.2V	20µA					"	6Q		"		"		"	"
		46	"									"	"					20µA	1.2V			"	7Q		"		"		"	"
		47	"									"	"							1.2V	20µA	"	8Q		"		"		"	"
V_{OL5}	3007	48-55	Same tests and terminal conditions as V_{OL3}, except I_{OL} = 7.8 mA.																						.26		.4		.26	"
I_{OS4}	3011	56	GND	GND	4.0V	4.0V			4.0V	4.0V		GND	4.0V		4.0V	4.0V			4.0V	4.0V		4.0V	1Q	-10	-135	-10	-135	-10	-135	mA
		57	"		"	"	GND		"	"		"	"		"	"			"	"		"	2Q	"	"	"	"	"	"	"
		58	"		"	"		GND	"	"		"	"		"	"			"	"		"	3Q	"	"	"	"	"	"	"
		59	"		"	"			"	"	GND	"	"		"	"			"	"		"	4Q	"	"	"	"	"	"	"
		60	"		"	"			"	"		"	"	GND	"	"			"	"		"	5Q	"	"	"	"	"	"	"
		61	"		"	"			"	"		"	"		"	"	GND		"	"		"	6Q	"	"	"	"	"	"	"
		62	"		"	"			"	"		"	"		"	"		GND	"	"		"	7Q	"	"	"	"	"	"	"
		63	"		"	"			"	"		"	"		"	"			"	"	GND	"	8Q	"	"	"	"	"	"	"

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TABLE III. Group A inspection for device type 03 – Continued.

Symbol	MIL-STD-883 method	Cases R, S, and 2	Terminal conditions 1/																			Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	$\overline{OE}$	1Q	1D	2D	2Q	3Q	3D	4D	4Q	GND	$\overline{E}$	5Q	5D	6D	6Q	7Q	7D	8D	8Q	V_{CC}		Min	Max	Min	Max	Min	Max	
I_{IH}	3010	64	6.0V		6.0V	6.0V			6.0V	6.0V		GND	6.0V		6.0V	6.0V			6.0V	6.0V		6.0V	$\overline{OE}$		.05		.1			µA
		65	"		"	"			"	"		"	"		"	"			"	"		"	1D		"		"			"
		66	"		"	"			"	"		"	"		"	"			"	"		"	2D		"		"			"
		67	"		"	"			"	"		"	"		"	"			"	"		"	3D		"		"			"
		68	"		"	"			"	"		"	"		"	"			"	"		"	4D		"		"			"
		69	"		"	"			"	"		"	"		"	"			"	"		"	$\overline{E}$		"		"			"
		70	"		"	"			"	"		"	"		"	"			"	"		"	5D		"		"			"
		71	"		"	"			"	"		"	"		"	"			"	"		"	6D		"		"			"
		72	"		"	"			"	"		"	"		"	"			"	"		"	7D		"		"			"
		73	"		"	"			"	"		"	"		"	"			"	"		"	8D		"		"			"
I_{IL}	3009	74	GND		GND	GND			GND	GND		"	GND		GND	GND			GND	GND		"	$\overline{OE}$		-.05		-.1			"
		75	"		"	"			"	"		"	"		"	"			"	"		"	1D		"		"			"
		76	"		"	"			"	"		"	"		"	"			"	"		"	2D		"		"			"
		77	"		"	"			"	"		"	"		"	"			"	"		"	3D		"		"			"
		78	"		"	"			"	"		"	"		"	"			"	"		"	4D		"		"			"
		79	"		"	"			"	"		"	"		"	"			"	"		"	$\overline{E}$		"		"			"
		80	"		"	"			"	"		"	"		"	"			"	"		"	5D		"		"			"
		81	"		"	"			"	"		"	"		"	"			"	"		"	6D		"		"			"
		82	"		"	"			"	"		"	"		"	"			"	"		"	7D		"		"			"
		83	"		"	"			"	"		"	"		"	"			"	"		"	8D		"		"			"
I_{OZH}		84	7/	6.0V	7/	7/	6.0V	6.0V	7/	7/	6.0V	"	4.2V	6.0V	7/	7/	6.0V	6.0V	7/	7/	6.0V	"	1Q		.2		2.0			"
		85	"		"	"			"	"		"	"		"	"			"	"		"	2Q		"		"			"
		86	"		"	"			"	"		"	"		"	"			"	"		"	3Q		"		"			"
		87	"		"	"			"	"		"	"		"	"			"	"		"	4Q		"		"			"
		88	"		"	"			"	"		"	"		"	"			"	"		"	5Q		"		"			"
		89	"		"	"			"	"		"	"		"	"			"	"		"	6Q		"		"			"
		90	"		"	"			"	"		"	"		"	"			"	"		"	7Q		"		"			"
		91	"		"	"			"	"		"	"		"	"			"	"		"	8Q		"		"			"
I_{OZL}		92	"	GND	7/	7/	GND	GND	7/	7/	GND	"	"	GND	7/	7/	GND	GND	7/	7/	GND	"	1Q		-.2		-2.0			"
		93	"		"	"			"	"		"	"		"	"			"	"		"	2Q		"		"			"
		94	"		"	"			"	"		"	"		"	"			"	"		"	3Q		"		"			"
		95	"		"	"			"	"		"	"		"	"			"	"		"	4Q		"		"			"
		96	"		"	"			"	"		"	"		"	"			"	"		"	5Q		"		"			"
		97	"		"	"			"	"		"	"		"	"			"	"		"	6Q		"		"			"
		98	"		"	"			"	"		"	"		"	"			"	"		"	7Q		"		"			"
		99	"		"	"			"	"		"	"		"	"			"	"		"	8Q		"		"			"
																								Subgroup 4 T_C = +25°C						
																								Min	Max					
C_{in}	3012	100	2/		2/	2/			2/	2/		GND			2/	2/			2/	2/		GND	$\overline{OE}$		10					pF
		101	"		"	"			"	"		"	"		"	"			"	"		"	1D		"					"
		102	"		"	"			"	"		"	"		"	"			"	"		"	2D		"					"
		103	"		"	"			"	"		"	"		"	"			"	"		"	3D		"					"
		104	"		"	"			"	"		"	"		"	"			"	"		"	4D		"					"
		105	"		"	"			"	"		"	"		"	"			"	"		"	5D		"					"
		106	"		"	"			"	"		"	"		"	"			"	"		"	6D		"					"
		107	"		"	"			"	"		"	"		"	"			"	"		"	7D		"					"
		108	"		"	"			"	"		"	"		"	"			"	"		"	8D		"					"
		109	"		"	"			"	"		"	2/		"	"			"	"		"	$\overline{E}$		"					"
C_O	3012	110	4.2V	2/			2/	2/			2/	"	GND	2/			2/	2/			2/	6.0V	1Q		20					"
		111	"		"	"			"	"		"	"		"	"			"	"		"	2Q		"					"
		112	"		"	"			"	"		"	"		"	"			"	"		"	3Q		"					"
		113	"		"	"			"	"		"	"		"	"			"	"		"	4Q		"					"
		114	"		"	"			"	"		"	"		"	"			"	"		"	5Q		"					"
		115	"		"	"			"	"		"	"		"	"			"	"		"	6Q		"					"
		116	"		"	"			"	"		"	"		"	"			"	"		"	7Q		"					"
		117	"		"	"			"	"		"	"		"	"			"	"		"	8Q		"					"

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TABLE III. Group A inspection for device type 03 – Continued.

Symbol	MIL-STD-883 method	Cases R, S, and 2	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 7 T_C = +25°C		Subgroup 8 T_C = +125°C		Subgroup 8 T_C = -55°C		
		Test no.	$\overline{OE}$	1Q	1D	2D	2Q	3Q	3D	4D	4Q	GND	$\overline{E}$	5Q	5D	6D	6Q	7Q	7D	8D	8Q	V_{CC}		Min	Max	Min	Max	Min	Max	
Truth table 3/	3014	118	B	H	A	A	H	H	A	A	H	GND	A	H	A	A	H	H	A	A	H	4.5V	All outputs 4/							
		119	"	L	B	B	L	L	B	B	L	"	A	L	B	B	L	L	B	B	L	"								
		120	"	L	B	B	L	L	B	B	L	"	B	L	B	B	L	L	B	B	L	"								
		121	"	L	A	A	L	L	A	A	L	"	B	L	A	A	L	L	A	A	L	"								
		122	"	H	A	A	H	H	A	A	H	"	A	H	A	A	H	H	A	A	H	"								
		123	"	H	A	A	H	H	A	A	H	"	B	H	A	A	H	H	A	A	H	"								
		124	"	H	B	B	H	H	B	B	H	"	B	H	B	B	H	H	B	B	H	"								
																								Subgroup 9 T_C = +25°C		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C		
																								Min	Max	Min	Max	Min	Max	
t_{PHL1}	3003 (Fig. 3)	125	GND	OUT	IN	IN	OUT	OUT	IN	IN	OUT	GND	IN	OUT	IN	IN	OUT	OUT	IN	IN	OUT	4.5V	1D to 1Q	6	26	6	35	6	26	ns
		126	"									"	"									"	2D to 2Q	"	"	"	"	"	"	"
		127	"									"	"									"	3D to 3Q	"	"	"	"	"	"	"
		128	"									"	"									"	4D to 4Q	"	"	"	"	"	"	"
		129	"									"	"									"	5D to 5Q	"	"	"	"	"	"	"
		130	"									"	"									"	6D to 6Q	"	"	"	"	"	"	"
		131	"									"	"									"	7D to 7Q	"	"	"	"	"	"	"
		132	"									"	"									"	8D to 8Q	"	"	"	"	"	"	"
t_{PLH1}	3003 (Fig. 3)	133-140	Same tests and terminal conditions and limits as specified above for at t_{PHL1}.																											
t_{PLH2}	3003 (Fig. 3)	141	GND	OUT	4.5V	4.5V	OUT	OUT	4.5V	4.5V	OUT	GND	IN	OUT	4.5V	4.5V	OUT	OUT	4.5V	4.5V	OUT	4.5V	$\overline{E}$ to 1Q	6	31	6	41	6	31	ns
		142	"									"	"									"	$\overline{E}$ to 2Q	"	"	"	"	"	"	"
		143	"									"	"									"	$\overline{E}$ to 3Q	"	"	"	"	"	"	"
		144	"									"	"									"	$\overline{E}$ to 4Q	"	"	"	"	"	"	"
		145	"									"	"									"	$\overline{E}$ to 5Q	"	"	"	"	"	"	"
		146	"									"	"									"	$\overline{E}$ to 6Q	"	"	"	"	"	"	"
		147	"									"	"									"	$\overline{E}$ to 7Q	"	"	"	"	"	"	"
		148	"									"	"									"	$\overline{E}$ to 8Q	"	"	"	"	"	"	"
t_{PHL2}	3003 (Fig. 3)	149	"	OUT	GND	GND	OUT	OUT	GND	GND	OUT	"	"	OUT	GND	GND	OUT	OUT	GND	GND	OUT	"	$\overline{E}$ to 1Q	"	"	"	"	"	"	"
		150	"									"	"									"	$\overline{E}$ to 2Q	"	"	"	"	"	"	"
		151	"									"	"									"	$\overline{E}$ to 3Q	"	"	"	"	"	"	"
		152	"									"	"									"	$\overline{E}$ to 4Q	"	"	"	"	"	"	"
		153	"									"	"									"	$\overline{E}$ to 5Q	"	"	"	"	"	"	"
		154	"									"	"									"	$\overline{E}$ to 6Q	"	"	"	"	"	"	"
		155	"									"	"									"	$\overline{E}$ to 7Q	"	"	"	"	"	"	"
		156	"									"	"									"	$\overline{E}$ to 8Q	"	"	"	"	"	"	"
t_{PHZ}	3003 (Fig. 3)	157	IN	OUT	4.5V	4.5V	OUT	OUT	4.5V	4.5V	OUT	"	4.5V	OUT	4.5V	4.5V	OUT	OUT	4.5V	4.5V	OUT	"	$\overline{OE}$ to 1Q	5	26	5	35	5	26	"
		158	"									"	"									"	$\overline{OE}$ to 2Q	"	"	"	"	"	"	"
		159	"									"	"									"	$\overline{OE}$ to 3Q	"	"	"	"	"	"	"
		160	"									"	"									"	$\overline{OE}$ to 4Q	"	"	"	"	"	"	"
		161	"									"	"									"	$\overline{OE}$ to 5Q	"	"	"	"	"	"	"
		162	"									"	"									"	$\overline{OE}$ to 6Q	"	"	"	"	"	"	"
		163	"									"	"									"	$\overline{OE}$ to 7Q	"	"	"	"	"	"	"
		164	"									"	"									"	$\overline{OE}$ to 8Q	"	"	"	"	"	"	"
t_{PLZ}	3003 (Fig. 3)	165	"	OUT	GND	GND	OUT	OUT	GND	GND	OUT	"	"	OUT	GND	GND	OUT	OUT	GND	GND	OUT	"	$\overline{OE}$ to 1Q	"	"	"	"	"	"	"
		166	"									"	"									"	$\overline{OE}$ to 2Q	"	"	"	"	"	"	"
		167	"									"	"									"	$\overline{OE}$ to 3Q	"	"	"	"	"	"	"
		168	"									"	"									"	$\overline{OE}$ to 4Q	"	"	"	"	"	"	"
		169	"									"	"									"	$\overline{OE}$ to 5Q	"	"	"	"	"	"	"
		170	"									"	"									"	$\overline{OE}$ to 6Q	"	"	"	"	"	"	"
		171	"									"	"									"	$\overline{OE}$ to 7Q	"	"	"	"	"	"	"
		172	"									"	"									"	$\overline{OE}$ to 8Q	"	"	"	"	"	"	"

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TABLE III. Group A inspection for device type 03 – Continued.

Symbol	MIL-STD-883 method	Cases R, S, and 2	Terminal conditions 1/																			Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 9 T_C = +25°C		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C		
		Test no.	$\overline{OE}$	1Q	1D	2D	2Q	3Q	3D	4D	4Q	GND	$\overline{E}$	5Q	5D	6D	6Q	7Q	7D	8D	8Q	V_{CC}		Min	Max	Min	Max	Min	Max	
t_{PZH}	3003 (Fig. 3)	173	IN	OUT	4.5V							GND	4.5V									4.5V	$\overline{OE}$ to 1Q	5	26	5	35	5	26	ns
		174	"			4.5V	OUT					"	"									"	$\overline{OE}$ to 2Q	"	"	"	"	"	"	"
		175	"					OUT	4.5V			"	"									"	$\overline{OE}$ to 3Q	"	"	"	"	"	"	"
		176	"							4.5V	OUT	"	"									"	$\overline{OE}$ to 4Q	"	"	"	"	"	"	"
		177	"									"	"	OUT	4.5V							"	$\overline{OE}$ to 5Q	"	"	"	"	"	"	"
		178	"									"	"			4.5V	OUT					"	$\overline{OE}$ to 6Q	"	"	"	"	"	"	"
		179	"									"	"					OUT	4.5V			"	$\overline{OE}$ to 7Q	"	"	"	"	"	"	"
		180	"									"	"							4.5V	OUT	"	$\overline{OE}$ to 8Q	"	"	"	"	"	"	"
t_{PZL}	3003 (Fig. 3)	181-188	Same test and terminal conditions and limits as specified above for t_{PLZ}.																											
t_{THL}	3004 (Fig. 3)	189	GND	OUT	IN							GND	4.5V									4.5V	1Q	2	12	2	16	2	12	ns
		190	"			IN	OUT					"	"									"	2Q	"	"	"	"	"	"	"
		191	"					OUT	IN			"	"									"	3Q	"	"	"	"	"	"	"
		192	"							IN	OUT	"	"									"	4Q	"	"	"	"	"	"	"
		193	"									"	"	OUT	IN							"	5Q	"	"	"	"	"	"	"
		194	"									"	"			IN	OUT					"	6Q	"	"	"	"	"	"	"
		195	"									"	"					OUT	IN			"	7Q	"	"	"	"	"	"	"
		196	"									"	"							IN	OUT	"	8Q	"	"	"	"	"	"	"
t_{TLH}	3004 (Fig. 3)	197-204	Same tests and terminal conditions and limits as specified above for t_{THL}.																											

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TABLE III. Group A inspection for device type 04.

Symbol	MIL-STD-883 method	Cases R, S, and 2	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	$\overline{OE}$	$\overline{Q0}$	D0	D1	$\overline{Q1}$	$\overline{Q2}$	D2	D3	$\overline{Q3}$	GND	$\overline{LE}$	$\overline{Q4}$	D4	D5	$\overline{Q5}$	$\overline{Q6}$	D6	D7	$\overline{Q7}$	V_{CC}		Min	Max	Min	Max	Min	Max	
V_{IC} (pos)		1	1mA		1mA							1/										GND	$\overline{OE}$	.4 1/	1.5					V
		2				1mA						"										"	D0	"	"					"
		3							1mA			"										"	D1	"	"					"
		4								1mA		"										"	D2	"	"					"
		5										"	1mA									"	D3	"	"					"
		6										"										"	$\overline{LE}$	"	"					"
		7										"			1mA							"	D4	"	"					"
		8										"				1mA						"	D5	"	"					"
		9										"							1mA			"	D6	"	"					"
		10										"								1mA			"	D7	"	"					"
V_{IC} (neg)		11	-1mA		-1mA							GND										1/	$\overline{OE}$		-1.5					"
		12				-1mA						"										"	D0		"					"
		13							-1mA			"										"	D1		"					"
		14								-1mA		"										"	D2		"					"
		15										"	-1mA									"	$\overline{D3}$		"					"
		16										"			-1mA							"	$\overline{LE}$		"					"
		17										"				-1mA						"	D4		"					"
		18										"							-1mA			"	D5		"					"
		19										"								-1mA			"	D6		"					"
		20										"									-1mA	"	D7		"					"
I_{CC}	3005	21	GND		6.0V	6.0V			6.0V	6.0V		"	6.0V		6.0V	6.0V			6.0V	6.0V		6.0V	V_{CC}		.2		20			µA
		22	GND		GND	GND			GND	GND		"	"		GND	GND			GND	GND		"	V_{CC}		.2		20			"
I_{CCZ}	3005	23	6.0V		6.0V	6.0V			6.0V	6.0V		"	"		GND	GND			GND	GND		"	V_{CC}		.1		10			"
		24	6.0V		GND	GND			GND	GND		"	"		6.0V	6.0V			6.0V	6.0V		"	V_{CC}		.1		10			"
V_{OH3}	3006	25	1.2V	-20µA	1.2V							"	4.2V									"	$\overline{Q0}$	5.95		5.95		5.95		V
		26	"	"		1.2V	-20µA					"	"									"	$\overline{Q1}$	"		"		"		"
		27	"	"				-20µA	1.2V			"	"									"	$\overline{Q2}$	"		"		"		"
		28	"	"						1.2V	-20µA	"	"									"	$\overline{Q3}$	"		"		"		"
		29	"	"								"	"	-20µA	1.2V							"	$\overline{Q4}$	"		"		"		"
		30	"	"								"	"			1.2V	-20µA					"	$\overline{Q5}$	"		"		"		"
		31	"	"								"	"					-20µA	1.2V			"	$\overline{Q6}$	"		"		"		"
		32	"	"								"	"							1.2V	-20µA	"	$\overline{Q7}$	"		"		"		"
V_{OH5}	3006	33	"	-7.8mA	1.2V							"	"									"	$\overline{Q0}$	5.36		5.2		5.36		"
		34	"	"		1.2V	-7.8mA					"	"									"	$\overline{Q1}$	"		"		"		"
		35	"	"				-7.8mA	1.2V			"	"									"	$\overline{Q2}$	"		"		"		"
		36	"	"						1.2V	-7.8mA	"	"									"	$\overline{Q3}$	"		"		"		"
		37	"	"								"	"	-7.8mA	1.2V							"	$\overline{Q4}$	"		"		"		"
		38	"	"								"	"			1.2V	-7.8mA					"	$\overline{Q5}$	"		"		"		"
		39	"	"								"	"					-7.8mA	1.2V			"	$\overline{Q6}$	"		"		"		"
		40	"	"								"	"							1.2V	-7.8mA	"	$\overline{Q7}$	"		"		"		"
V_{OL3}	3007	41	"	20µA	4.2V							"	"									"	$\overline{Q0}$		.05		.05		.05	V
		42	"	"		4.2V	20µA					"	"									"	$\overline{Q1}$		"		"		"	"
		43	"	"				20µA	4.2V			"	"									"	$\overline{Q2}$		"		"		"	"
		44	"	"						4.2V	20µA	"	"									"	$\overline{Q3}$		"		"		"	"
		45	"	"								"	"	20µA	4.2V							"	$\overline{Q4}$		"		"		"	"
		46	"	"								"	"			4.2V	20µA					"	$\overline{Q5}$		"		"		"	"
		47	"	"								"	"					20µA	4.2V			"	$\overline{Q6}$		"		"		"	"
		48	"	"								"	"							4.2V	20µA	"	$\overline{Q7}$		"		"		"	"
V_{OL5}	3007	49	"	7.8mA	4.2V							"	"									"	$\overline{Q0}$		.26		.4		.26	"
		50	"	"		4.2V	7.8mA					"	"									"	$\overline{Q1}$		"		"		"	"
		51	"	"				7.8mA	4.2V			"	"									"	$\overline{Q2}$		"		"		"	"
		52	"	"						4.2V	7.8mA	"	"									"	$\overline{Q3}$		"		"		"	"
		53	"	"								"	"	7.8mA	4.2V							"	$\overline{Q4}$		"		"		"	"
		54	"	"								"	"			4.2V	7.8mA					"	$\overline{Q5}$		"		"		"	"
		55	"	"								"	"					7.8mA	4.2V			"	$\overline{Q6}$		"		"		"	"
		56	"	"								"	"							4.2V	7.8mA	"	$\overline{Q7}$		"		"		"	"

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TABLE III. Group A inspection for device type 04 – Continued.

Symbol	MIL-STD-883 method	Cases R, S, and 2	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	$\overline{OE}$	$\overline{Q0}$	D0	D1	$\overline{Q1}$	$\overline{Q2}$	D2	D3	$\overline{Q3}$	GND	$\overline{LE}$	$\overline{Q4}$	D4	D5	$\overline{Q5}$	$\overline{Q6}$	D6	D7	$\overline{Q7}$	V_{CC}		Min	Max	Min	Max	Min	Max	
I_{OS4}	3011	57	GND		GND	GND			GND	GND		GND	4V		GND	GND			GND	GND		4V	$\overline{Q0}$	-10	-135	-10	-135	-10	-135	mA
		58	"									"	"									"	$\overline{Q1}$	"	"	"	"	"	"	"
		59	"									"	"									"	$\overline{Q2}$	"	"	"	"	"	"	"
		60	"									"	"									"	$\overline{Q3}$	"	"	"	"	"	"	"
		61	"									"	"									"	$\overline{Q4}$	"	"	"	"	"	"	"
		62	"									"	"									"	$\overline{Q5}$	"	"	"	"	"	"	"
		63	"									"	"									"	$\overline{Q6}$	"	"	"	"	"	"	"
		64	"									"	"									"	$\overline{Q7}$	"	"	"	"	"	"	"
I_{OZH}		65	7/	6V	7/	7/	6V	6V	7/	7/	6V	"	4.2V	6V	7/	7/	6V	6V	7/	7/	6V	6V	$\overline{Q0}$		-.2		-2.0			µA
		66	"									"	"									"	$\overline{Q1}$		"		"			"
		67	"									"	"									"	$\overline{Q2}$		"		"			"
		68	"									"	"									"	$\overline{Q3}$		"		"			"
		69	"									"	"									"	$\overline{Q4}$		"		"			"
		70	"									"	"									"	$\overline{Q5}$		"		"			"
		71	"									"	"									"	$\overline{Q6}$		"		"			"
		72	"									"	"									"	$\overline{Q7}$		"		"			"
I_{OZL}		73	"	GND	7/	7/	GND	GND	7/	7/	GND	"	"	GND	7/	7/	GND	GND	7/	7/	GND	"	$\overline{Q0}$		.2		2.0			"
		74	"									"	"									"	$\overline{Q1}$		"		"			"
		75	"									"	"									"	$\overline{Q2}$		"		"			"
		76	"									"	"									"	$\overline{Q3}$		"		"			"
		77	"									"	"									"	$\overline{Q4}$		"		"			"
		78	"									"	"									"	$\overline{Q5}$		"		"			"
		79	"									"	"									"	$\overline{Q6}$		"		"			"
		80	"									"	"									"	$\overline{Q7}$		"		"			"
I_{IH}	3010	81	6V		6V	6V			6V	6V		"	6V		6V	6V			6V	6V		"	$\overline{OE}$		.05		.1			"
		82										"										"	D0		"		"			"
		83										"										"	D1		"		"			"
		84										"										"	D2		"		"			"
		85										"										"	D3		"		"			"
		86										"										"	$\overline{LE}$		"		"			"
		87										"										"	D4		"		"			"
		88										"										"	D5		"		"			"
		89										"										"	D6		"		"			"
		90										"										"	D7		"		"			"
I_{IL}	3009	91	GND		GND	GND			GND	GND		"	GND		GND	GND			GND	GND		"	$\overline{OE}$		-.05		-.1			"
		92										"										"	D0		"		"			"
		93										"										"	D1		"		"			"
		94										"										"	D2		"		"			"
		95										"										"	D3		"		"			"
		96										"										"	$\overline{LE}$		"		"			"
		97										"										"	D4		"		"			"
		98										"										"	D5		"		"			"
		99										"										"	D6		"		"			"
		100										"										"	D7		"		"			"
																								Subgroup 4 T_C = +25°C						
																								Min	Max					
C_{in}	3012	101	2/		2/	2/			2/	2/		"			2/	2/			2/	2/		GND	$\overline{OE}$		10					pF
		102										"										"	D0		"					"
		103										"										"	D1		"					"
		104										"										"	D2		"					"
		105										"										"	D3		"					"
		106										"										"	D4		"					"
		107										"										"	D5		"					"
		108										"										"	D6		"					"
		109										"										"	D7		"					"
		110										"	2/									"	$\overline{LE}$		"					"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 04 – Continued.

Symbol	MIL-STD-883 method	Cases R, S, and 2	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 4 T_C = +25°C					
		Test no.	$\overline{OE}$	$\overline{Q0}$	D0	D1	$\overline{Q1}$	$\overline{Q2}$	D2	D3	$\overline{Q3}$	GND	$\overline{LE}$	$\overline{Q4}$	D4	D5	$\overline{Q5}$	$\overline{Q6}$	D6	D7	$\overline{Q7}$	V_{CC}		Min	Max				
C_O	3012	111	6V	2/			2/					GND										6V	$\overline{Q0}$		20					pF
		112	"					2/				"										"	Q1		"					"
		113	"									"										"	$\overline{Q2}$		"					"
		114	"								2/	"										"	$\overline{Q3}$		"					"
		115	"									"		2/								"	$\overline{Q4}$		"					"
		116	"									"					2/					"	$\overline{Q5}$		"					"
		117	"									"						2/				"	$\overline{Q6}$		"					"
		118	"									"									2/	"	Q7		"					"
																								Subgroup 7 T_C = +25°C		Subgroup 8 T_C = +125°C		Subgroup 8 T_C = -55°C		
																								Min	Max	Min	Max	Min	Max	
Truth table 3/	3014	119	B	L	A	B	H	L	A	B	H	GND	A	L	A	B	H	L	A	B	H	4.5V	All outputs 4/							
		120	"	H	B	A	L	H	B	A	L	"	A	H	B	A	L	H	B	A	L	"	"							
		121	"	H	A	B	L	H	B	A	L	"	B	H	A	B	L	H	A	B	L	"	"							
		122	"	H	B	A	L	H	A	B	H	"	B	L	A	B	L	H	A	B	H	"	"							
		123	"	L	A	B	H	L	A	B	L	"	A	H	B	A	H	L	B	A	L	"	"							
		124	"	H	B	A	L	H	B	A	L	"	A	L	A	B	L	H	A	B	H	"	"							
		125	"	L	A	B	H	L	B	A	H	"	A	L	B	A	H	L	B	A	L	"	"							
		126	"	L	B	A	H	L	A	B	H	"	B	L	A	B	H	L	B	A	H	"	"							
		127	"	L	B	A	H	L	B	A	H	"	B	L	B	A	H	L	B	A	H	"	"							
																								Subgroup 9 T_C = +25°C		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C		
																								Min	Max	Min	Max	Min	Max	
t_{PLH1}	3003 (Fig. 3)	128	0.0V	OUT	IN							GND	4.5V									4.5V	D0 to $\overline{Q0}$	6	29	6	39	6	29	ns
		129	"			IN	OUT	OUT	IN	IN	OUT	"	"									"	D1 to $\overline{Q1}$	"	"	"	"	"	"	"
		130	"									"	"	OUT	IN							"	D2 to $\overline{Q2}$	"	"	"	"	"	"	"
		131	"									"	"			IN	OUT					"	D3 to $\overline{Q3}$	"	"	"	"	"	"	"
		132	"									"	"					OUT	IN			"	D4 to $\overline{Q4}$	"	"	"	"	"	"	"
		133	"									"	"							IN	OUT	"	D5 to $\overline{Q5}$	"	"	"	"	"	"	"
		134	"									"	"									"	D6 to $\overline{Q6}$	"	"	"	"	"	"	"
		135	"									"	"									"	D7 to $\overline{Q7}$	"	"	"	"	"	"	"
t_{PHL1}	3003 (Fig. 3)	136-143	Same tests, terminal conditions, and limits as specified above for t_{PLH1}.																											
t_{PLH2}	3003 (Fig. 3)	144	0.0V	OUT	0.0V							GND	IN									4.5V	$\overline{LE}$ to $\overline{Q0}$	6	31	6	41	6	31	ns
		145	"			0.0V	OUT	OUT	0.0V	0.0V	OUT	"	"									"	$\overline{LE}$ to $\overline{Q1}$	"	"	"	"	"	"	"
		146	"									"	"	OUT	0.0V							"	$\overline{LE}$ to $\overline{Q2}$	"	"	"	"	"	"	"
		147	"									"	"			0.0V	OUT					"	$\overline{LE}$ to $\overline{Q3}$	"	"	"	"	"	"	"
		148	"									"	"					OUT	0.0V			"	$\overline{LE}$ to $\overline{Q4}$	"	"	"	"	"	"	"
		149	"									"	"							0.0V	OUT	"	$\overline{LE}$ to $\overline{Q5}$	"	"	"	"	"	"	"
		150	"									"	"									"	$\overline{LE}$ to $\overline{Q6}$	"	"	"	"	"	"	"
		151	"									"	"									"	$\overline{LE}$ to $\overline{Q7}$	"	"	"	"	"	"	"
t_{PHL2}	3003 (Fig. 3)	152	"	OUT	4.5V							"	"									"	$\overline{LE}$ to $\overline{Q0}$	"	"	"	"	"	"	"
		153	"			4.5V	OUT	OUT	4.5V	4.5V	OUT	"	"									"	$\overline{LE}$ to $\overline{Q1}$	"	"	"	"	"	"	"
		154	"									"	"	OUT	4.5V							"	$\overline{LE}$ to $\overline{Q2}$	"	"	"	"	"	"	"
		155	"									"	"			4.5V	OUT					"	$\overline{LE}$ to $\overline{Q3}$	"	"	"	"	"	"	"
		156	"									"	"					OUT	4.5V			"	$\overline{LE}$ to $\overline{Q4}$	"	"	"	"	"	"	"
		157	"									"	"							4.5V	OUT	"	$\overline{LE}$ to $\overline{Q5}$	"	"	"	"	"	"	"
		158	"									"	"									"	$\overline{LE}$ to $\overline{Q6}$	"	"	"	"	"	"	"
		159	"									"	"									"	$\overline{LE}$ to $\overline{Q7}$	"	"	"	"	"	"	"
t_{PHZ}	3003 (Fig. 3)	160	IN	OUT	0.0V							"	4.5V									"	$\overline{OE}$ to $\overline{Q0}$	5	26	5	35	5	26	"
		161	"			0.0V	OUT	OUT	0.0V	0.0V	OUT	"	"									"	$\overline{OE}$ to $\overline{Q1}$	"	"	"	"	"	"	"
		162	"									"	"	OUT	0.0V							"	$\overline{OE}$ to $\overline{Q2}$	"	"	"	"	"	"	"
		163	"									"	"			0.0V	OUT					"	$\overline{OE}$ to $\overline{Q3}$	"	"	"	"	"	"	"
		164	"									"	"					OUT	0.0V			"	$\overline{OE}$ to $\overline{Q4}$	"	"	"	"	"	"	"
		165	"									"	"							0.0V	OUT	"	$\overline{OE}$ to $\overline{Q5}$	"	"	"	"	"	"	"
		166	"									"	"									"	$\overline{OE}$ to $\overline{Q6}$	"	"	"	"	"	"	"
		167	"									"	"									"	$\overline{OE}$ to $\overline{Q7}$	"	"	"	"	"	"	"

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TABLE III. Group A inspection for device type 04 – Continued.

Symbol	MIL-STD-883 method	Cases R, S, and 2	Terminal conditions 1/																			Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 9 T_C = +25°C		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C		
		Test no.	$\overline{OE}$	$\overline{Q0}$	D0	D1	$\overline{Q1}$	$\overline{Q2}$	D2	D3	$\overline{Q3}$	GND	$\overline{LE}$	$\overline{Q4}$	D4	D5	$\overline{Q5}$	$\overline{Q6}$	D6	D7	$\overline{Q7}$	V_{CC}		Min	Max	Min	Max	Min	Max	
t_{PLZ}	3003 (Fig. 3)	168	IN	OUT	4.5V							GND	4.5V									4.5V	$\overline{OE}$ to $\overline{Q0}$	5	26	5	35	5	26	ns
		169	"			4.5V	OUT					"	"									"	$\overline{OE}$ to $\overline{Q1}$	"	"	"	"	"	"	"
		170	"					OUT	4.5V			"	"									"	$\overline{OE}$ to $\overline{Q2}$	"	"	"	"	"	"	"
		171	"							4.5V	OUT	"	"									"	$\overline{OE}$ to $\overline{Q3}$	"	"	"	"	"	"	"
		172	"									"	"	OUT	4.5V							"	$\overline{OE}$ to $\overline{Q4}$	"	"	"	"	"	"	"
		173	"									"	"			4.5V	OUT					"	$\overline{OE}$ to $\overline{Q5}$	"	"	"	"	"	"	"
		174	"									"	"					OUT	4.5V			"	$\overline{OE}$ to $\overline{Q6}$	"	"	"	"	"	"	"
		175	"									"	"							4.5V	OUT	"	$\overline{OE}$ to $\overline{Q7}$	"	"	"	"	"	"	"
t_{PZH}	3003 (Fig. 3)	176	"	OUT	0.0V							"	"									"	$\overline{OE}$ to $\overline{Q0}$	"	"	"	"	"	"	"
		177	"			0.0V	OUT					"	"									"	$\overline{OE}$ to $\overline{Q1}$	"	"	"	"	"	"	"
		178	"					OUT	0.0V			"	"									"	$\overline{OE}$ to $\overline{Q2}$	"	"	"	"	"	"	"
		179	"							0.0V	OUT	"	"									"	$\overline{OE}$ to $\overline{Q3}$	"	"	"	"	"	"	"
		180	"									"	"	OUT	0.0V							"	$\overline{OE}$ to $\overline{Q4}$	"	"	"	"	"	"	"
		181	"									"	"			0.0V	OUT					"	$\overline{OE}$ to $\overline{Q5}$	"	"	"	"	"	"	"
		182	"									"	"					OUT	0.0V			"	$\overline{OE}$ to $\overline{Q6}$	"	"	"	"	"	"	"
		183	"									"	"							0.0V	OUT	"	$\overline{OE}$ to $\overline{Q7}$	"	"	"	"	"	"	"
t_{PZL}	3003 (Fig. 3)	184	"	OUT	4.5V							"	"									"	$\overline{OE}$ to $\overline{Q0}$	"	"	"	"	"	"	"
		185	"			4.5V	OUT					"	"									"	$\overline{OE}$ to $\overline{Q1}$	"	"	"	"	"	"	"
		186	"					OUT	4.5V			"	"									"	$\overline{OE}$ to $\overline{Q2}$	"	"	"	"	"	"	"
		187	"							4.5V	OUT	"	"									"	$\overline{OE}$ to $\overline{Q3}$	"	"	"	"	"	"	"
		188	"									"	"	OUT	4.5V							"	$\overline{OE}$ to $\overline{Q4}$	"	"	"	"	"	"	"
		189	"									"	"			4.5V	OUT					"	$\overline{OE}$ to $\overline{Q5}$	"	"	"	"	"	"	"
		190	"									"	"					OUT	4.5V			"	$\overline{OE}$ to $\overline{Q6}$	"	"	"	"	"	"	"
		191	"									"	"							4.5V	OUT	"	$\overline{OE}$ to $\overline{Q7}$	"	"	"	"	"	"	"
t_{THL}	3004 (Fig. 3)	192	0.0V	OUT	IN							"	"									"	$\overline{Q0}$	2	12	2	16	2	12	"
		193	"			IN	OUT					"	"									"	$\overline{Q1}$	"	"	"	"	"	"	"
		194	"					OUT	IN			"	"									"	$\overline{Q2}$	"	"	"	"	"	"	"
		195	"							IN	OUT	"	"									"	$\overline{Q3}$	"	"	"	"	"	"	"
		196	"									"	"	OUT	IN							"	$\overline{Q4}$	"	"	"	"	"	"	"
		197	"									"	"			IN	OUT					"	$\overline{Q5}$	"	"	"	"	"	"	"
		198	"									"	"					OUT	IN			"	$\overline{Q6}$	"	"	"	"	"	"	"
		199	"									"	"							IN	OUT	"	$\overline{Q7}$	"	"	"	"	"	"	"
t_{TLH}	3004 (Fig. 3)	200-207	Same tests, terminal conditions, and limits as specified above for t_{THL}.																											

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TABLE III. Group A inspection for device type 05.

Symbol	MIL-STD-883 method	Cases R, S, and 2	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	$\overline{OE}$	D0	D1	D2	D3	D4	D5	D6	D7	GND	$\overline{LE}$	$\overline{Q7}$	$\overline{Q6}$	$\overline{Q5}$	$\overline{Q4}$	$\overline{Q3}$	$\overline{Q2}$	$\overline{Q1}$	$\overline{Q0}$	V_{CC}		Min	Max	Min	Max	Min	Max	
V_{IC} (pos)		1	1mA									1/										GND	$\overline{OE}$	.4 1/	1.5					V
		2		1mA								"										"	D0	"	"					"
		3			1mA							"										"	D1	"	"					"
		4				1mA						"										"	D2	"	"					"
		5					1mA					"										"	D3	"	"					"
		6						1mA				"										"	D4	"	"					"
		7							1mA			"										"	D5	"	"					"
		8								1mA		"										"	D6	"	"					"
		9									1mA	"										"	D7	"	"					"
		10										"	1mA									"	$\overline{LE}$	"	"					"
V_{IC} (neg)		11	-1mA									GND										1/	$\overline{OE}$		-1.5					"
		12		-1mA								"										"	D0		"					"
		13			-1mA							"										"	D1		"					"
		14				-1mA						"										"	D2		"					"
		15					-1mA					"										"	D3		"					"
		16						-1mA				"										"	D4		"					"
		17							-1mA			"										"	D5		"					"
		18								-1mA		"										"	D6		"					"
		19									-1mA	"										"	D7		"					"
		20										"	-1mA									"	$\overline{LE}$		"					"
I_{CC}	3005	21	GND	6.0V	6.0V	6.0V	6.0V	6.0V	6.0V	6.0V	6.0V	"	6.0V									6.0V	V_{CC}		.2		20			µA
		22	GND	GND	GND	GND	GND	GND	GND	GND	GND	"	6.0V									6.0V	V_{CC}		.2		20			µA
I_{CCZ}	3005	23	6.0V	6.0V	6.0V	6.0V	6.0V	6.0V	6.0V	6.0V	6.0V	"	"									6.0V	V_{CC}		.1		10			µA
V_{OH3}	3006	24	1.2V	1.2V								GND	4.2V								-20µA	6.0V	$\overline{Q0}$	5.95		5.95		5.95		V
		25	"		1.2V							"	"									"	$\overline{Q1}$	"		"		"		"
		26	"			1.2V						"	"						-20µA			"	$\overline{Q2}$	"		"		"		"
		27	"				1.2V					"	"					-20µA				"	$\overline{Q3}$	"		"		"		"
		28	"					1.2V				"	"				-20µA					"	$\overline{Q4}$	"		"		"		"
		29	"						1.2V			"	"			-20µA						"	$\overline{Q5}$	"		"		"		"
		30	"							1.2V		"	"		-20µA							"	$\overline{Q6}$	"		"		"		"
		31	"								1.2V	"	"	-20µA								"	$\overline{Q7}$	"		"		"		"
V_{OH5}	3006	32	"	1.2V								"	"								-7.8mA	"	$\overline{Q0}$	5.48		5.2		5.48		"
		33	"		1.2V							"	"						-7.8mA			"	$\overline{Q1}$	"		"		"		"
		34	"			1.2V						"	"					-7.8mA				"	$\overline{Q2}$	"		"		"		"
		35	"				1.2V					"	"				-7.8mA					"	$\overline{Q3}$	"		"		"		"
		36	"					1.2V				"	"			-7.8mA						"	$\overline{Q4}$	"		"		"		"
		37	"						1.2V			"	"		-7.8mA							"	$\overline{Q5}$	"		"		"		"
		38	"							1.2V		"	"	-7.8mA								"	$\overline{Q6}$	"		"		"		"
		39	"								1.2V	"	"									"	$\overline{Q7}$	"		"		"		"
V_{OL3}	3007	40	"	4.2V								"	"								20µA	"	$\overline{Q0}$		.05		.05		.05	V
		41	"		4.2V							"	"						20µA			"	$\overline{Q1}$		"		"		"	"
		42	"			4.2V						"	"					20µA				"	$\overline{Q2}$		"		"		"	"
		43	"				4.2V					"	"				20µA					"	$\overline{Q3}$		"		"		"	"
		44	"					4.2V				"	"			20µA						"	$\overline{Q4}$		"		"		"	"
		45	"						4.2V			"	"		20µA							"	$\overline{Q5}$		"		"		"	"
		46	"							4.2V		"	"	20µA								"	$\overline{Q6}$		"		"		"	"
		47	"								4.2V	"	"									"	$\overline{Q7}$		"		"		"	"
V_{OL5}	3007	48	"	4.2V								"	"								7.8mA	"	$\overline{Q0}$		.26		.4		.26	"
		49	"		4.2V							"	"						7.8mA			"	$\overline{Q1}$		"		"		"	"
		50	"			4.2V						"	"					7.8mA				"	$\overline{Q2}$		"		"		"	"
		51	"				4.2V					"	"				7.8mA					"	$\overline{Q3}$		"		"		"	"
		52	"					4.2V				"	"			7.8mA						"	$\overline{Q4}$		"		"		"	"
		53	"						4.2V			"	"		7.8mA							"	$\overline{Q5}$		"		"		"	"
		54	"							4.2V		"	"	7.8mA								"	$\overline{Q6}$		"		"		"	"
		55	"								4.2V	"	"									"	$\overline{Q7}$		"		"		"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 05 - Continued.

Symbol	MIL-STD-883 method	Cases R, S, and 2	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	$\overline{OE}$	D0	D1	D2	D3	D4	D5	D6	D7	GND	$\overline{LE}$	$\overline{Q7}$	$\overline{Q6}$	$\overline{Q5}$	$\overline{Q4}$	$\overline{Q3}$	$\overline{Q2}$	$\overline{Q1}$	$\overline{Q0}$	V_{CC}		Min	Max	Min	Max	Min	Max	
I_{OS4}	3011	56	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	4V									4 V	$\overline{Q0}$	-10	-135	-10	-135	-10	-135	mA
		57	"									"	"									"	$\overline{Q1}$	"	"	"	"	"	"	"
		58	"									"	"									"	$\overline{Q2}$	"	"	"	"	"	"	"
		59	"									"	"									"	$\overline{Q3}$	"	"	"	"	"	"	"
		60	"									"	"									"	$\overline{Q4}$	"	"	"	"	"	"	"
		61	"									"	"									"	$\overline{Q5}$	"	"	"	"	"	"	"
		62	"									"	"									"	$\overline{Q6}$	"	"	"	"	"	"	"
		63	"									"	"									"	$\overline{Q7}$	"	"	"	"	"	"	"
I_{OZH}		64	7/	7/	7/	7/	7/	7/	7/	7/	7/	"	4.2V									6 V	$\overline{Q0}$		-.2		-2.0			µA
		65	"									"	"									"	$\overline{Q1}$		"		"			"
		66	"									"	"									"	$\overline{Q2}$		"		"			"
		67	"									"	"									"	$\overline{Q3}$		"		"			"
		68	"									"	"									"	$\overline{Q4}$		"		"			"
		69	"									"	"									"	$\overline{Q5}$		"		"			"
		70	"									"	"									"	$\overline{Q6}$		"		"			"
		71	"									"	"	6 V								"	$\overline{Q7}$		"		"			"
I_{OZL}		72	"	7/	7/	7/	7/	7/	7/	7/	7/	"	"									"	$\overline{Q0}$		.2		2.0			"
		73	"									"	"									"	$\overline{Q1}$		"		"			"
		74	"									"	"									"	$\overline{Q2}$		"		"			"
		75	"									"	"									"	$\overline{Q3}$		"		"			"
		76	"									"	"									"	$\overline{Q4}$		"		"			"
		77	"									"	"									"	$\overline{Q5}$		"		"			"
		78	"									"	"									"	$\overline{Q6}$		"		"			"
		79	"									"	"	GND								"	$\overline{Q7}$		"		"			"
I_{IH}	3010	80	6 V	6 V	6 V	6V	6V	6V	6V	6V	6V	"										"	$\overline{OE}$		.05		.1			"
		81										"										"	D0		"		"			"
		82										"										"	D1		"		"			"
		83										"										"	D2		"		"			"
		84										"										"	D3		"		"			"
		85										"										"	D4		"		"			"
		86										"										"	D5		"		"			"
		87										"										"	D6		"		"			"
		88										"										"	D7		"		"			"
		89										"	6 V									"	$\overline{LE}$		"		"			"
I_{IL}	3009	90	GND	GND	GND	GND	GND	GND	GND	GND	GND	"										"	$\overline{OE}$		-.05		-.1			"
		91										"										"	D0		"		"			"
		92										"										"	D1		"		"			"
		93										"										"	D2		"		"			"
		94										"										"	D3		"		"			"
		95										"										"	D4		"		"			"
		96										"										"	D5		"		"			"
		97										"										"	D6		"		"			"
		98										"										"	D7		"		"			"
		99										"	GND									"	$\overline{LE}$		"		"			"
																								Subgroup 4 T_C = +25°C						
																								Min	Max					
C_{in}	3012	100	2/	2/	2/	2/	2/	2/	2/	2/	2/	GND										GND	$\overline{OE}$		10					pF
		101										"										"	D0		"					"
		102										"										"	D1		"					"
		103										"										"	D2		"					"
		104										"										"	D3		"					"
		105										"										"	D4		"					"
		106										"										"	D5		"					"
		107										"										"	D6		"					"
		108										"										"	D7		"					"
		109										"	2/									"	$\overline{LE}$		"					"

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TABLE III. Group A inspection for device type 05 – Continued.

Symbol	MIL-STD-883 method	Cases R, S, and 2	Terminal conditions 1/																			Measured terminal	Test limits						Unit
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 4 T_C = +25°C					
		Test no.	$\overline{OE}$	D0	D1	D2	D3	D4	D5	D6	D7	GND	$\overline{LE}$	$\overline{Q7}$	$\overline{Q6}$	$\overline{Q5}$	$\overline{Q4}$	$\overline{Q3}$	$\overline{Q2}$	$\overline{Q1}$	$\overline{Q0}$	V_{CC}		Min	Max				
C_O	3012	110	6 V									GND										$\overline{Q0}$		20					pF
		111	"									"										$\overline{Q1}$		"					"
		112	"									"										$\overline{Q2}$		"					"
		113	"									"										$\overline{Q3}$		"					"
		114	"									"										$\overline{Q4}$		"					"
		115	"									"										$\overline{Q5}$		"					"
		116	"									"										$\overline{Q6}$		"					"
		117	"									"										$\overline{Q7}$		"					"
														Subgroup 7 T_C = +25°C		Subgroup 8 T_C = +125°C		Subgroup 8 T_C = -55°C										
														Min	Max	Min	Max	Min	Max									
Truth table 3/	3014	118	B	A	B	A	B	A	B	A	B	GND	A	H	L	H	L	H	L	H	L	4.5 V	All Outputs 4/						
		119	"	B	A	B	A	B	A	B	A	"	A	L	H	L	H	L	H	L	H	"	"						
		120	"	A	B	A	B	A	B	A	B	"	B	L	H	L	H	L	H	L	H	"	"						
		121	"	A	B	A	B	A	B	A	B	"	B	H	L	L	H	L	H	L	H	"	"						
		122	"	B	A	B	A	B	A	B	A	"	A	L	H	H	L	H	L	H	L	"	"						
		123	"	A	B	A	B	A	B	A	B	"	A	H	L	L	H	L	H	L	H	"	"						
		124	"	B	A	B	A	B	A	B	A	"	B	L	H	H	L	H	L	H	L	"	"						
		125	"	A	B	A	B	A	B	A	B	"	B	H	L	H	L	H	L	H	L	"	"						
		126	"	B	A	B	A	B	A	B	A	"	A	L	H	L	H	L	H	L	H	"	"						
														Subgroup 9 T_C = +25°C		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C										
														Min	Max	Min	Max	Min	Max									
t_{PLH1}	3003 (Fig. 3)	127	0.0V	IN	IN	IN	IN	IN	IN	IN	IN	GND	4.5 V	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	4.5 V	D0 to $\overline{Q0}$	4	31	4	41	4	31	ns
		128	"									"	"									"	D1 to $\overline{Q1}$	"	"	"	"	"	"	"
		129	"									"	"									"	D2 to $\overline{Q2}$	"	"	"	"	"	"	"
		130	"									"	"									"	D3 to $\overline{Q3}$	"	"	"	"	"	"	"
		131	"									"	"									"	D4 to $\overline{Q4}$	"	"	"	"	"	"	"
		132	"									"	"									"	D5 to $\overline{Q5}$	"	"	"	"	"	"	"
		133	"									"	"									"	D6 to $\overline{Q6}$	"	"	"	"	"	"	"
		134	"									"	"									"	D7 to $\overline{Q7}$	"	"	"	"	"	"	"
t_{PHL1}	3003 (Fig. 3)	135-142	Same tests, terminal conditions and limits as specified above for t_{PLH1}.																										
t_{PLH2}	3003 (Fig. 3)	143	0.0V	0.0V	0.0V	0.0V	0.0V	0.0V	0.0V	0.0V	0.0V	GND	IN	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	4.5 V	$\overline{LE}$ to $\overline{Q0}$	4	31	4	41	4	31	ns
		144	"									"	"									"	$\overline{LE}$ to $\overline{Q1}$	"	"	"	"	"	"	"
		145	"									"	"									"	$\overline{LE}$ to $\overline{Q2}$	"	"	"	"	"	"	"
		146	"									"	"									"	$\overline{LE}$ to $\overline{Q3}$	"	"	"	"	"	"	"
		147	"									"	"									"	$\overline{LE}$ to $\overline{Q4}$	"	"	"	"	"	"	"
		148	"									"	"									"	$\overline{LE}$ to $\overline{Q5}$	"	"	"	"	"	"	"
		149	"									"	"									"	$\overline{LE}$ to $\overline{Q6}$	"	"	"	"	"	"	"
		150	"									"	"									"	$\overline{LE}$ to $\overline{Q7}$	"	"	"	"	"	"	"
t_{PHL2}	3003 (Fig. 3)	151	"	4.5V	4.5V	4.5V	4.5V	4.5V	4.5V	4.5V	4.5V	"	"	"	"	"	"	"	"	"	"	"	$\overline{LE}$ to $\overline{Q0}$	"	"	"	"	"	"	"
		152	"									"	"									"	$\overline{LE}$ to $\overline{Q1}$	"	"	"	"	"	"	"
		153	"									"	"									"	$\overline{LE}$ to $\overline{Q2}$	"	"	"	"	"	"	"
		154	"									"	"									"	$\overline{LE}$ to $\overline{Q3}$	"	"	"	"	"	"	"
		155	"									"	"									"	$\overline{LE}$ to $\overline{Q4}$	"	"	"	"	"	"	"
		156	"									"	"									"	$\overline{LE}$ to $\overline{Q5}$	"	"	"	"	"	"	"
		157	"									"	"									"	$\overline{LE}$ to $\overline{Q6}$	"	"	"	"	"	"	"
		158	"									"	"									"	$\overline{LE}$ to $\overline{Q7}$	"	"	"	"	"	"	"
t_{PHZ}	3003 (Fig. 3)	159	IN	0.0V	0.0V	0.0V	0.0V	0.0V	0.0V	0.0V	0.0V	GND	4.5 V	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	$\overline{OE}$ to $\overline{Q0}$	3	26	3	35	3	26	ns
		160	"									"	"									"	$\overline{OE}$ to $\overline{Q1}$	"	"	"	"	"	"	"
		161	"									"	"									"	$\overline{OE}$ to $\overline{Q2}$	"	"	"	"	"	"	"
		162	"									"	"									"	$\overline{OE}$ to $\overline{Q3}$	"	"	"	"	"	"	"
		163	"									"	"									"	$\overline{OE}$ to $\overline{Q4}$	"	"	"	"	"	"	"
		164	"									"	"									"	$\overline{OE}$ to $\overline{Q5}$	"	"	"	"	"	"	"
		165	"									"	"									"	$\overline{OE}$ to $\overline{Q6}$	"	"	"	"	"	"	"
		166	"									"	"									"	$\overline{OE}$ to $\overline{Q7}$	"	"	"	"	"	"	"

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TABLE III. Group A inspection for device type 05 – Continued.

Symbol	MIL-STD-883 method	Cases R, S, and 2	Terminal conditions 1/																			Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 9 T_C = +25°C		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C		
		Test no.	$\overline{OE}$	D0	D1	D2	D3	D4	D5	D6	D7	GND	$\overline{LE}$	$\overline{Q7}$	$\overline{Q6}$	$\overline{Q5}$	$\overline{Q4}$	$\overline{Q3}$	$\overline{Q2}$	$\overline{Q1}$	$\overline{Q0}$	V_{CC}		Min	Max	Min	Max	Min	Max	
t_{PLZ}	3003 (Fig. 3)	167	IN	4.5V	4.5V	4.5V	4.5V	4.5V	4.5V	4.5V	4.5V	GND	4.5 V	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	4.5 V	$\overline{OE}$ to $\overline{Q0}$	3	26	3	35	3	26	ns
		168	"									"	"									"	$\overline{OE}$ to $\overline{Q1}$	"	"	"	"	"	"	"
		169	"									"	"									"	$\overline{OE}$ to $\overline{Q2}$	"	"	"	"	"	"	"
		170	"									"	"									"	$\overline{OE}$ to $\overline{Q3}$	"	"	"	"	"	"	"
		171	"									"	"									"	$\overline{OE}$ to $\overline{Q4}$	"	"	"	"	"	"	"
		172	"									"	"									"	$\overline{OE}$ to $\overline{Q5}$	"	"	"	"	"	"	"
		173	"									"	"									"	$\overline{OE}$ to $\overline{Q6}$	"	"	"	"	"	"	"
		174	"									"	"									"	$\overline{OE}$ to $\overline{Q7}$	"	"	"	"	"	"	"
t_{PZH}	3003 (Fig. 3)	175	"	0.0V	0.0V	0.0V	0.0V	0.0V	0.0V	0.0V	0.0V	"	"	"	"	"	"	"	"	"	"	"	$\overline{OE}$ to $\overline{Q0}$	4	26	4	35	4	26	"
		176	"									"	"									"	$\overline{OE}$ to $\overline{Q1}$	"	"	"	"	"	"	"
		177	"									"	"									"	$\overline{OE}$ to $\overline{Q2}$	"	"	"	"	"	"	"
		178	"									"	"									"	$\overline{OE}$ to $\overline{Q3}$	"	"	"	"	"	"	"
		179	"									"	"									"	$\overline{OE}$ to $\overline{Q4}$	"	"	"	"	"	"	"
		180	"									"	"									"	$\overline{OE}$ to $\overline{Q5}$	"	"	"	"	"	"	"
		181	"									"	"									"	$\overline{OE}$ to $\overline{Q6}$	"	"	"	"	"	"	"
		182	"									"	"									"	$\overline{OE}$ to $\overline{Q7}$	"	"	"	"	"	"	"
t_{PZL}	3003 (Fig. 3)	183	"	4.5V	4.5V	4.5V	4.5V	4.5V	4.5V	4.5V	4.5V	"	"	"	"	"	"	"	"	"	"	"	$\overline{OE}$ to $\overline{Q0}$	"	"	"	"	"	"	"
		184	"									"	"									"	$\overline{OE}$ to $\overline{Q1}$	"	"	"	"	"	"	"
		185	"									"	"									"	$\overline{OE}$ to $\overline{Q2}$	"	"	"	"	"	"	"
		186	"									"	"									"	$\overline{OE}$ to $\overline{Q3}$	"	"	"	"	"	"	"
		187	"									"	"									"	$\overline{OE}$ to $\overline{Q4}$	"	"	"	"	"	"	"
		188	"									"	"									"	$\overline{OE}$ to $\overline{Q5}$	"	"	"	"	"	"	"
		189	"									"	"									"	$\overline{OE}$ to $\overline{Q6}$	"	"	"	"	"	"	"
		190	"									"	"									"	$\overline{OE}$ to $\overline{Q7}$	"	"	"	"	"	"	"
t_{THL}	3004 (Fig. 3)	191	0.0V	IN	IN	IN	IN	IN	IN	IN	IN	"	"	"	"	"	"	"	"	"	"	"	$\overline{Q0}$	2	12	2	16	2	12	ns
		192	"									"	"									"	$\overline{Q1}$	"	"	"	"	"	"	"
		193	"									"	"									"	$\overline{Q2}$	"	"	"	"	"	"	"
		194	"									"	"									"	$\overline{Q3}$	"	"	"	"	"	"	"
		195	"									"	"									"	$\overline{Q4}$	"	"	"	"	"	"	"
		196	"									"	"									"	$\overline{Q5}$	"	"	"	"	"	"	"
		197	"									"	"									"	$\overline{Q6}$	"	"	"	"	"	"	"
		198	"									"	"									"	$\overline{Q7}$	"	"	"	"	"	"	"
t_{TLH}	3004 (Fig. 3)	199-206	Same tests, terminal conditions and limits as specified above for t_{THL}.																											

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TABLE III. Group A inspection for device type 06.

Symbol	MIL-STD-883 method	Cases R, S, and 2	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	$\overline{OE}$	1D	2D	3D	4D	5D	6D	7D	8D	GND	$\overline{ENC}$	8Q	7Q	6Q	5Q	4Q	3Q	2Q	1Q	V_{CC}		Min	Max	Min	Max	Min	Max	
V_{IC} (pos)		1	1mA									1/										GND	$\overline{OE}$	.4 1/	1.5					V
		2		1mA								"										"	1D	"	"					"
		3			1mA							"										"	2D	"	"					"
		4				1mA						"										"	3D	"	"					"
		5					1mA					"										"	4D	"	"					"
		6						1mA				"										"	5D	"	"					"
		7							1mA			"										"	6D	"	"					"
		8								1mA		"										"	7D	"	"					"
		9									1mA	"										"	8D	"	"					"
		10										"	1mA									"	$\overline{ENC}$	"	"					"
V_{IC} (neg)		11	-1mA									GND										1/	$\overline{OE}$		-1.5					"
		12		-1mA								"										"	1D		"					"
		13			-1mA							"										"	2D		"					"
		14				-1mA						"										"	3D		"					"
		15					-1mA					"										"	4D		"					"
		16						-1mA				"										"	5D		"					"
		17							-1mA			"										"	6D		"					"
		18								-1mA		"										"	7D		"					"
		19									-1mA	"										"	8D		"					"
		20										"	-1mA									"	$\overline{ENC}$		"					"
I_{CC}	3005	21	GND	6.0V	6.0V	6.0V	6.0V	6.0V	6.0V	6.0V	6.0V	"	GND									6.0V	V_{CC}		.2		20			µA
		22	GND	GND	GND	GND	GND	GND	GND	GND	GND	"	6.0V									"	V_{CC}		.2		20			µA
I_{CCZ}	3005	23	6.0V	6.0V	6.0V	6.0V	6.0V	6.0V	6.0V	6.0V	6.0V	"	6.0V									"	V_{CC}		.1		10			µA
V_{OH3}	3006	24	1.2V								4.2V	GND	4.2V	-20µA								"	8Q	5.95		5.95		5.95		V
		25	"							4.2V	"	"	"		-20µA							"	7Q	"		"		"		"
		26	"						4.2V	"	"	"	"			-20µA						"	6Q	"		"		"		"
		27	"					4.2V	"	"	"	"	"				-20µA					"	5Q	"		"		"		"
		28	"				4.2V	"	"	"	"	"	"					-20µA				"	4Q	"		"		"		"
		29	"			4.2V	"	"	"	"	"	"	"						-20µA			"	3Q	"		"		"		"
		30	"		4.2V	"	"	"	"	"	"	"	"							-20µA		"	2Q	"		"		"		"
		31	"	4.2V	"	"	"	"	"	"	"	"	"								-20µA	"	1Q	"		"		"		"
V_{OH5}	3006	32-39	Same tests and terminal conditions as V_{OH3}, except I_{OH} = -7.8 mA.																					5.48		5.2		5.48		"
V_{OL3}	3007	40	1.2V								1.2V	GND	4.2V	20µA								"	8Q		.05		.05		.05	"
		41	"							1.2V	"	"	"		20µA							"	7Q		"		"		"	"
		42	"						1.2V	"	"	"	"			20µA						"	6Q		"		"		"	"
		43	"					1.2V	"	"	"	"	"				20µA					"	5Q		"		"		"	"
		44	"				1.2V	"	"	"	"	"	"					20µA				"	4Q		"		"		"	"
		45	"			1.2V	"	"	"	"	"	"	"						20µA			"	3Q		"		"		"	"
		46	"		1.2V	"	"	"	"	"	"	"	"							20µA		"	2Q		"		"		"	"
		47	"	1.2V	"	"	"	"	"	"	"	"	"								20µA	"	1Q		"		"		"	"
V_{OL5}	3007	48-55	Same tests and terminal conditions as V_{OL3}, except I_{OL} = 7.8 mA.																						.26		.4		.26	"
I_{OS4}	3011	56	GND	4.0V	4.0V	4.0V	4.0V	4.0V	4.0V	4.0V	4.0V	GND	4.0V	GND								4.0V	8Q	-10	-135	-10	-135	-10	-135	mA
		57	"	"	"	"	"	"	"	"	"	"	"		GND							"	7Q	"	"	"	"	"	"	"
		58	"	"	"	"	"	"	"	"	"	"	"			GND						"	6Q	"	"	"	"	"	"	"
		59	"	"	"	"	"	"	"	"	"	"	"				GND					"	5Q	"	"	"	"	"	"	"
		60	"	"	"	"	"	"	"	"	"	"	"					GND				"	4Q	"	"	"	"	"	"	"
		61	"	"	"	"	"	"	"	"	"	"	"						GND			"	3Q	"	"	"	"	"	"	"
		62	"	"	"	"	"	"	"	"	"	"	"							GND		"	2Q	"	"	"	"	"	"	"
		63	"	"	"	"	"	"	"	"	"	"	"								GND	"	1Q	"	"	"	"	"	"	"
I_{IH}	3010	64		6.0V								"										6.0V	1Q		.05		.1			µA
		65			6.0V							"										"	2Q		"		"			"
		66				6.0V						"										"	3Q		"		"			"
		67					6.0V					"										"	4Q		"		"			"
		68						6.0V				"										"	5Q		"		"			"
		69							6.0V			"										"	6Q		"		"			"
		70								6.0V		"										"	7Q		"		"			"
		71									6.0V	"										"	8Q		"		"			"
		72	6.0V									"										"	$\overline{OE}$		"		"			"
		73										"	6.0V									"	$\overline{ENC}$		"		"			"

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TABLE III. Group A inspection for device type 06 – Continued.

Symbol	MIL-STD-883 method	Cases R, S, and 2	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	$\overline{OE}$	1D	2D	3D	4D	5D	6D	7D	8D	GND	$\overline{ENC}$	8Q	7Q	6Q	5Q	4Q	3Q	2Q	1Q	V_{CC}		Min	Max	Min	Max	Min	Max	
I_{IL}	3009	74		GND								GND										6.0V	1D		-.05		-.1			µA
		75			GND							"										"	2D		"		"			"
		76				GND						"										"	3D		"		"			"
		77					GND					"										"	4D		"		"			"
		78						GND				"										"	5D		"		"			"
		79							GND			"										"	6D		"		"			"
		80								GND		"										"	7D		"		"			"
		81									GND	"										"	8D		"		"			"
		82	GND									"										"	$\overline{OE}$		"		"			"
		83										"	GND									"	$\overline{ENC}$		"		"			"
I_{OZH}		84	7/	7/								GND	4.2V	6.0V								"	8Q		.2		2.0			µA
		85	"		7/							"	"		6.0V							"	7Q		"		"			"
		86	"			7/						"	"			6.0V						"	6Q		"		"			"
		87	"				7/					"	"				6.0V					"	5Q		"		"			"
		88	"					7/				"	"					6.0V				"	4Q		"		"			"
		89	"						7/			"	"						6.0V				"	3Q		"		"			"
		90	"							7/		"	"							6.0V		"	2Q		"		"			"
		91	"								7/	"	"								6.0V	"	1Q		"		"			"
I_{OZL}		92	"	7/								"	"	GND								"	8Q		-.2		-2.0			"
		93	"		7/							"	"		GND							"	7Q		"		"			"
		94	"			7/						"	"			GND						"	6Q		"		"			"
		95	"				7/					"	"				GND					"	5Q		"		"			"
		96	"					7/				"	"					GND				"	4Q		"		"			"
		97	"						7/			"	"						GND				"	3Q		"		"			"
		98	"							7/		"	"							GND		"	2Q		"		"			"
		99	"								7/	"	"								GND	"	1Q		"		"			"
																								Subgroup 4 T_C = +25°C						
																								Min	Max					
C_{in}	3012	100	2/																			GND	$\overline{OE}$		10					pF
		101		2/								"										"	1D		"					"
		102			2/							"										"	2D		"					"
		103				2/						"										"	3D		"					"
		104					2/					"										"	4D		"					"
		105						2/				"										"	5D		"					"
		106							2/			"										"	6D		"					"
		107								2/		"										"	7D		"					"
		108									2/	"										"	8D		"					"
		109										"	2/									"	$\overline{ENC}$		"					"
C_O	3012	110	6.0V									"	6.0V	2/	2/	2/	2/	2/	2/	2/	2/	6.0V	8Q		20					"
		111	"									"	"									"	7Q		"					"
		112	"									"	"									"	6Q		"					"
		113	"									"	"									"	5Q		"					"
		114	"									"	"									"	4Q		"					"
		115	"									"	"									"	3Q		"					"
		116	"									"	"									"	2Q		"					"
		117	"									"	"									"	1Q		"					"
																								Subgroup 7 T_C = +25°C		Subgroup 8 T_C = +125°C		Subgroup 8 T_C = -55°C		
																								Min	Max	Min	Max	Min	Max	
Truth table 3/	3014	118	B	B	B	B	B	B	B	B	B	GND	A	L	L	L	L	L	L	L	L	4.5V	All outputs 4/							
		119	"	A	A	A	A	A	A	A	A	"	A	H	L	H	H	H	H	H	H	"								
		120	"	B	B	B	B	B	B	B	B	"	B	L	L	L	L	L	L	L	L	"								
		121	"	B	A	A	A	A	A	A	A	"	B	L	L	L	L	L	L	L	L	"								
		122	"	A	B	A	A	A	A	A	A	"	A	H	H	H	H	H	H	H	H	"								
		123	"	A	A	B	A	A	A	A	A	"	B	H	H	H	H	H	H	H	H	"								
		124	"	A	A	A	B	A	A	A	A	"	B	H	H	H	H	H	H	H	H	"								
		125	"	A	A	A	A	B	A	A	A	"	B	H	H	H	H	H	H	H	H	"								
		126	"	A	A	A	A	A	B	B	B	"	A	L	L	L	L	L	L	L	L	"								

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TABLE III. Group A inspection for device type 06 – Continued.

Symbol	MIL-STD-883 method	Cases R, S, and 2	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 9 T_C = +25°C		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C		
		Test no.	$\overline{OE}$	1D	2D	3D	4D	5D	6D	7D	8D	GND	$\overline{ENC}$	8Q	7Q	6Q	5Q	4Q	3Q	2Q	1Q	V_{CC}		Min	Max	Min	Max	Min	Max	
t_{PHL1}	3003 (Fig. 3)	127	GND	IN	IN	IN	IN	IN	IN	IN	IN	GND	IN									4.5V	1Q to 1Q	4	31	4	41	4	31	ns
		128	"									"	"									"	2Q to 2Q	"	"	"	"	"	"	"
		129	"									"	"									"	3Q to 3Q	"	"	"	"	"	"	"
		130	"									"	"					OUT	OUT			"	4Q to 4Q	"	"	"	"	"	"	"
		131	"									"	"				OUT					"	5Q to 5Q	"	"	"	"	"	"	"
		132	"									"	"			OUT						"	6Q to 6Q	"	"	"	"	"	"	"
		133	"									"	"		OUT							"	7Q to 7Q	"	"	"	"	"	"	"
		134	"									"	"	OUT								"	8Q to 8Q	"	"	"	"	"	"	"
t_{PLH1}	3003 (Fig. 3)	135-142	Same tests, terminal conditions and limits as specified above for t_{PHL1}.																											
t_{PHL2}	3003 (Fig. 3)	143	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	IN									4.5V	ENC to 1Q	4	31	4	41	4	31	ns
		144	"									"	"									"	ENC to 2Q	"	"	"	"	"	"	"
		145	"									"	"									"	ENC to 3Q	"	"	"	"	"	"	"
		146	"									"	"					OUT	OUT			"	ENC to 4Q	"	"	"	"	"	"	"
		147	"									"	"				OUT					"	ENC to 5Q	"	"	"	"	"	"	"
		148	"									"	"			OUT						"	ENC to 6Q	"	"	"	"	"	"	"
		149	"									"	"		OUT							"	ENC to 7Q	"	"	"	"	"	"	"
		150	"									"	"	OUT								"	ENC to 8Q	"	"	"	"	"	"	"
t_{PLH2}	3003 (Fig. 3)	151-158	Same tests, terminal conditions and limits as specified above for t_{PHL2}.																											
t_{PHZ}	3003 (Fig. 3)	159	IN	4.5V	4.5V	4.5V	4.5V	4.5V	4.5V	4.5V	4.5V	GND	4.5V									"	$\overline{OE}$ to 1Q	3	26	3	35	3	26	ns
		160	"									"	"									"	$\overline{OE}$ to 2Q	"	"	"	"	"	"	"
		161	"									"	"									"	$\overline{OE}$ to 3Q	"	"	"	"	"	"	"
		162	"									"	"					OUT	OUT			"	$\overline{OE}$ to 4Q	"	"	"	"	"	"	"
		163	"									"	"				OUT					"	$\overline{OE}$ to 5Q	"	"	"	"	"	"	"
		164	"									"	"			OUT						"	$\overline{OE}$ to 6Q	"	"	"	"	"	"	"
		165	"									"	"		OUT							"	$\overline{OE}$ to 7Q	"	"	"	"	"	"	"
		166	"									"	"	OUT								"	$\overline{OE}$ to 8Q	"	"	"	"	"	"	"
t_{PZH}	3003 (Fig. 3)	167	"	4.5V	4.5V	4.5V	4.5V	4.5V	4.5V	4.5V	4.5V	"	"									"	$\overline{OE}$ to 1Q	4	"	4	"	4	"	"
		168	"									"	"									"	$\overline{OE}$ to 2Q	"	"	"	"	"	"	"
		169	"									"	"									"	$\overline{OE}$ to 3Q	"	"	"	"	"	"	"
		170	"									"	"					OUT	OUT			"	$\overline{OE}$ to 4Q	"	"	"	"	"	"	"
		171	"									"	"				OUT					"	$\overline{OE}$ to 5Q	"	"	"	"	"	"	"
		172	"									"	"			OUT						"	$\overline{OE}$ to 6Q	"	"	"	"	"	"	"
		173	"									"	"		OUT							"	$\overline{OE}$ to 7Q	"	"	"	"	"	"	"
		174	"									"	"	OUT								"	$\overline{OE}$ to 8Q	"	"	"	"	"	"	"
t_{PLZ}	3003 (Fig. 3)	175	"	0.0V	0.0V	0.0V	0.0V	0.0V	0.0V	0.0V	0.0V	"	"									"	$\overline{OE}$ to 1Q	3	"	3	"	3	"	"
		176	"									"	"									"	$\overline{OE}$ to 2Q	"	"	"	"	"	"	"
		177	"									"	"									"	$\overline{OE}$ to 3Q	"	"	"	"	"	"	"
		178	"									"	"					OUT	OUT			"	$\overline{OE}$ to 4Q	"	"	"	"	"	"	"
		179	"									"	"				OUT					"	$\overline{OE}$ to 5Q	"	"	"	"	"	"	"
		180	"									"	"			OUT						"	$\overline{OE}$ to 6Q	"	"	"	"	"	"	"
		181	"									"	"		OUT							"	$\overline{OE}$ to 7Q	"	"	"	"	"	"	"
		182	"									"	"	OUT								"	$\overline{OE}$ to 8Q	"	"	"	"	"	"	"
t_{PZL}	3003 (Fig. 3)	183	"	GND	GND	GND	GND	GND	GND	GND	GND	"	"									"	$\overline{OE}$ to 1Q	4	"	4	"	4	"	"
		184	"									"	"									"	$\overline{OE}$ to 2Q	"	"	"	"	"	"	"
		185	"									"	"									"	$\overline{OE}$ to 3Q	"	"	"	"	"	"	"
		186	"									"	"					OUT	OUT			"	$\overline{OE}$ to 4Q	"	"	"	"	"	"	"
		187	"									"	"				OUT					"	$\overline{OE}$ to 5Q	"	"	"	"	"	"	"
		188	"									"	"			OUT						"	$\overline{OE}$ to 6Q	"	"	"	"	"	"	"
		189	"									"	"		OUT							"	$\overline{OE}$ to 7Q	"	"	"	"	"	"	"
		190	"									"	"	OUT								"	$\overline{OE}$ to 8Q	"	"	"	"	"	"	"
		191	"									"	"									"								

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TABLE III. Group A inspection for device type 06 – Continued.

Symbol	MIL-STD-883 method	Cases R, S, and 2	Terminal conditions 1/																			Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 9 T_C = +25°C		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C		
		Test no.	$\overline{OE}$	1D	2D	3D	4D	5D	6D	7D	8D	GND	$\overline{ENC}$	8Q	7Q	6Q	5Q	4Q	3Q	2Q	1Q	V_{CC}		Min	Max	Min	Max	Min	Max	
t_{THL}	3004 (Fig. 3)	192	GND	IN	IN	IN	IN	IN	IN	IN	IN	GND	4.5V	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	4.5V	1Q	2	12	2	16	2	12	ns
		193	"									"	"									"	2Q	"	"	"	"	"	"	"
		194	"									"	"									"	3Q	"	"	"	"	"	"	"
		195	"									"	"									"	4Q	"	"	"	"	"	"	"
		196	"									"	"									"	5Q	"	"	"	"	"	"	"
		197	"									"	"									"	6Q	"	"	"	"	"	"	"
		198	"									"	"									"	7Q	"	"	"	"	"	"	"
		199	"									"	"									"	8Q	"	"	"	"	"	"	"
t_{TLH}	3004 (Fig. 3)	200-207	Same tests, terminal conditions and limits as specified above for t_{THL}.																											

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TABLE III. Group A inspection for device type 53.

Symbol	MIL-STD-883 method	Cases R, S, and 2	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	$\overline{OE}$	1Q	1D	2D	2Q	3Q	3D	4D	4Q	GND	$\overline{E}$	5Q	5D	6D	6Q	7Q	7D	8D	8Q	V_{CC}		Min	Max	Min	Max	Min	Max	
V_{IC} (pos)		1	1mA		1mA							1/										GND	$\overline{OE}$	.4 1/	1.5					V
		2				1mA						"										"	1D	"	"					"
		3							1mA			"										"	2D	"	"					"
		4								1mA		"										"	3D	"	"					"
		5										"	1mA									"	4D	"	"					"
		6										"										"	$\overline{E}$	"	"					"
		7										"			1mA							"	5D	"	"					"
		8										"				1mA						"	6D	"	"					"
		9										"							1mA			"	7D	"	"					"
		10										"								1mA			"	8D	"	"					"
V_{IC} (neg)		11	-1mA		-1mA							GND										1/	$\overline{OE}$		-1.5					"
		12				-1mA						"										"	1D		"					"
		13							-1mA			"										"	2D		"					"
		14								-1mA		"										"	3D		"					"
		15										"	-1mA									"	4D		"					"
		16										"										"	$\overline{E}$		"					"
		17										"			-1mA							"	5D		"					"
		18										"				-1mA						"	6D		"					"
		19										"							-1mA			"	7D		"					"
		20										"								-1mA			"	8D		"					"
I_{CC}	3005	21	GND		5.5V	5.5V			5.5V	5.5V		"	5.5V		5.5V	5.5V			5.5V	5.5V		5.5V	V_{CC}		.2		20			µA
		22	GND		GND	GND			GND	GND		"	"		GND	GND			GND	GND		"	V_{CC}		.2		20			"
I_{CCZ}	3005	23	5.5V		5.5V	5.5V			5.5V	5.5V		"	"		5.5V	5.5V			5.5V	5.5V		"	V_{CC}		.1		10			"
$I_{CC}\Delta$	3005	24	2.4V									"	"									"	V_{CC}		3.0		3.0	3.0		mA
		25	0.8V									"	2.4V									"	V_{CC}		3.0		3.0	3.0		"
		26	0.8V		0.8V	0.8V			0.8V	0.8V		"	5.5V		0.8V	0.8V			0.8V	2.4V		"	V_{CC}		3.0		3.0	3.0		"
I_{OZH}		27	7/	5.5V	7/							"	5.5V									"	1Q		.2		2.0			µA
		28	"			7/	5.5V					"	"									"	2Q		"		"			"
		29	"					5.5V	7/			"	"									"	3Q		"		"			"
		30	"							7/	5.5V	"	"									"	4Q		"		"			"
		31	"									"	"	5.5V	7/							"	5Q		"		"			"
		32	"									"	"			7/	5.5V					"	6Q		"		"			"
		33	"									"	"					5.5V	7/			"	7Q		"		"			"
		34	"									"	"							7/	5.5V	"	8Q		"		"			"
I_{OZL}		35	"	GND	7/							"	"									"	1Q		-.2		-2.0			"
		36	"			7/	GND					"	"									"	2Q		"		"			"
		37	"					GND	7/			"	"									"	3Q		"		"			"
		38	"							7/	GND	"	"									"	4Q		"		"			"
		39	"									"	"	GND	7/							"	5Q		"		"			"
		40	"									"	"			7/	GND					"	6Q		"		"			"
		41	"									"	"					GND	7/			"	7Q		"		"			"
		42	"									"	"							7/	GND	"	8Q		"		"			"
V_{OH6}	3006	43	0.8V	-20µA	2.0V							"	2.0V									4.5V	1Q	4.4		4.4		4.4		V
		44	"			2.0V	-20µA					"	"									"	2Q	"		"		"		"
		45	"					-20µA	2.0V			"	"									"	3Q	"		"		"		"
		46	"							2.0V	-20µA	"	"									"	4Q	"		"		"		"
		47	"									"	"	-20µA	2.0V							"	5Q	"		"		"		"
		48	"									"	"			2.0V	-20µA					"	6Q	"		"		"		"
		49	"									"	"					-20µA	2.0V			"	7Q	"		"		"		"
		50	"									"	"							2.0V	-20µA	"	8Q	"		"		"		"
V_{OH7}	3006	51-58	Same tests and terminal conditions as V_{OH6}, except I_{OH} = -6.0 mA.																			5.5 V		3.98		3.7		3.98		"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 53 – Continued.

Symbol	MIL-STD-883 method	Cases R, S, and 2	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	$\overline{OE}$	1Q	1D	2D	2Q	3Q	3D	4D	4Q	GND	$\overline{E}$	5Q	5D	6D	6Q	7Q	7D	8D	8Q	V_{CC}		Min	Max	Min	Max	Min	Max	
V_{OL6}	3007	59	0.8V	20µA	0.8V	0.8V	20µA	20µA	0.8V	0.8V	20µA	GND	2.0V	20µA	0.8V	0.8V	20µA	20µA	0.8V	0.8V	20µA	4.5V	1Q		0.1		0.1		0.1	V
		60	"									"	"									"	2Q		"		"		"	"
		61	"									"	"									"	3Q		"		"		"	"
		62	"									"	"									"	4Q		"		"		"	"
		63	"									"	"									"	5Q		"		"		"	"
		64	"									"	"									"	6Q		"		"		"	"
		65	"									"	"									"	7Q		"		"		"	"
		66	"									"	"									"	8Q		"		"		"	"
V_{OL7}	3007	67-74	Same tests and terminal conditions as V_{OL6}, except I_{OL} = 6.0 mA.																			5.5 V			.25		.40		.25	"
I_{OS4}	3011	75	GND	GND	4.0V	4.0V	GND	GND	4.0V	4.0V	GND	GND	4.0V	GND	4.0V	4.0V	GND	GND	4.0V	4.0V	GND	4.0V	1Q	-10	-135	-10	-135	-10	-135	mA
		76	"		"	"			"	"		"	"		"	"			"	"		"	2Q	"	"	"	"	"	"	"
		77	"		"	"			"	"		"	"		"	"			"	"		"	3Q	"	"	"	"	"	"	"
		78	"		"	"			"	"		"	"		"	"			"	"		"	4Q	"	"	"	"	"	"	"
		79	"		"	"			"	"		"	"		"	"			"	"		"	5Q	"	"	"	"	"	"	"
		80	"		"	"			"	"		"	"		"	"			"	"		"	6Q	"	"	"	"	"	"	"
		81	"		"	"			"	"		"	"		"	"			"	"		"	7Q	"	"	"	"	"	"	"
		82	"		"	"			"	"		"	"		"	"			"	"		"	8Q	"	"	"	"	"	"	"
I_{IH}	3010	83	5.5V		5.5V	5.5V			5.5V	5.5V		"	5.5V		5.5V	5.5V			5.5V	5.5V		5.5V	$\overline{OE}$		.05		.1			µA
		84										"										"	1D		"		"			"
		85										"										"	2D		"		"			"
		86										"										"	3D		"		"			"
		87										"										"	4D		"		"			"
		88										"										"	$\overline{E}$		"		"			"
		89										"										"	5D		"		"			"
		90										"										"	6D		"		"			"
		91										"										"	7D		"		"			"
		92										"										"	8D		"		"			"
I_{IL}	3009	93	GND		GND	GND			GND	GND		"	GND		GND	GND			GND	GND		"	$\overline{OE}$		-.05		-.1			"
		94										"										"	1D		"		"			"
		95										"										"	2D		"		"			"
		96										"										"	3D		"		"			"
		97										"										"	4D		"		"			"
		98										"										"	$\overline{E}$		"		"			"
		99										"										"	5D		"		"			"
		100										"										"	6D		"		"			"
		101										"										"	7D		"		"			"
		102										"										"	8D		"		"			"
																								Subgroup 4 T_C = +25°C						
																								Min	Max					
C_{in}	3012	103	2/		2/	2/			2/	2/		GND	2/		2/	2/			2/	2/		GND	$\overline{OE}$		10					pF
		104										"										"	1D		"					"
		105										"										"	2D		"					"
		106										"										"	3D		"					"
		107										"										"	4D		"					"
		108										"										"	5D		"					"
		109										"										"	6D		"					"
		110										"										"	7D		"					"
		111										"										"	8D		"					"
		112										"	2/									"	$\overline{E}$		"					"
C_O	3012	113	2.0V	2/			2/	2/			2/	GND	GND	2/			2/	2/			2/	5.5V	1Q							"
		114	"									"	"									"	2Q							"
		115	"									"	"									"	3Q							"
		116	"									"	"									"	4Q							"
		117	"									"	"									"	5Q							"
		118	"									"	"									"	6Q							"
		119	"									"	"									"	7Q							"
		120	"									"	"									"	8Q							"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 53 – Continued.

Symbol	MIL-STD-883 method	Cases R, S, and 2	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 7 T_C = +25°C		Subgroup 8 T_C = +125°C		Subgroup 8 T_C = -55°C		
		Test no.	$\overline{OE}$	1Q	1D	2D	2Q	3Q	3D	4D	4Q	GND	$\overline{E}$	5Q	5D	6D	6Q	7Q	7D	8D	8Q	V_{CC}		Min	Max	Min	Max	Min	Max	
Truth table 3/	3014	121	B	H	A	A	H	H	A	A	H	GND	A	H	A	A	H	H	A	A	H	4.5V	All outputs 4/							
		122	"	L	B	B	L	L	B	B	L	"	A	L	B	B	L	L	B	B	L	"	"							
		123	"	L	B	A	L	L	B	A	L	"	B	L	A	A	L	L	A	A	L	"	"							
		124	"	L	A	A	L	L	A	A	L	"	B	L	A	A	L	L	A	A	L	"	"							
		125	"	H	A	A	H	H	A	A	H	"	A	H	A	A	H	H	A	A	H	"	"							
		126	"	H	A	A	H	H	A	A	H	"	B	H	A	A	H	H	A	A	H	"	"							
		127	"	H	B	B	H	H	B	B	H	"	B	H	B	B	H	H	B	B	H	"	"							
																								Subgroup 9 T_C = +25°C		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C		
																								Min	Max	Min	Max	Min	Max	
t_{PHL1}	3003 (Fig. 3)	128	GND	OUT	IN	IN	OUT	OUT	IN	IN	OUT	GND	IN	OUT	IN	IN	OUT	OUT	IN	IN	OUT	4.5V	1D to 1Q	6	31	6	41	6	31	ns
		129	"									"	"									"	2D to 2Q	"	"	"	"	"	"	"
		130	"									"	"									"	3D to 3Q	"	"	"	"	"	"	"
		131	"									"	"									"	4D to 4Q	"	"	"	"	"	"	"
		132	"									"	"									"	5D to 5Q	"	"	"	"	"	"	"
		133	"									"	"									"	6D to 6Q	"	"	"	"	"	"	"
		134	"									"	"									"	7D to 7Q	"	"	"	"	"	"	"
		135	"									"	"									"	8D to 8Q	"	"	"	"	"	"	"
t_{PLH1}	3003 (Fig. 3)	136-143	Same tests, terminal conditions and limits as specified above for t_{PHL1}.																											
t_{PLH2}	3003 (Fig. 3)	144	GND	OUT	4.5V	4.5V	OUT	OUT	4.5V	4.5V	OUT	GND	IN	OUT	4.5V	4.5V	OUT	OUT	4.5V	4.5V	OUT	4.5V	$\overline{E}$ to 1Q	6	31	6	41	6	31	ns
		145	"									"	"									"	$\overline{E}$ to 2Q	"	"	"	"	"	"	"
		146	"									"	"									"	$\overline{E}$ to 3Q	"	"	"	"	"	"	"
		147	"									"	"									"	$\overline{E}$ to 4Q	"	"	"	"	"	"	"
		148	"									"	"									"	$\overline{E}$ to 5Q	"	"	"	"	"	"	"
		149	"									"	"									"	$\overline{E}$ to 6Q	"	"	"	"	"	"	"
		150	"									"	"									"	$\overline{E}$ to 7Q	"	"	"	"	"	"	"
		151	"									"	"									"	$\overline{E}$ to 8Q	"	"	"	"	"	"	"
t_{PHL2}	3003 (Fig. 3)	152	"	OUT	GND	GND	OUT	OUT	GND	GND	OUT	"	"	OUT	GND	GND	OUT	OUT	GND	GND	OUT	"	$\overline{E}$ to 1Q	"	"	"	"	"	"	"
		153	"									"	"									"	$\overline{E}$ to 2Q	"	"	"	"	"	"	"
		154	"									"	"									"	$\overline{E}$ to 3Q	"	"	"	"	"	"	"
		155	"									"	"									"	$\overline{E}$ to 4Q	"	"	"	"	"	"	"
		156	"									"	"									"	$\overline{E}$ to 5Q	"	"	"	"	"	"	"
		157	"									"	"									"	$\overline{E}$ to 6Q	"	"	"	"	"	"	"
		158	"									"	"									"	$\overline{E}$ to 7Q	"	"	"	"	"	"	"
		159	"									"	"									"	$\overline{E}$ to 8Q	"	"	"	"	"	"	"
t_{PHZ}	3003 (Fig. 3)	160	IN	OUT	4.5V	4.5V	OUT	OUT	4.5V	4.5V	OUT	"	4.5V	OUT	4.5V	4.5V	OUT	OUT	4.5V	4.5V	OUT	"	$\overline{OE}$ to 1Q	5	"	5	"	5	"	"
		161	"									"	"									"	$\overline{OE}$ to 2Q	"	"	"	"	"	"	"
		162	"									"	"									"	$\overline{OE}$ to 3Q	"	"	"	"	"	"	"
		163	"									"	"									"	$\overline{OE}$ to 4Q	"	"	"	"	"	"	"
		164	"									"	"									"	$\overline{OE}$ to 5Q	"	"	"	"	"	"	"
		165	"									"	"									"	$\overline{OE}$ to 6Q	"	"	"	"	"	"	"
		166	"									"	"									"	$\overline{OE}$ to 7Q	"	"	"	"	"	"	"
		167	"									"	"									"	$\overline{OE}$ to 8Q	"	"	"	"	"	"	"
t_{PLZ}	3003 (Fig. 3)	168	"	OUT	GND	GND	OUT	OUT	GND	GND	OUT	"	4.5V	OUT	GND	GND	OUT	OUT	GND	GND	OUT	"	$\overline{OE}$ to 1Q	"	"	"	"	"	"	"
		169	"									"	"									"	$\overline{OE}$ to 2Q	"	"	"	"	"	"	"
		170	"									"	"									"	$\overline{OE}$ to 3Q	"	"	"	"	"	"	"
		171	"									"	"									"	$\overline{OE}$ to 4Q	"	"	"	"	"	"	"
		172	"									"	"									"	$\overline{OE}$ to 5Q	"	"	"	"	"	"	"
		173	"									"	"									"	$\overline{OE}$ to 6Q	"	"	"	"	"	"	"
		174	"									"	"									"	$\overline{OE}$ to 7Q	"	"	"	"	"	"	"
		175	"									"	"									"	$\overline{OE}$ to 8Q	"	"	"	"	"	"	"

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TABLE III. Group A inspection for device type 53 – Continued.

Symbol	MIL-STD-883 method	Cases R, S, and 2	Terminal conditions 1/																			Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 9 T_C = +25°C		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C		
		Test no.	$\overline{OE}$	1Q	1D	2D	2Q	3Q	3D	4D	4Q	GND	$\overline{E}$	5Q	5D	6D	6Q	7Q	7D	8D	8Q	V_{CC}		Min	Max	Min	Max	Min	Max	
t_{PZH}	3003 (Fig. 3)	176	IN	OUT	4.5V	4.5V	OUT	OUT	4.5V	4.5V	OUT	GND	4.5V	OUT	4.5V	4.5V	OUT	OUT	4.5V	4.5V	OUT	4.5V	$\overline{OE}$ to 1Q	5	31	5	41	5	31	ns
		177	"									"	"									"	$\overline{OE}$ to 2Q	"	"	"	"	"	"	"
		178	"									"	"									"	$\overline{OE}$ to 3Q	"	"	"	"	"	"	"
		179	"									"	"									"	$\overline{OE}$ to 4Q	"	"	"	"	"	"	"
		180	"									"	"									"	$\overline{OE}$ to 5Q	"	"	"	"	"	"	"
		181	"									"	"									"	$\overline{OE}$ to 6Q	"	"	"	"	"	"	"
		182	"									"	"									"	$\overline{OE}$ to 7Q	"	"	"	"	"	"	"
		183	"									"	"									"	$\overline{OE}$ to 8Q	"	"	"	"	"	"	"
t_{PZL}	3003 (Fig. 3)	184-191	Same tests, terminal conditions and limits as specified above for t_{PLZ}.																											
t_{THL}	3004 (Fig. 3)	192	GND	OUT	IN	IN	OUT	OUT	IN	IN	OUT	GND	4.5V	OUT	IN	IN	OUT	OUT	IN	IN	OUT	4.5V	1Q	2	12	2	16	2	12	ns
		193	"									"	"									"	2Q	"	"	"	"	"	"	"
		194	"									"	"									"	3Q	"	"	"	"	"	"	"
		195	"									"	"									"	4Q	"	"	"	"	"	"	"
		196	"									"	"									"	5Q	"	"	"	"	"	"	"
		197	"									"	"									"	6Q	"	"	"	"	"	"	"
		198	"									"	"									"	7Q	"	"	"	"	"	"	"
		199	"									"	"									"	8Q	"	"	"	"	"	"	"
t_{TLH}	3004 (Fig. 3)	200-208	Same tests, terminal conditions and limits as specified above for t_{THL}.																											

1/ Input pins not designated shall be "high" level logic or "low" level logic, or may be left open provided they do not influence the outcome of the measurement. Output pins not designated shall be tied to the loads or left open provided they do not influence the outcome of the measurement. Exceptions are as follows:

a. V_{IC}(pos) tests, the GND terminal shall be open and the minimum limit of 0.4V applies only where test equipment limitations do not allow for the GND pin to be open during test.
b. V_{IC}(neg) tests, the V_{CC} terminal shall be open.
c. I_{CC} tests, the output terminal shall be open.

2/ See 4.4.1c.

3/ Tests shall be performed in sequence, attributes data only.

4/ H > 2.5V; L < 2.5V; A = 3.7V; B = 0.4V, except for device 53; A = 2.4 V; B = 0.4 V.

5/ $\overline{EN}$ must be held at logic "1" while changing the address lines (A0, A1, A2) to select the next output bit.

6/ Apply 4.5 V / 0 V pulse (4.5 V momentary) prior to each test.

7/ Three-state output conditions are required. For I_{OZL}, set output to high state. For I_{OZH}, set output to low state. Set input pins to $V_{IL} = V_{IL}$ (max) and $V_{IH} = V_{IH}$ (min), as required.

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4.5 Methods of inspection. Methods of inspection shall be specified and as follows:

4.5.1 Voltage and current. Unless otherwise specified, all voltages given are referenced to the microcircuit GND terminal. Currents given are conventional current and positive when flowing into the referenced terminal.

4.5.2 Burn-in and life test cool down procedures. When the burn-in and life tests are completed and prior to removal of bias voltages, the devices under test (DUT) shall be cooled to within 10°C of their power stable condition at room temperature; then, electrical parameter end-point measurements shall be performed.

TABLE IV. Delta limits at 25°C.

Parameter 1/	Device types	
	All	
I_{CC}		±30 nA

1/ The above parameters shall be recorded before and after the required burn-in and life tests to determine deltas (Δ).

4.5.3 Quiescent supply current (I_{CC} test). When performing quiescent supply current measurements (I_{CC}), the meter shall be placed so that all currents flow through the meter.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements are as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. PIN and compliance identifier, if applicable (see 1.2).

c. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

d. Requirements for certificate of compliance, if applicable.

e. Requirements for notification of change of product or process to contracting activity in addition to notification to the qualifying activity, if applicable.

f. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

g. Requirements for product assurance options.

h. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements should not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

i. Requirements for "JAN" marking.

j. Packaging requirements (see 5.1).

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6.3 Superseding information. The requirements of MIL-M-38510 have been superseded to take advantage of the available Qualified Manufacturer Listing (QML) system provided by MIL-PRF-38535. Previous references to MIL-M-38510 in this document have been replaced by appropriate references to MIL-PRF-38535. All technical requirements now consist of this specification and MIL-PRF-38535. The MIL-M-38510 specification sheet number and PIN have been retained to avoid adversely impacting existing government logistics systems and contractors parts lists.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List QML-38535 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DSCC-VQ, P.O. Box 3990, Columbus, Ohio 43218-3990.

6.5 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-PRF-38535, MIL-HDBK-1331, and as follows:

T_C ..... Case temperature.
C_{IN}..... Input terminal-to-GND capacitance.
GND ..... Ground zero voltage potential.
I_{CC} ..... Quiescent supply current.
T_A ..... Free air temperature.
V_{CC} ..... Positive supply voltage.
C_{PD}..... Power dissipation capacitance.

6.6 Logistic support. Lead materials and finishes (see 3.4) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class S for National Aeronautics and Space Administration or class B for Department of Defense (see 1.2.2), lead material and finish A (see 3.4). Longer length leads and lead forming should not affect the part number.

6.7 Data reporting. When specified in the purchase order or contract, a copy of the following data, as applicable, will be supplied.

a. Attributes data for all screening tests (see 4.2) and variables data for all static burn-in, dynamic burn-in, and steady-state life tests (see 3.6).

b. A copy of each radiograph.

c. The technology conformance inspection (TCI) data (see 4.4).

d. Parameter distribution data on parameters evaluated during burn-in (see 3.6).

e. Final electrical parameters data (see 4.2d).

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6.8 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-35810 device types and may have slight physical variations in relation to case size. The presence of this information should not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-PRF-38535.

Military device typez	Generic-industry type
01	54HC75
02	54HC259
03	54HC373
04	54HC533
05	54HC563
06	54HC573
53	54HCT373

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5962-2005-021)

Review activities:
Army - MI, SM
Navy - AS, CG, MC, SH, TD
Air Force – 03, 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST Online database at http://assist.daps.dla.mil.

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